# Yang–Mills on a finite torus along Bałaban's route: a verified reduction

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**Abstract**

Bałaban's renormalization-group construction of four-dimensional SU(2) lattice Yang–Mills theory rests on the control of the running coupling, his Theorem 2, whose proof was never published. We prove that a weak form of Theorem 2 (two-sided bounds on $1/g_k^2$ with an additive constant), which is enough for Bałaban's Theorem 1, follows from one uniform inequality (H2) on a two-step minimizer map together with named fixed-level and history inputs; the shooting lemmas, the arithmetic of Theorem K, and the bridge lemmas of the certifier are formalized in Lean 4. We prove that, on a fixed torus, a uniform moment bound (H-U) gives compactness of the lattice Schwinger functions of the centred plaquette density, and that every limit point satisfies the torus forms of the Osterwalder–Schrader axioms OS0–OS3. The bound (H-U) is reduced, not proved: it rests on sourced versions of Bałaban's bounds, on a hypothesis (FX) that is now reduced to a holomorphy bound (H-hol) for Bałaban's increments together with history bounds and three named inputs of the large-field step, and on a bridge from the source form to (H-U), and one dependency table records the status of each block. Numerically, (H2) holds in floating point with margins of 31% and 58% at $L = 3$ and $L = 5$, and a rigorous interval enclosure holds at one point at $L = 3$, and at $L = 13$, inside Bałaban's range $L > 11$, the one-step bound $\sup \rho_1^2 = 4.37 < 13$ holds in floating point (not yet verified); (H2) is certified at no $L$, and the result is not a solution of the Clay Millennium problem.

## Contents

**1 Introduction** **3**
1.1 The problem . . . 3
1.2 Bałaban's programme and the missing Theorem 2 . . . 3
1.3 The two targets and the main results . . . 4
1.4 Status conventions . . . 5
1.5 Relation to other work . . . 6
1.6 Organization . . . 7

**2 Setup and notation** **7**
2.1 Lattice, measure, and scales . . . 7
2.2 Range of $L$ . . . 7
2.3 The coupling sequence . . . 8
2.4 One-loop notation . . . 8
2.5 Observables and Schwinger functions . . . 8
2.6 Names . . . 9

**3 The coupling flow and Theorem 2** **9**
3.1 What Bałaban's Theorem 1 actually needs . . . 9
3.2 The weak form proved conditionally . . . 10
3.3 Literal (0.31) . . . 10
3.4 Shooting lemmas . . . 10
3.4.1 Lemma D′ (shooting without early positivity) [PROVED+VERIFIED; LEAN] . . . 11
3.4.2 Lemma D″ (tail-sum form; no index $J$) [PROVED+VERIFIED; LEAN] . . . 11
3.4.3 Lemma D″ meets Balaban's proof: the cross-scale sweep [PROVED, UNVERIFIED (SWEEP); 4 OF 19 TABLE ROWS VERIFIED] . . . 12
3.4.4 Lemma D and Lemma E [PROVED+VERIFIED; LEAN] . . . 13
3.4.5 History dependence (L5) [OPEN] . . . 13
3.5 The exact structure of the one-loop sums . . . 13
3.5.1 Telescoping [PROVED+VERIFIED, CORRECTED SCOPE] . . . 14
3.5.2 Anchor-free identity [PROVED+VERIFIED] . . . 14
3.5.3 The constant $c$ [PROVED+VERIFIED] . . . 14
3.5.4 The decomposition $S_k = ck \log L + \kappa_W^F - \kappa_k$ . . . 14
3.5.5 Level-uniform symbol facts used below [PROVED+VERIFIED] . . . 15
3.6 Bounded window sums: two formally distinct conditional routes . . . 15
3.6.1 Route K: Theorem K [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS] . . . 16
3.6.2 Route B: Theorem $B_{PW}$ [PROVED+VERIFIED, CONDITIONAL (LOGIC OF THEOREM $B_{PW}$)] . . . 17
3.6.3 Route B inputs after `BETA_FP_U` [PROVED, UNVERIFIED; NO VERIFIER FILE] . . . 18
3.6.4 Comparison . . . 18
3.7 The hypothesis (B2) of Route K, and the single uniform inequality (H2) . . . 19
3.7.1 Proved . . . 19
3.7.2 Refuted, and the failed one-step bound . . . 20
3.7.3 (H2) and the two-step theorems [PROVED+VERIFIED, CONDITIONAL ON (H2)] 20
3.7.4 One-step sufficiency [ARITHMETIC PROVED; INPUTS NUMERICAL] . . . 21
3.7.5 The precise remaining inequality . . . 21
3.8 Conclusion of the reduction . . . 21
3.9 The convergence route: what it adds, and its status . . . 22
3.9.1 Components and status . . . 22
3.9.2 Early steps and $j_*$ . . . 23

**4 The finite-torus existence theorem** **23**
4.1 What "exists" means here . . . 23
4.2 Hypothesis delivered by the blocks . . . 24
4.3 Theorem (finite-torus existence, compactness form) . . . 24
4.4 Theorem (nontriviality), conditional on (B-NT) . . . 25
4.5 Proof modulo blocks . . . 25
4.5.1 (B-T2) Theorem 2: the running coupling . . . 25
4.5.2 (B-S) Sourced bounds E14-a..g . . . 25
4.5.3 (B-N1) Bridge: source-form bounds ⇒ (H-U) [OPEN] . . . 26
4.5.4 (B-C) Compactness [PROVED HERE, GIVEN (H-U)] . . . 26
4.5.5 (B-REG) Regularity (OS0) [PROVED HERE, GIVEN (H-U)] . . . 27
4.5.6 (B-TI) Translation and hypercubic invariance (OS1) [PROVED HERE, GIVEN (H-U)] . . . 27

4.5.7 (B-RP) Reflection positivity (OS2) . . . 28
4.5.8 (OS3) Symmetry and reality [PROVED HERE] . . . 28
4.5.9 (B-NT) Nontriviality: what is needed . . . 29
4.6 Dependency table . . . 29
4.7 What the theorem does not give . . . 38
4.8 Corrections after verification . . . 39

**5 Numerical results** **39**
5.1 First increment $\beta_1(0)$ . . . 40
5.2 Later increments ($L = 2$ only) . . . 40
5.3 Quantities of the Route K proof . . . 41
5.4 Values of (H2) [NUMERICAL; NONE CERTIFIED] . . . 42
5.5 One-step values . . . 43
5.6 A rigorous point enclosure at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED] . . . 43
5.7 Certification plan and $j_0$ [DESIGNED, NOT DONE] . . . 44

**6 What remains** **44**
6.1 Certification . . . 44
6.2 Fixed-level and structural inputs of Theorem 2w . . . 45
6.3 Items of the sources block . . . 46
6.4 Verification debt . . . 47
6.5 Open for stronger statements . . . 47

**A Lean formalization** **47**

**B Index of results by status** **48**

**C Label map** **52**

**D Revision history** **54**

# 1 Introduction

## 1.1 The problem

The Clay problem on quantum Yang–Mills theory asks for a four-dimensional quantum gauge theory with compact simple gauge group that satisfies axioms at least as strong as those of Osterwalder and Schrader [19, 20], and that has a mass gap [15]. One constructive approach starts from Wilson's lattice action on a torus with lattice spacing $\varepsilon$ and takes $\varepsilon \to 0$. The bare coupling $g_0(\varepsilon)$ must then go to zero at the rate that asymptotic freedom prescribes, so that the effective coupling at a fixed physical scale stays fixed. This paper treats the first limit only: lattice spacing to zero on a fixed torus. It does not treat the infinite-volume limit or the mass gap (Section 4.7).

## 1.2 Bałaban's programme and the missing Theorem 2

Between 1984 and 1989 Bałaban built a renormalization-group (RG) analysis of lattice gauge theories [1, 2, 3, 5, 4, 6, 7, 8, 10, 9, 11, 12]. His main output for $d = 4$ is Theorem 1 of CMP 122 II: uniform bounds on the effective densities at every RG scale, under the hypothesis that the sequence

of effective couplings stays small. The hypothesis was to be discharged by his Theorem 2, which controls the running coupling. The two statements read as follows.

Theorem 2 (CMP 109 (1987), p. 259): "Let $d = 4$, $G = SU(2)$, and let $\gamma$ be a sufficiently small positive constant, then for a sufficiently small positive $g$ there exists a bare coupling constant $g_0 = g_0(\varepsilon, g)$ such that the sequence of the effective coupling constants $g_k$ is contained in the interval $]0, \gamma]$, and $g_K = g$. Moreover, there exist constants $\beta, \beta'$, $0 < \beta \leq \beta'$, such that $g_k$ satisfy the inequality $1/g^2 + \beta \log(L^k \varepsilon)^{-1} \leq 1/g_k^2 \leq 1/g^2 + \beta' \log(L^k \varepsilon)^{-1}$. (0.31)" Balaban adds that the proof "will be given in a separate paper". It was never published (CMP 122 II p. 355, verbatim: "has not been published yet, so we have Theorem 1 with the assumption").

Theorem 1 (CMP 122 II p. 355): "If the sequence of the effective coupling constants is contained in an interval $]0, \gamma]$ with a sufficiently small positive $\gamma$, then the effective densities $\rho_k$ have the form, and satisfy all the conditions and bounds, described in Sect. 2 [III]." The bound (0.1) there reads $\chi_k \exp[-(1/g_k^2) A(U_k) - E_- |T_\eta|] \leq \rho_k \leq \exp E_+ |T_\eta|$ (corrected reading, `BALABAN_EXCERPTS_CHECK`[1] item 3).

Without Theorem 2, the hypothesis of Theorem 1 is an assumption on the unknown coupling flow. Every continuum statement built on Bałaban's bounds inherits this assumption.

### 1.3 The two targets and the main results

The paper has two targets. The first target is a proof of enough of Theorem 2 to remove the coupling hypothesis from Theorem 1. The second target is a finite-torus existence theorem for the continuum limit in compactness form, with the Osterwalder–Schrader properties proved on the limit. Both results are reductions: each holds under named hypotheses, and the paper records the status of every hypothesis.

**Theorem A** (weak Theorem 2; proved in Section 3)**.** *Let $G = SU(2)$, $d = 4$, and $L$ odd, $L > 11$ (or $L$ odd, $L \geq 3$, with Bałaban's inductive assumption (H0) and his construction of $\beta_j(g)$ at this $L$ as explicit hypotheses).*

*(i) Assume the uniform inequality (H2) (Section 3.7.3), the fixed-level inputs (PI), (F), $(M_j)$, V1, (G2) (Section 3.6.1), the history hypothesis L5 (Section 3.4.5), and the $K$-independence of $\beta_j(g)$ (item 9 of Section 6). Then the conclusions (b), (c′), (S) of Theorem 3.1 (Theorem 2w) hold. Status of the implication:* [PROVED+VERIFIED, CONDITIONAL]*, by Route K (Theorem 3.8 and Lemma 3.4).*

*(ii) Given the conclusions of Theorem 2w, Bałaban's Theorem 1 holds for the bare coupling $g_0(K, g)$, for every $K$ and every $g \in ]0, g_1]$, without his coupling assumption. Status:* [PROVED, UNVERIFIED (SWEEP), CONDITIONAL ON ROUTE K OR ROUTE B INPUTS, ON L5, AND ON BALABAN'S PUBLISHED PROOFS] *(Section 3.8).*

*A second, formally distinct route (Route B, Theorem $B_{\mathrm{PW}}$ in Section 3.6.2) also gives (i), under its own input set (items B1–B7 of Section 6.2). Both routes rest on (H2).*

**Theorem B** (finite-torus existence; proved in Section 4)**.** *Let $G = SU(2)$, $d = 4$, $L$ odd and large, and $\ell \in 2\mathbb{N}$ fixed. Let $g_0 = g_0(\varepsilon, g)$ be a shooting solution whose effective couplings satisfy $g_k \in ]0, \gamma]$ for $0 \leq k \leq K$ and $g_K = g$. Assume the uniform bound (H-U) of Definition 4.1. Then the lattice Schwinger functions of the centred plaquette density are relatively compact as $\varepsilon = L^{-K} \to 0$, and every limit family satisfies the torus forms of (OS0)–(OS3). This is Theorem 4.2,* [PROVED MODULO NAMED BLOCKS]*. Nontriviality is the separate Theorem 4.3, conditional on (NT).*

[1] Project files named in this paper live in the `notes/` directory of the `yang-mills` repository [14].

The hypothesis (H-U) is the output of three blocks: Theorem 2 (B-T2, that is, Theorem A), sourced versions of Bałaban's bounds (B-S), and a bridge from the source form to (H-U) (B-N1). The dependency table, Section 4.6, gives the status of each block and is the single status table of the paper. Section 6 lists what remains.

What the paper does not give: literal (0.31) at every $K$ (Section 3.3), the limit of $\beta_j(0)$ and a rate (Section 3.9), uniqueness of the limit, infinite volume, rotation invariance, and a mass gap (Section 4.7). Theorem B is a compactness ("weak-existence") result. Under Jaffe–Witten footnote 2 it is not an admissible solution of the Millennium problem (Section 4.1).

### 1.4 Status conventions

Every result carries a status tag in small capitals. The tags mean the following.

- [PROVED+VERIFIED]: proved in a project file and confirmed by an independent verifier file.
- [PROVED+VERIFIED, CONDITIONAL ON $X$]: the implication is confirmed; $X$ is open or assumed.
- [PROVED, UNVERIFIED]: proved in a project file; no verifier has graded it yet.
- [LEAN]: formalized in Lean 4 + Mathlib, no `sorry`, axioms propext, Classical.choice, Quot.sound only.
- [VERIFIED QUOTE]: read on the page image of Balaban's paper (`reports/BALABAN_EXCERPTS_CHECK.md` or a verifier file).
- [NUMERICAL, REPRODUCED]: floating-point result reproduced by a verifier. [NUMERICAL, RUN BY THE VERIFIER]: run once, by a verifier. [NUMERICAL]: not reproduced. None is certified.
- [ARGUED]: a plausibility argument, not a proof. [OPEN]: no proof. [REFUTED]: shown false.
- [BALABAN]: an input taken from Balaban's published papers, used as stated there.
- [NOT WRITTEN]: a proof plan; no project file contains the proof.
- [CITED]: a standard result; its text was not re-read. [CITED, SOURCE CHECKED]: the source was read, the transfer to the lattice is standard and not proved here.
- [PROVED HERE]: a short proof given in full in this paper (Section 4). [HYPOTHESIS]: assumed, and owned by a named block. [FORMAL]: a formal step whose analytic justification is not written. [DERIVED SKETCH]: a route with steps listed, not a proof.
- [INTERVAL ENCLOSURE, VERIFIED]: a rigorous interval-arithmetic enclosure, rerun by a verifier.
- [VERIFIED REDUCTION]: a reduction confirmed by a verifier file; the inputs stay open.
- [PROVED MODULO NAMED BLOCKS]: proved given the hypotheses owned by the named blocks.

A compound tag such as [PROVED+VERIFIED, CONDITIONAL ON (H2)] combines these meanings. Tags refer to the project files cited next to them. In [VERIFIED QUOTE], a quote of a source other than Bałaban's papers was checked verbatim against a copy of that source in the project. Project files are cited as `NAME.md`; a section sign after a file name (for example `KAPPA_BOUNDED_VERIFY` §4) refers to a section of that file, not of this paper. "Grade $n$" and "item $n$" after a verifier file name refer to the numbered verdicts of that file. The paper as a whole is not verified as one document.

### 1.5 Relation to other work

Two recent works overlap with this paper. The record of how their content reached this project is given below; it is kept in the form fixed when the overlap was found.

Related work 1. E. Sussman and L. Alpöge (with Claude; README: "written largely by LLM ... Novel mathematical ideas should be attributed to Claude (under the guidance of LA)"), "The Yang–Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0" (LaTeX, files `theorem_0_complete_account_raw_{master,I,II,III}.tex`, 424k lines), and "SU(2) Yang–Mills theory in four dimensions via Bałaban's renormalisation group: construction, axioms and non-triviality" (`ym4_manuscript_zeroth_draft.pdf`, dated 6 October 2026) [25, 24]. Repository: `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit `1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310` (2026-10-27T05:45:28-05:00), accessed 2026-10-27. Repository created 2026-10-06. Disclaimer in README: "This repository contains work-in-progress, written largely by LLM. Its content has not been satisfactorily vetted. All claims should be understood as tentative, pending verification." Suggested keys: SA26-T0 (account), SA26-YM4 (draft).

Related work 2. OpenAI, "The canonical massive continuum limit of the two-dimensional O(3) model" (October 4, 2026), `github.com/openai/math`, family 215 (preprints folder; read locally from the PDF, `reports/ONLINE_SWEEP.md` §1.2; text in `reports/o3_continuum.txt`) [18]. Suggested key: OAI26-O3.

*How content from these works reached this project.* The record (file times, `PLAN.md`, read lists; `THEOREM2_PROOF_v2_VERIFY` §I) shows the following.

1. Section 3 was written without reading the Sussman–Alpöge repository. Only the citation block above (`reports/SUSSMAN_ALPOGE_MAP.md` §5.1) was read.

2. First channel. On the night of 2026-10-27, `reports/ONLINE_SWEEP.md` §2 (about 20:20) read the Sussman–Alpöge repository directly. It reported that they prove a weak two-sided flow with an additive constant, never need per-step positivity, use a first-passage cutoff, and invoke only containment $g_k \le \gamma$. This is the first project file with Sussman–Alpöge content.

3. Second channel. `reports/SUSSMAN_ALPOGE_MAP.md` (about 20:28) was written by a separate agent under a firewall. Through it, the bounded-$\kappa$ idea entered the project: the weak form of Theorem 2 needs only $\kappa_k$ bounded, not convergence (their Lemma 4.19 structure; `PLAN.md` line 146). `KAPPA_BOUNDED.md` (about 20:39) received this idea through its prompt. Its read list excludes the map and ONLINE_SWEEP.

4. Lemma D′ and the cross-scale sweep predate any Sussman–Alpöge content in project files. The shooting-without-positivity lemma and the band with additive constant appear in `THEOREM2_PROOF_VERIFY` Point 5(b) (about 20:05) and `LEMMA_DPRIME.md` (about 20:10). `LEMMA_DPRIME_SWEEP.md` (about 20:16–20:23) overlaps ONLINE_SWEEP in time, but its read list is LEMMA_DPRIME and its verifier only; it did not read ONLINE_SWEEP. Lemma D″ is in `KAPPA_BOUNDED.md`, written after the map.

5. Theorem K. Its result form, $|S_k - ck \log L| \le \kappa_*$, matches Sussman–Alpöge Lemma 4.2(b), $|\sum_l \beta_l^0 - kc_0| \le c_2$ (as reported in `PLAN.md` line 146; cited there, not proved). The proof in this project (Prop. 1.1, Lemma 2.1, Prop. 2.2 of `KAPPA_BOUNDED`) was written by firewalled agents from Balaban's papers and project files.

6. O(3) paper. Route B's doubling comparison follows the template of OAI26-O3 Lemma 6.1 (doubling difference; `BETA_FP_PW` uses the "O(3) Lemma 6.1 form"). The idea of a cou-

pling band in place of per-step positivity also appears there (ONLINE_SWEEP §1.2(a)). `BETA_FP_PW` read ONLINE_SWEEP §1.2, §5, §8 and o3_continuum Lemma 6.1, and did not read ONLINE_SWEEP §2.

7. Route B (other than the O(3) template), the numerics, and the symbol results of Sections 3.5.5 and 3.7 were derived by firewalled agents from Balaban's papers and project files.

In short: two agents read the Sussman–Alpöge repository, the web sweep (item 2) and the mapping agent (item 3), and neither wrote a derivation. The derivation agents were firewalled; their read lists exclude the repository, the map, and ONLINE_SWEEP §2. One idea crossed the firewall: the bounded-$\kappa$ route reached `KAPPA_BOUNDED.md` through the map (item 3). Their public, timestamped account predates this one. Any write-up must cite SA26-T0, SA26-YM4, and OAI26-O3, and state these overlaps.

### 1.6 Organization

Section 2 fixes the notation for both parts of the paper. Section 3 proves the reduction of Theorem 2 (Theorem A). Section 4 proves the finite-torus existence theorem modulo named blocks (Theorem B) and contains the dependency table. Section 5 collects the numerical results. Section 6 lists what remains. Appendix A lists the Lean formalization, Appendix B indexes the results by status, Appendix C maps the numbering of this paper to the names used in the project files, and Appendix D records the revision history.

## 2 Setup and notation

This section fixes the notation for the whole paper. Sections 2.1–2.4 serve the coupling flow of Section 3. Section 2.5 serves the existence theorem of Section 4. Section 2.6 lists the names that would otherwise collide.

### 2.1 Lattice, measure, and scales

*Notation.* $L$ is odd, $L > 11$, fixed (Balaban, CMP 109 p. 256 [8]). $K \in \mathbb{N}$, $\varepsilon = L^{-K}$. The physical side $\ell \in \mathbb{N}$ is in units of the unit lattice. $T_\ell^4 = (\mathbb{R}/\ell\mathbb{Z})^4$ and $T_\varepsilon = (\varepsilon\mathbb{Z}/\ell\mathbb{Z})^4$, so $N_\varepsilon = \ell L^K$ sites per direction. The Wilson action is $A(U) = \sum_p P_p(U)$, $P_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U(\partial p) \in [0,2]$, over unoriented plaquettes $p$. The measure is $\mu_\varepsilon = Z_\varepsilon^{-1} \exp(-g_0^{-2} A(U)) \prod_b dU_b$ (Haar). $x_p$ is the centre of $p$.

The bare coupling $g_0 = g_0(\varepsilon, g)$ is any solution of the shooting problem of Lemma 3.4 (Lemma D″) with terminal coupling $g_K = g$. Lemma D″ gives existence, not uniqueness. Any choice works below.

### 2.2 Range of $L$

Balaban assumes $L$ odd, $L > 11$, as a standing assumption for the whole of CMP 109 (pp. 251–252; OCR line 124, verbatim confirmed in `THEOREM2_PROOF_VERIFY` Step 6) [8]. His construction of $\beta_j(g)$ for $g > 0$ and his inductive assumption (H0) are therefore available only at odd $L > 11$. Every statement below that uses (H0) or $\beta_j(g)$ at $g > 0$ is stated for odd $L > 11$, or carries "(H0) and Balaban's construction at this $L$" as an explicit hypothesis. Statements about the one-loop quantities $\beta_j(0)$, $S_k$, and the symbols $\sigma_j$ hold for odd $L \geq 3$.

### 2.3 The coupling sequence

$\varepsilon = L^{-K}$, so $\log(L^k\varepsilon)^{-1} = (K-k)\log L$. $u_k := 1/g_k^2$. Recursion (CMP 109 (0.18)/(0.20)): $u_{k+1} = u_k - \beta_{k+1}(g_k)$. The increment $\beta_{k+1}$ depends on the whole history $g_0, \dots, g_k$ (CMP 109 p. 298, [VERIFIED QUOTE]). $\beta_j(0)$ is the one-loop (Gaussian) increment; it is history-independent by definition.

Shooting notation: $u_k = 1/g_k^2$, $t = 1/g^2$, $T(k,K) := \sum_{i=k+1}^{K} \beta_i(g_{i-1})$. Clamp $\varphi(u) := \max(u, 1/\gamma^2)^{-1/2}$.

Here $\gamma$ is the upper bound for the couplings in Bałaban's Theorem 1. Bałaban's small constant $\beta_0$ (1/7 in CMP 122 II p. 389) is not a coupling (Section 3.1). In Lemma 3.6 (Lemma E) the symbols $\beta_0$, $\beta_0'$ instead denote lower and upper bounds on $\beta_j(0)$, as in `LEMMAS.md`.

### 2.4 One-loop notation

The letter $K$ has two uses. As an integer it is the number of RG steps, $\varepsilon = L^{-K}$. As a vector $K \in \mathrm{BZ} = [-\pi, \pi]^4$ it is a momentum in the Brillouin zone of the coarse unit lattice; then $k_l = (K + 2\pi l)/L$ are the fine momenta over $K$. The meaning is always clear from context. The other one-loop objects are defined where they are first used:

- $S_k$, the $p \to 0$ $F^2$-coefficient of the composed $k$-step one-loop integral, and its telescoping into $\beta_j(0)$: Section 3.5.1;
- $c = 11N^2/(12\pi^2)$ and $b := c\log L$: Section 3.5.3;
- the $f^m$ Taylor symbols $\sigma_j^{(m)}(K)$ of the coarse kernel at level $j$, and $d_j$, $\kappa_j$, $\kappa_W^F$: Section 3.5.4;
- $\Delta_M$ (Gaussian RG kernel at block size $M = L^j$), $\Delta_\infty$, and the bordered matrix $\mathfrak{B}_j(K)$: Sections 3.5.5 and 3.9;
- the one-step minimizer map $H_j(K)$, $\rho_1$, $\rho_2$, and the inequality (H2): Section 3.7;
- the plane-wave quantities $\beta_{j+1}(p)$, $\Gamma^{ex}$, $\theta_*$ of Route B: Section 3.6.2.

### 2.5 Observables and Schwinger functions

*Observable* (class C1 of `SOURCES.md` §1.2). $\varphi_p := P_p - c_\varepsilon$, $c_\varepsilon := \langle P_p \rangle_{\mu_\varepsilon}$. By hypercubic invariance (Section 4.5.6) $c_\varepsilon$ does not depend on $p$.

*Test space.* $\mathcal{D}_n := \{F \in C^\infty(T^{4n}) : \text{every derivative of } F \text{ vanishes on every diagonal } \{x_i = x_j\}, i \neq j\}$. The norm is $\|F\|_m := \max_{|\alpha|\le m} \sup |\partial^\alpha F|$. On the compact torus this is the Osterwalder–Schrader (OS) Schwartz seminorm without weights.

*Lattice Schwinger functions* (`SOURCES.md` §1.3). $S_{n,\varepsilon}(F) := \sum_{p_1,\dots,p_n} F(x_{p_1}, \dots, x_{p_n}) \langle \varphi_{p_1} \cdots \varphi_{p_n} \rangle_{\mu_\varepsilon}$, $F \in \mathcal{D}_n$. These are moments, not cumulants (`SOURCES_VERIFY` item 5(a)). No field renormalization factor is inserted: `SOURCES_VERIFY` item 8 gives $B_K = \prod c_j$ bounded above and below ($c_j = 1 + O(g_{j-1}^3)$, $\sum g_j^3 < \infty$), so a bounded factor $B_K^n$ changes only the normalization of the limit.

*Scaling check* (`SOURCES_VERIFY` item 4, derived there). $\langle \varphi_p \varphi_{p'} \rangle \sim \varepsilon^8 g^4 |x_p - x_{p'}|^{-8}$ at tree level. So $S_{2,\varepsilon}(F) = O(1)$ for $F \in \mathcal{D}_2$ with no rescaling, while $\sum_p f(x_p)\varphi_p$ has variance $\sim \varepsilon^{-4}$. The limit object is a family of distributions off the diagonals, not a random field.

*Extra hypothesis found while writing the source.* $\ell$ is even (sufficient for the reflection argument; see Section 4.8, item 1). For a reflection of the periodic lattice to exchange two halves with both seams being link planes, $N_\varepsilon$ must be even. With $L$ odd, $N_\varepsilon = \ell L^K$ is even iff $\ell$ is even.

### 2.6 Names

Hypotheses are named by short labels in parentheses, such as (H2), (PI), or (FX); the dependency table (Section 4.6) lists each one with its status. Three names of the project files collide in one document and are changed here; Appendix C records the old names.

- The plane-wave theorem of `BETA_FP_PW` (its Theorem 3.2, called "Theorem B" there) is called Theorem $\mathrm{B_{PW}}$ (Theorem 3.9), so that "Theorem B" means the main Theorem B.
- The three global conditions of the dependency table, called (G1)–(G3) in `ASSEMBLY.md`, are called (GC1)–(GC3). The labels (G1) (constraint conditions of Proposition 3.7) and (G2) (per-cell infinite-volume densities) keep their meaning from the Theorem 2 files.
- The final sourced bound of `SOURCES_D4D5.md`, called (S) there, is called ($\mathrm{S_{src}}$). The label (S) means the smallness condition of Section 3.1.

## 3 The coupling flow and Theorem 2

This section proves Theorem A. Section 3.1 lists what Bałaban's proof of Theorem 1 uses of the coupling sequence; the answer is weaker than his (0.31). Section 3.2 states the weak form, Theorem 2w, that supplies exactly this. Sections 3.4–3.7 prove Theorem 2w from bounded window sums of the one-loop increments, give two routes to these window sums, and reduce the uniform part of both routes to the single inequality (H2). Section 3.8 assembles the result. Its output, the trajectory conditions (b), (c′), (S), is the block (B-T2) of Section 4.

### 3.1 What Bałaban's Theorem 1 actually needs

Source: `LEMMA_DPRIME_SWEEP` §2, §7. The statements of Theorems 1 and 2 are quoted in Section 1.2.

The quoted hypothesis of Theorem 1 is only $g_k \in ]0, \gamma]$. The proof uses more. CMP 119 Theorem 1 (p. 262) assumes "(I.0.33)", a misnumbered reference to (0.31) of CMP 109. Its working form is CMP 119 p. 255–256 (2.6)–(2.9), for scales $m < n$:

- (2.6) $g_n \le (1 + g_n^2 \beta'(n-m))^{1/2} g_m$ and $g_m \le (1+\beta_0) g_n$;
- (2.7) $(\log g_n^{-2})^p \le (1+\beta_0)(\log g_m^{-2})^p$ and $(\log g_m^{-2})^p \le (1 + g_n^2\beta'(n-m))^{\beta_0} (\log g_n^{-2})^p$;
- (2.8), (2.9) the same for $\varepsilon_j = g_j p_0(g_j)$ and $R_j$.

Here $\beta_0$ is Balaban's small constant (1/7 in CMP 122 II p. 389), not a coupling. [VERIFIED QUOTE] for (2.6)–(2.9): page images, `LEMMA_DPRIME_SWEEP` §1; the sweep itself is single-author, see Section 3.4.3.

In $u$-variables these follow from two pairwise bounds:

- (P-up) $u_m \le u_n + B'(n-m)$, with no additive constant;
- (P-low) $u_m \ge u_n + \beta(n-m) - D'$, with $\beta > 0$, $D' \ge 0$ fixed,

plus the smallness condition

- (S) $\gamma^{-2} \ge c_{sw} D'$, $c_{sw} = \max(2, 1/(1-(1+\beta_0)^{-2})) = 64/15$ at $\beta_0 = 1/7$, and $\gamma \le \gamma_*(\beta_0, \beta, B', p_0, p_1, r, \nu, \kappa_0)$ (the conditions that literal (0.31) also needs). $\gamma_*$ is free of the $D'$-ratio conditions, but it depends on $R$ through $B'$ (it contains $\gamma^2 B' \le (1+\beta_0)^2 - 1$ with $B' = 5b/4 + 2R$). The order of constants in Section 3.8 ($R$ before $\gamma$) handles this.

$\beta_0$ must be fixed before $\gamma$.

### 3.2 The weak form proved conditionally

**Theorem 3.1** (Theorem 2w, weak form)**.** *Let $G = SU(2)$, $d = 4$, $L$ odd, $L > 11$ (equivalently: $L$ odd $\geq 3$ with (H0) and Balaban's construction of $\beta_j(g)$ at this $L$ as an explicit hypothesis). There are $\gamma > 0$, $g_1 \in ]0, \gamma]$, $\beta > 0$, $B' \geq \beta$, $D' \geq 0$ such that for every $K \geq 1$ and every $g \in ]0, g_1]$ there is a bare coupling $g_0 = g_0(K, g)$ with*

- *(b) $g_k \in ]0, \gamma]$ for $0 \leq k \leq K$, and $g_K = g$;*
- *(c′) for all $0 \leq m \leq n \leq K$: $u_n + \beta(n - m) - D' \leq u_m \leq u_n + B'(n - m)$;*
- *(S) holds for $\gamma$, $D'$.*

*Consequence: Balaban's (0.31) holds up to the additive constant $D'$ on the lower side, with constants $\beta/\log L$ and $B'/\log L$. The two-sided bound fixes $1/g_k^2$ up to constants, as required.*

Status: [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS OF SECTION 3.6] by Route K (Theorem K + Lemma D″). (B2) is reduced to the single inequality (H2) of Section 3.7 [PROVED+VERIFIED REDUCTION: `B2_TWOSTEP_VERIFY` GRADE 3]. Second route: Route B (`BETA_FP_PW` Theorem 3.2 + Lemma D″). The logic of Theorem $B_{PW}$ is [PROVED+VERIFIED, CONDITIONAL] (`BETA_FP_PW_VERIFY` §6 items 1–4). Its inputs (U), (DF), (T1) from `BETA_FP_U` are [PROVED, UNVERIFIED]; the route is conditional on (U), ($B\theta$), and the plane-wave defects. `BETA_FP_U` reduces (U) to (T1) + (H2) [PROVED, UNVERIFIED]. Both routes also need (H0) at this $L$ and L5 (Section 3.4.5). See Section 3.6.

### 3.3 Literal (0.31)

- (L1) Literal (0.31) for every $K \geq K_0$ and every $g \in ]0, \gamma]$. Needs eventual uniform positivity: there are $J$ and $\beta_* > 0$ with $\beta_j(0) \geq \beta_*$ for all $j > J$, plus $\sup_j |\beta_j(0)| < \infty$ (Lemma 3.2 (d)). [PROVED+VERIFIED; LEAN] as an implication. Theorem K and Theorem $B_{PW}$ do NOT give eventual positivity: they bound window sums, not single increments. Eventual positivity follows from the convergence route (Section 3.9), which is open.
- (L2) Literal (0.31) for every $K \geq 1$. Needs $\inf_j \beta_j(0) > 0$ and $\sup_j \beta_j(0) < \infty$ (Lemmas D + E). If $\beta_j(0) \to \beta_\infty > 0$, then "every $\beta_j(0) > 0$" is equivalent to $\inf_j \beta_j(0) > 0$ (Lemma D′ Step 8). [PROVED+VERIFIED; LEAN] as an implication. Inputs open.
- Literal (0.31) is not needed for Theorem 1: Section 3.1 shows the proof uses (c′) + (S). Literal (0.31) is used directly in one place only, the alternative choice of $\varepsilon_0$ in CMP 119 p. 248, which Balaban does not adopt.
- Necessity. Literal (0.31) at every $K \geq 1$ forces $\beta_K(0) \geq \beta \log L > 0$ for every $K$ (window $k = K - 1$, let $g \to 0$ using (H0); so $L$ odd $> 11$). [PROVED, UNVERIFIED]; elementary; `THEOREM2_PROOF_VERIFY` Point 5(a) agrees. So certified early steps are needed only for (L2), not for Theorem 1.

### 3.4 Shooting lemmas

The shooting lemmas turn bounds on the increments $\beta_j$ into a trajectory with prescribed terminal coupling. Lemma 3.2 needs eventual positivity of $\beta_j(0)$; Lemma 3.4 needs only bounded window sums, and is the one used for Theorem 2w. The notation is that of Section 2.3. Lemmas D′, D″, D, E are abstract statements about a sequence of functions $\beta_j$. Applied to Balaban's $\beta_j(g)$, they need $L$ odd $> 11$ (or (H0) at this $L$) and L5.

### 3.4.1 Lemma D′ (shooting without early positivity) [PROVED+VERIFIED; LEAN]

Sources: `LEMMA_DPRIME.md` §1–§2; `LEMMA_DPRIME_VERIFY.md` grades 1, 2, 5 (CONFIRMED). Step 7 of that file also holds grades 3, 4, 6, which are CORRECTED (Remark 1 overclaim, misreads in §5, ledger); this chapter uses none of them except through the sweep (Section 3.4.3). Lean: `lean/YmLean/LemmaDPrime.lean` (`YmLean.lemma_DPrime`, 405 lines), statement check `lean/VERIFY_LemmaDPrime.md`.

**Lemma 3.2** (Lemma D′)**.** Hypotheses. *(i)* $\beta_{k+1}$ *continuous on* $[0,\gamma_0]$ *and* $|\beta_{k+1}(g)-\beta_{k+1}(0)| \le C_1 g$, $C_1$ *independent of* $k$, $K$. *(ii)* $\sup_j |\beta_j(0)| \le B$. *(iii′) there is* $J$ *with* $\beta_j(0) \ge \beta_\infty/2$ *for all* $j > J$ *(this is all the proof uses of "*$\beta_j(0) \to \beta_\infty > 0$*").*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, \beta_\infty/(4C_1))$; $B' := B + \beta_\infty/4$; $D := J \cdot B'$; $D' := J(B + \beta_\infty/2)$; $g_1 := (1/\gamma^2 + D)^{-1/2}$; $K_0 := J + \lceil 8D'/\beta_\infty \rceil$.

Conclusions.

- *(a)* $T(k,K) \ge -D$ *for all* $k < K$ *along any couplings in* $]0,\gamma]$.
- *(b) For every* $K \ge 1$ *and* $g \in ]0, g_1]$ *there is* $g_0$ *with all* $g_k \in ]0,\gamma]$ *and* $g_K = g$.
- *(c)* $u_K + (\beta_\infty/4)(K-k) - D' \le u_k \le u_K + B'(K-k)$. *The upper side has no constant.*
- *(d) For* $K \ge K_0$*: literal (0.31) with lower constant* $\beta_\infty/8$*, and shooting for every* $g \in ]0,\gamma]$. *If every* $\beta_j(0) > 0$*: literal (0.31) for all* $K$ *(Step 8 = Lemmas D + E).*

*Proof idea.* Clamped flow $\tilde{u}_{k+1} = \tilde{u}_k - \beta_{k+1}(\varphi(\tilde{u}_k))$ is continuous in $u_0$. Its tail sums lie in $[-D, B'(K-k)]$ for every $u_0$, because clamped couplings lie in $]0,\gamma]$. IVT on $[t-D, t+B'K]$ gives $\tilde{u}_K = t$. Then $\tilde{u}_k \ge t - D \ge 1/\gamma^2$ for $g \le g_1$, so the clamp is never active. The bound is derived for the clamped flow first; there is no circularity. For history-dependent $\beta$ this proof needs $\beta_{k+1}$ defined and continuous on the whole cube $]0,\gamma]^{k+1}$; see L5 in Section 3.4.5.

*Lean.* The Lean hypotheses are weaker (ContinuousOn instead of $C^1$; (iii′) instead of convergence). Step 8 is `lemma_DPrime_step8` (`lean/YmLean/LemmaDE.lean`). Lean covers one-argument $\beta_j(g)$ only, not the history form.

*Remark* 3.3 (Remark 4 of `LEMMA_DPRIME.md`). Without $g \le g_1$, (b) can fail at small $K$ ($\beta_1 = \beta_2 = \beta_3 = -0.8$, $K = 3$, $g = \gamma$). [PROVED+VERIFIED]

### 3.4.2 Lemma D″ (tail-sum form; no index $J$) [PROVED+VERIFIED; LEAN]

Source: `KAPPA_BOUNDED.md` §5.3; `KAPPA_BOUNDED_VERIFY` §6 (CONFIRMED: "No $J$ needed"); hand rederivation in `THEOREM2_PROOF_v2_VERIFY` §A.

**Lemma 3.4** (Lemma D″)**.** Hypotheses. *(i) as in 2.1, and the window bound*

$$\textit{(5.1)}\quad T_0(k,K) := \sum_{i=k+1}^{K} \beta_i(0) \in [\, b(K-k) - 2R,\ b(K-k) + 2R\,] \quad \textit{for all } 0 \le k < K,$$

*with* $b := c \log L > 0$ *and some* $R \ge 0$. *(Route K gives* $R = \kappa_*$*; Route B gives* $R = C_B$*.)*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, b/(4C_1))$; $\beta := 3b/4$; $B' := 5b/4 + 2R$; $D = D' := 2R$; $g_1 := (1/\gamma^2 + D)^{-1/2}$.

Conclusions. *(b) of 2.1, and the pairwise bound*

$$\textit{(c′)}\quad u_n + \beta(n-m) - D' \le u_m \le u_n + B'(n-m), \quad 0 \le m \le n \le K,$$

*upper side without constant.*

*Proof.* $|T - T_0| \le C_1\gamma(K-k) \le (b/4)(K-k)$. Add (5.1). Steps 2–5 of Lemma D′ use only continuity and the bracket $\tilde{T} \in [-D, B'(K-k)]$; this bracket now comes from (5.1). The pairwise form is the same window argument with $K \to n$. The proof is the clamped-flow proof of 2.1; for history-dependent $\beta$ the L5 caveat of Section 3.4.5 applies.

Lean: `lean/YmLean/LemmaDDoublePrime.lean` (`YmLean.lemma_DDoublePrime`, 146 lines, no `sorry`, axioms propext, Classical.choice, Quot.sound; `lean/README.md`). It proves (a″), (b) with the clamp inactive on the solution, (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, and (d) below. The Lean hypotheses are weaker than the note: continuity instead of $C^1$; any $\gamma$ with $0 < \gamma \le \gamma_0$ and $C_1\gamma \le b/4$; $R \ge 0$ derived; $K = 0$ allowed; (c′) and (d) hold for every trajectory with couplings in $]0, \gamma]$. One-argument $\beta$ only (L5 not formalized).

(d) [LEAN]: for $n - m \ge 8R/b$, $u_n + (b/2)(n-m) \le u_m \le u_n + B'(n-m)$. So literal (0.31) from (5.1) alone holds, with lower constant $b/2$, only for windows with $n - m \ge 8R/b$; short windows near $K$ are not controlled (see Section 3.3).

### 3.4.3 Lemma D″ meets Balaban's proof: the cross-scale sweep [PROVED, UNVERIFIED (SWEEP); 4 OF 19 TABLE ROWS VERIFIED]

Source: `LEMMA_DPRIME_SWEEP.md` (all 12 Balaban files; OCR grep plus page image for every non-single-scale hit). `LEMMA_DPRIME_VERIFY` Steps 4–5 checked comparisons A–E on page images (CMP 122 I p. 192, p. 198 (1.94); II p. 381 ×2, p. 386) and corrected two misreadings of A (factor $(1+\beta_0)^2(k-j)^{\beta_0}$; $\delta'_j = g_j A_1 p_1(g_j)$). The five comparisons map to four table rows: A = row 11, E = row 12, C and D = row 16, B = row 18.

*Result.* 19 cross-scale uses of the coupling sequence (table in SWEEP §7), all in CMP 119 and CMP 122 I–II; none in CMP 95, 96, 98, 99, 102, 109, 116. Each is covered by (b) + (c′) + (S) of Section 3.1:

- items 1, 7, 10, 12, 17 need (P-up) with no constant (e.g. (2.6) line 1 is exactly (P-up));
- items 2, 3, 11, 13, 15, 16, 18 need (P-low) with fixed multiplicative slack $(1+\beta_0)$ or $(1+\beta_0)^2$; item 2 ((2.6) line 2) gives the constant 64/15; item 11 needs only $2.42D'$;
- items 4 ((2.8)–(2.9), $\varepsilon$ and $R$ ratios), 6 (CMP 119 (3.7), (3.8), (3.12), (3.13)), and 9 (CMP 122 I (1.52)), the last two of the form $\varepsilon_{k+1} \le (1+\beta_0)\varepsilon_k$, use both sides; they follow as items 1 and 3 do (item 4 from items 1–3);
- item 14 (CMP 122 I p. 200, $R_{k-N_0+1}$ vs $R_k$) uses the lower side; it follows as item 3 does;
- item 5 (CMP 119 (2.31), (2.46): $\sum_j g_j^{\kappa_0} \le g_n^{\kappa_0 - 6}$) needs linear growth (P-low) and $\gamma^{-2} \ge 2D'$;
- item 0 is CMP 119 Theorem 1 assuming (0.31); its proof works with (2.6)–(2.9);
- item 8 (CMP 119 p. 284) is in a sketch, not load-bearing.

Caveats [OPEN]: (1) the sweep is a grep plus page images, not a line-by-line read; (2) "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) was not located; (3) Balaban says $\beta_0$ "can be chosen arbitrarily small, if $g$ is sufficiently small"; under (P-low) the floor is $\beta_0 \gtrsim D\gamma^2/2$, so $\beta_0$ must be fixed first; no step was found that needs $\beta_0 \to 0$; (4) the sweep has no independent verifier beyond rows 11, 12, 16, 18; (5) `BALABAN_EXCERPTS_CHECK` item 11 lists readings not checked on page images: Π (4.37)/(5.1), CMP 119 (2.42), the (2.12)/(2.23) ambiguity, the "$L$ odd" page, the Sect. 3 estimates, and the 1989 bound lists.

### 3.4.4 Lemma D and Lemma E [PROVED+VERIFIED; LEAN]

Sources: `LEMMAS.md` (corrected), `LEMMAS_DE_VERIFY.md` §1.4, §1.5, §2.4; `lean/YmLean/LemmaDE.lean` (`lemma_D`, `lemma_E`, `lemma_DE`).

**Lemma 3.5** (Lemma D)**.** *If $b \le \beta_j(g) \le b'$ on $]0,\gamma]$ with $b > 0$, then for every $K$ and $g \in ]0,\gamma]$ shooting succeeds and $t + b(K-k) \le u_k \le t + b'(K-k)$.*

This is literal (0.31). An earlier draft used a wrong first IVT interval, which the verifier refuted; the corrected interval is $[t+bK,\ t+b'K]$. The nested-interval variant for $\beta$ defined only on admissible histories (`LEMMAS_DE_VERIFY` §1.5) is [PROVED+VERIFIED], not in Lean.

**Lemma 3.6** (Lemma E)**.** *If $\beta_0 \le \beta_j(0) \le \beta_0'$ and $|\beta_j(g) - \beta_j(0)| \le Cg^2$, then for $\gamma \le \sqrt{\beta_0/(2C)}$: $\beta_0/2 \le \beta_j(g) \le \beta_0' + \beta_0/2$.*

The verifier checked this $Cg^2$ form; the $C^1$ form ($C_1 g$, $\gamma \le \beta_0/(2C_1)$) is the same two-line proof [PROVED, TRIVIAL VARIANT]. Lean has the $Cg^2$ form.

### 3.4.5 History dependence (L5) [OPEN]

Balaban's $\beta_{k+1}$ depends on $(g_0, \dots, g_k)$. Lemma D′ Remark 3 (`LEMMA_DPRIME.md`) states the cube-continuity version (same proof). The gap is as follows (`THEOREM2_PROOF_v2_VERIFY` §F).

- *Domain.* Balaban defines $\beta_{k+1}$ only on histories for which his Theorem 1 induction has run to step $k$ (`LEMMAS_DE_VERIFY` §2 item 1). By the sweep (§2.3), that induction uses (2.6)–(2.9), that is (c′) + (S), on the history, not only $g_i \in ]0,\gamma]$.
- *The new gap.* The clamped shooting of Lemmas 3.2 and 3.4 evaluates $\beta$ on clamped histories $\tilde{g}_i = \varphi(\tilde{u}_i)$. A clamped history can violate the growth side of (c′): from $\max(a,c) - \max(b,c) \ge \min(a-b, 0)$ one gets only $u_m \ge u_n - D'$, not $u_m \ge u_n + \beta(n-m) - D'$. So the clamped shooting evaluates $\beta$ on histories outside Balaban's domain. The clamp proof is valid only under the cube hypothesis, which Balaban does not supply.
- *A variant that avoids it* [NOT WRITTEN]. Use a nested-interval proof in the style of `LEMMAS_DE_VERIFY` §1.5, with domain $D_k = \{u_0, \dots, u_k \ge \gamma^{-2}\}$ of true (unclamped) trajectories. On such trajectories, (c′) up to step $k$ follows from the tail-sum bound (a″) of Lemma D″, and (S) holds by the choice of $\gamma$. So every evaluation of $\beta$ stays inside Balaban's domain. At each step $k$, keep the set of $u_0$ whose trajectory stays in $D_k$; show by IVT, step by step, that this set contains a closed interval on which $u_k$ sweeps a range that contains the next target interval. For Lemma D (all increments positive) this is written. For Lemma D″ the increments can be negative, so the nested intervals need new envelopes, built from the window bounds $[b(n-m) - 2R - (b/4)(n-m), \dots]$. The adaptation is plausible; it is not written and not verified.
- Lean is one-argument only.

## 3.5 The exact structure of the one-loop sums

The window sums of the shooting lemma are sums of one-loop increments $\beta_j(0)$. This section expresses these sums as level-$k$ quantities of a coarse Gaussian theory and isolates the remainder $\kappa_k$ whose boundedness is the content of Theorem K. It also lists the symbol facts, uniform in the level, that Section 3.6 uses.

### 3.5.1 Telescoping [PROVED+VERIFIED, CORRECTED SCOPE]

$S_k := p \to 0$ $F^2$-coefficient of the composed $k$-step gauge-fixed one-loop integral (U1 (1.2)). Then $S_k = \sum_{j=1}^{k} \beta_j(0)$, $S_0 = 0$ (`BETA_FP` Prop. 1.1; `BETA_FP_VERIFY` item 1). Scope: exact for the $p \to 0$ coefficient at fixed $k$, not for $\log Z_k$ at fixed $p$ (the plane wave is off-shell).

### 3.5.2 Anchor-free identity [PROVED+VERIFIED]

Source: `BETA_FP_LEMMA43` Prop. 2.1; `BETA_FP_LEMMA43_VERIFY` items 1, 3–4 (hand rederivation; lattice check with the real CMP 109 averaging at random SU(2) backgrounds, relative difference $10^{-14}$).

**Proposition 3.7** (anchor-free identity; Prop. 2.1 of `BETA_FP_LEMMA43`)**.** *Under (G1) = $\{\Delta D = 0,\ \tilde{Q}D = D_c \iota_a^T\}$:*

$$\log Z_\Lambda(V) = \big[\log\det X - \tfrac{1}{2}\log\det K\big] - \big[\log\det(D_c^T D_c) - \tfrac{1}{2}\log\det(\Delta_\Lambda(V) + D_c D_c^T)\big].$$

In words: constrained one-loop = fine Wilson one-loop in background Feynman gauge minus the one-loop of the coarse Gaussian theory with kernel $\Delta_\Lambda(V)$. All point-anchor objects cancel; the constant is exactly 0.

- $\tilde{Q}D = D_c\iota_a^T$ is exact at any background (lattice $10^{-15}$). $\Delta D = 0$ is load-bearing (raw Hessian gives $O(1)$ errors).
- At the constant abelian field $V_f$ the background is critical, so $\Delta D = 0$ is exact and Prop. 2.1 holds exactly (`BETA_FP_H_VERIFY` §2, CONFIRMED). At a plane wave it holds up to the defect $\mathfrak{d}_Y$ (`BETA_FP_PW` Lemma 2.3) [PROVED, UNVERIFIED].
- Per-cell infinite-volume densities only: on a torus the constant zero modes must be removed. This is input (G2) [FORMAL].

### 3.5.3 The constant $c$ [PROVED+VERIFIED]

$c = 11N^2/(12\pi^2) = 2N \cdot b_0/(8\pi^2)$, $b_0 = 11N/3$; SU(2): $c = 0.371511$. Normalization $1/g_B^2 = 2N/g^2$ (`R2_VERIFY`; `BETA_FP_LEMMA43_VERIFY` §5). The log coefficient of the fine piece at constant field is $c_2 = c/2$ per $\ln(1/\varphi)$, checked against Savvidy, arXiv:1910.00654 eq. (1.4) [23] (`BETA_FP_H_VERIFY` §4) = input (F) [CITED, SOURCE CHECKED; LATTICE = CONTINUUM LOG COEFFICIENT IS STANDARD, NOT PROVED IN PROJECT FILES].

$b := c\log L$: 0.25751 ($L = 2$), 0.40815 (3), 0.51502 (4), 0.59793 (5), 0.72293 (7).

### 3.5.4 The decomposition $S_k = ck\log L + \kappa_W^F - \kappa_k$

- As bookkeeping the split is empty: any $c'$ gives an exact identity if $\kappa_k$ is defined as the remainder (`U1_VERIFY`). Boundedness of a remainder-defined $\kappa_k$ is a restatement, not progress.
- Content comes from defining $\kappa_k$ by the coarse theory alone. `KAPPA_BOUNDED` Prop. 1.1 [PROVED+VERIFIED, CONDITIONAL ON (PI), (F), ($M_j$), V1, (G2)]: with the Wilson level $j = 0$ as reference,

$$S_j = cj\log L - d_j, \qquad \kappa_j = \kappa_W^F + d_j,$$

$$d_j := -\frac{1}{2n_2}\int_0^1 dt \int_{\mathrm{BZ}} \frac{dK}{(2\pi)^4}\ \mathrm{tr}\big[b_{t,2}\,\delta\sigma_j^{(0)} + b_{t,1}\,\delta\sigma_j^{(1)} + b_{t,0}\,\delta\sigma_j^{(2)}\big],$$

where $\delta\sigma_j^{(m)} = \sigma_j^{(m)} - \sigma_0^{(m)}$ are differences of the $f^m$ Taylor symbols of the coarse kernel at level $j$ and level 0, and $b_{t,m}$ are the Moyal inverse coefficients of the interpolated coarse operator. No limit symbols $\sigma_\infty$ are used.

- Checks [NUMERICAL, REPRODUCED]: $\rho(1) = \kappa_W^F$ (Lüscher–Weisz (5.7) [16]); $L = 2$: $d_1 = -0.372$, $\kappa_1 = 1.114$; $L = 3$: $d_1 = c\log 3 - \beta_1 = -0.256 \pm 0.01$ (`KAPPA_BOUNDED_VERIFY` §4: fits give $-0.254 \cdots - 0.258$). This checks the definition of $d_j$ through $S_j$; it does not evaluate (1.1) from the symbols independently.

- Cor. 3.1 of `BETA_FP_LEMMA43` ($\mathcal{C}(\Lambda) = c\log\Lambda + \kappa_W^F - \rho(\Lambda)$) is exact as an identity. "The whole $\log\Lambda$ sits in the fine piece" is equivalent to $\rho(\Lambda) = o(\log\Lambda)$. It is not a separate fact (`BETA_FP_LEMMA43_VERIFY` item 2; an earlier overclaim is removed).

- Given convergence $\beta_j(0) \to \beta_\infty$, Cesàro gives $\beta_\infty = c\log L - \lim\kappa_k/k$. So the hypothesis (H′) "$\kappa_k = o(k)$" of the first version of these notes is EQUIVALENT to $\beta_\infty = c\log L$, not weaker (`THEOREM2_PROOF_VERIFY` Point 4). Theorem K (Theorem 3.8) gives $\kappa_k$ bounded, hence $\kappa_k = o(k)$ trivially.

#### 3.5.5 Level-uniform symbol facts used below [PROVED+VERIFIED]

Source: `U3_AW.md`, `U3_AW_VERIFY.md` (grades 1–4 CONFIRMED); `U3_REPAIRS_VERIFY`.

- O5/A0: $\Delta_M$ real-analytic on a fixed strip, $|\partial^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, uniform to $K = 0$ (odd $L$, centered blocks). Prop. B: $c_0|\hat{K}|^{-2}P_T' \le P_T' S_M P_T' \le C_1|\hat{K}|^{-2}P_T'$, $c_0 = (2/\pi)^{12}$, $C_1 \le 1363$. Existential constants.

- Prop. E (evenness): $\sigma_j(f;-K) = \sigma_j(f;K)$ in the midpoint trivialization, all $f$, $j$. Tested in $d = 2$ only.

- Theorem W: $\sigma_j^{(1)}(0) = \tau_c = (3ie/2)(E_{10} - E_{01})$ for every $j \ge 0$, from the exact Ward identity $\Delta_j(f)D_f = 0$. Numerically 0.750000, 0.75015, 0.75051 at $j = 0, 1, 2$ vs 0.75 ($d = 2$ proxy).

- Theorem A3: $|\sigma_j^{(1)}(K) - \tau_c| \le C_A|K|^2$, real $K$, uniform in $j$ ($C^0$ only).

- Prop. A2: $\sigma_j^{(m)}$, $m \le 2$, analytic on the same strip at every fixed $j$ (not uniform in $j$ for $m = 1$ derivatives, $m = 2$).

- Lemma M: Maxwell coefficient exactly 1 at every level.

### 3.6 Bounded window sums: two formally distinct conditional routes

This section gives the two routes to the window bound (5.1) that Lemma 3.4 needs.

Both routes give the window bound (5.1) of Lemma 3.4, $|S_K - S_k - b(K-k)| \le 2R$, with $b = c\log L$. Lemma D″ then gives Theorem 2w (at odd $L > 11$, or with (H0) at this $L$ as a hypothesis, and with L5). Neither route uses convergence of $\beta_j(0)$, a rate, (N), or certified early steps. The two routes have formally distinct named hypotheses. Their mathematical content is not independent: both reduce their uniform-in-level content to the same inequality (H2) on the two-step minimizer map (Section 3.7), plus A3-type facts (`BETA_FP_U` §3.5, §4.3 [PROVED, UNVERIFIED]; Route K side: `B2_TWOSTEP_VERIFY` grade 3 [PROVED+VERIFIED]).

### 3.6.1 Route K: Theorem K [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS]

Source: `KAPPA_BOUNDED.md`; `KAPPA_BOUNDED_VERIFY.md` (items (i)–(v) and (vii) CONFIRMED; (vi) CORRECTED in reasoning, conclusion unchanged).

**Theorem 3.8** (Theorem K)**.** *$L$ odd $\geq 3$. Assume (B2) (Section 3.7; (B2) $\Leftarrow$ (H2) [PROVED+VERIFIED]) and the fixed-level inputs below. Then $\sup_k |d_k| \leq \kappa_* < \infty$, so $\sup_k |\kappa_k| \leq |\kappa_W^F| + \kappa_*$ and $|S_k - ck \log L| \leq \kappa_*$ for all $k$. Window sums satisfy (5.1) with $R = \kappa_*$. (Theorem K is a statement about one-loop quantities. Its use in Lemma D′ for Balaban's $\beta_j(g)$ needs $L$ odd $> 11$ or (H0) at this $L$.)*

*Proof structure.*

1. Prop. 1.1 (Section 3.5.4): $S_j = cj \log L - d_j$. [PROVED+VERIFIED, CONDITIONAL ON (PI), (F), $(M_j)$, V1, (G2)]
2. Lemma 2.1: the only $K$-derivative of $\sigma^{(1)}$ in $d_j$ enters linearly with $f^0$ coefficients. Integration by parts moves it onto $f^0$ data; boundary terms vanish (periodic up to a common constant conjugation; inner sphere $O(r^2)$). So $\sigma^{(1)}$ is needed only in $C^0$, which Theorem A3 gives. [PROVED+VERIFIED]
3. Prop. 2.2: with (E0) $f^0$ bounds (Prop. B, O5/A0, Lemma M), (E1) Theorem A3 + Theorem W, and (B2), every term of the $d_j$ integrand is $\leq C|K|^{-2}$, uniformly in $j$ and $t$. In $d = 4$ this has two powers of infrared margin. $|d_j| \leq \kappa_* := (C(L)/n_2)(1 + Y_1^2 + C_A(1 + Y_1) + Y_2) \cdot I_2$. [PROVED+VERIFIED, CONDITIONAL ON (B2)]
4. Infrared: the degree $-4$ part of the coarse $f^2$ integrand is the same at every level (Lemma M + Theorem W), so the $\tau_c \log$ cancels exactly in level differences. [PROVED+VERIFIED]

Constants are existential; $\kappa_*$ is not evaluated.

Fixed-level inputs of Route K (needed at each $j$ separately; no uniformity in $j$):

- (PI) probe independence: plane-wave $p^2$ coefficient = constant-field $f^2$ coefficient [ARGUED; PER-CELL LIMIT FOR THE LINEAR POTENTIAL NOT WRITTEN].
- (F) fine log coefficient $c_2$ at constant field [CITED, SOURCE CHECKED], Section 3.5.3.
- $(M_j)$ the formal $f^2$ coefficient $d_j$ equals the true one at fixed $j$. The resolvent step of the first argument was false (Nielsen–Olesen zero at $k_\perp^2 = |ef|$; `BETA_FP_H_VERIFY` §5). `BETA_FP_M` repairs it with a direct $\log|\det|$ bound given spectral bounds (S1)–(S3) [OPEN ON THE LATTICE]; Prop. 4.4 there is [ARGUED]; `BETA_FP_M` has no verifier file. The $t$-subdivision of KAPPA §1.3 removes the small-$\varepsilon$ requirement [ARGUED, CONFIRMED AS PLAUSIBLE]. Status: [ARGUED].
- V1: the Maxwell coefficient of the per-step determinant equals $\beta_{j+1}(0)$, i.e. order-$g^0$ completeness of CMP 109 (1.3)–(1.5) [OPEN; SUPPORTED AT ORDER $p^2$ BY `U3_VERIFY` §6].
- (G2) per-cell infinite-volume densities [FORMAL].
- (H0) Balaban's inductive assumption: $\beta_{j+1}$ smooth on $[0, \gamma]$, "uniformly bounded on this interval together with all derivatives" (CMP 109 pp. 263–264, (1.22)) [BALABAN; AVAILABLE AT ODD $L > 11$ ONLY]. CMP 116 has no theorem statement and no $C^m$-in-$g$ bound in words (`BALABAN_EXCERPTS_CHECK` item 10); the smoothness in $g$ is CMP 109's inductive assumption, asserted by CMP 109 Theorem 3, whose proof CMP 116 completes.

- L5 history-dependent $\beta$, its domain, and its continuity (Section 3.4.5) [OPEN].

#### 3.6.2 Route B: Theorem $\mathbf{B}_{\mathrm{PW}}$ [PROVED+VERIFIED, CONDITIONAL (LOGIC OF THEOREM $\mathrm{B}_{\mathrm{PW}}$)]

Source: `BETA_FP_PW.md` §1–§5 (Theorem 3.2 = Theorem $\mathrm{B}_{\mathrm{PW}}$, Theorem 5.1). `BETA_FP_PW_VERIFY.md` §6 grades it: items 1–4 CONFIRMED (doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B as conditional logic; Lemma 4.1, conditional), items 5 (comparison) and 6 (numerics) CORRECTED. The tag follows the decision recorded in `PLAN.md` (2026-10-28): the logic of Theorem $\mathrm{B}_{\mathrm{PW}}$ is [PROVED+VERIFIED, CONDITIONAL] on its hypotheses, citing `BETA_FP_PW_VERIFY` §6 items 1–4. Its inputs (U), (DF), (T1) from `BETA_FP_U` stay [PROVED, UNVERIFIED]. The file rederives the Lemma 1.3 arithmetic and the Lemma 2.3 / Prop. 2.2 algebra correctly, confirms the Prop. 2.2 determinant identity on a $4^4$ lattice ($4\mathrm{e}{-13}$), and notes that identity 3.0 is a tautology ($\Gamma^{ex} := \hat{\Pi} - S$), so all content sits in $(B\theta)/(C\theta)$.

**Theorem 3.9** (Theorem $\mathrm{B}_{\mathrm{PW}}$; Theorem 3.2 of `BETA_FP_PW`)**.** *$L$ odd $\geq 3$. Assume*

- *(U) there are $p_* > 0$, $C_U$ with $|\beta_{j+1}(p) - \beta_{j+1}(0)| \leq C_U p^2$ for all $j \geq 0$, $0 < p \leq p_*$* [HYPOTHESIS; ARGUED FROM UNIFORM ANALYTICITY IN $p$ OF THE ONE-STEP PLANE-WAVE DATA PLUS ONE-STEP O6]*;*
- *($B\theta$) there is $\theta_* \in (0, \min(p_*, 1))$ with $B_* := \sup_k |\Gamma^{ex}(\theta_*; L^k)| < \infty$, where $\Gamma^{ex}$ is the coarse remainder at fixed coarse momentum $\theta_*$* [HYPOTHESIS]*;*
- *Lüscher–Weisz (5.7) [16]* [CITED].

*Then $|S_k - ck \log L| \leq C_B := B_* + c_2 \ln(1/\theta_*^2) + |\kappa_W^F| + \sup_{p \leq \theta_*} |\eta(p)| + C_U \theta_*^2/(L^2 - 1)$, and window sums satisfy (5.1) with $R = C_B$. $\theta_* < 1$ is required so that $c_2 \ln(1/\theta_*^2) \geq 0$; otherwise the term must be replaced by its absolute value (`BETA_FP_PW_VERIFY` §3).*

*Key step.* Lemma 1.3: $|S(\theta; L^k) - S_k| \leq C_U \theta^2/(L^2 - 1)$ from (U). The massless coarse side is then needed only at a fixed $\theta_* > 0$, where there is no infrared problem. No (W), (PI), (M), IR weights, or spectral theory.

What ($B\theta$) reduces to (`BETA_FP_PW` §4) [ARGUED]: Lemma 4.1(i) (trivial-background convergence, from O5, odd $L$) [PROVED]; vertex bounds at fixed $\theta_*$ (the uniform bounds cited by U3 Theorem 7.1 to Balaban; the fixed-$\theta$ data hypothesis ($D\theta$) is needed only in its "bounded" form); and boundedness in $k$ of the defect $\mathfrak{d} = \mathfrak{d}_Y + \mathfrak{d}_{gf}$.

- $\mathfrak{d}_Y$ (Prop. 2.1 defect at the non-stationary plane wave): $O(a\Lambda^{-2})$ relative at fixed $\theta$, from zero capacity of point anchors in $d = 4$ [ARGUED; REGION $|K| \lesssim \Lambda^{-1}$ NOT WRITTEN].
- $\mathfrak{d}_{gf}$ (composite gauge vs $D_k^T$ gauge at the plane wave): $O(\theta^2)$ relative; uniformity in $k$ not shown [ARGUED]. Data: 0.953 (direct) vs 0.9595 (recursion) at $L = 2$, $p_0 = 0.2$; difference below the tabulation error, so unresolved.

*Identification.* Route B's $\beta_j(0)$ is the per-level plane-wave increment (the quantity the code computes). It matches Balaban's second-moment $\beta$ up to the expansion point (fine wave vs constrained minimizer), argued to change $S$ by $O(\theta^2)$ relative (`SOURCES_D1_VERIFY` item 8) [ARGUED]. This identification and the $\mathfrak{d}_{gf}$ bound need expansion-point bounds uniform in $k$. So the (PI)-type content of Route K is not removed in Route B; it moves into a uniform-in-$k$ input (`BETA_FP_PW_VERIFY` §3).

Three points from `BETA_FP_PW_VERIFY` §3 (item 5 of its §6 grades them CORRECTED):

1. $\theta_* < 1$ is required (above).

2. The $\beta$ identification (B4) is a uniform-in-$k$ input, not a fixed-level input.

3. (U) is a uniform-in-$j$ analyticity of the level data. It plausibly needs the plane-wave analogue of a uniform $C^1$ bound on $\sigma^{(1)}$: uniform $p$-smoothness of the $a^1$ vertex. That is the same uniform first-order vertex smoothness as Route K's open core ($C1\sigma$).

#### 3.6.3 Route B inputs after `BETA_FP_U` [PROVED, UNVERIFIED; NO VERIFIER FILE]

Source: `BETA_FP_U.md` §2–§4. It reduces the two uniform hypotheses of Theorem $\mathrm{B_{PW}}$.

- (U) $\Leftarrow$ (T1) + (H2) (Theorem 2.4, odd $L \geq 3$, with A0, A1, Lemma 2.2). (T1): $\sup_j \sup_q |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of Theorem A3 [ARGUED]. (H2) is the inequality of Section 3.7 (`BETA_FP_U` writes it as $\tilde{h}_{(2)} < L$).

- ($B\theta$): $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ exactly (Prop. 3.1). $\gamma_{rec}$ is bounded in $k$ given Theorem 2.4 (Prop. 3.2). $\mathfrak{d}_F$ is bounded under (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$ [HYPOTHESIS, ARGUED]. `BETA_FP_U` corrects `BETA_FP_PW` §4.3(b): $\mathfrak{d}_F$ is $O(1)$ relative at fixed $\theta$, not $O(\theta^2)$; only boundedness is used.

- Corollary 4.1: given (T1), (H2), (DF), LW (5.7), $|S_k - ck \log L| \leq C_B$, so (5.1) holds with $R = C_B$.

- Theorem 2w by Route B then needs in addition A6 (plane-wave $\beta$ = Balaban's $\beta$ up to the expansion point, uniform in $k$) [ARGUED] and (H0) (Lemma D″ (i)).

Route B hypothesis set: (T1), (H2), (DF), LW (5.7), A6, (H0) (plus L5, (G2), odd $L$, as for Route K). The fixed-level inputs (PI), ($M_j$), V1 of Route K do not appear; (DF) and A6 replace them.

#### 3.6.4 Comparison

| | Route K | Route B |
|---|---|---|
| uniform-in-level input | (B2): $\sup_j \sup_K \|\sigma^{(2)}_j\| < \infty$, $\Leftarrow$ (H2) [PROVED+VERIFIED] | (U), ($B\theta$), and the $\beta$ identification and $\mathfrak{d}_{gf}$ bound, both with $k$-uniform expansion-point bounds; (U) $\Leftarrow$ (T1) + (H2), ($B\theta$) $\Leftarrow$ Theorem 2.4 + (DF) [PROVED, UNVERIFIED] |
| single uniform inequality | (H2) | (H2) |
| fixed-level inputs | (PI), (F), ($M_j$), V1, (G2) | LW (5.7), $\mathfrak{d}_Y$, $\theta_* < 1$ |
| uses (W), Theorem A3 | yes | no; `BETA_FP_U` uses (T1), the plane-wave analogue of A3 [ARGUED] |

| | | |
|---|---|---|
| verification | verified (conditional) | logic of Theorem $B_{PW}$ verified (conditional): `BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED [PROVED+VERIFIED, CONDITIONAL]; inputs from `BETA_FP_U` [PROVED, UNVERIFIED], no verifier file |
| common | (H0) at odd $L > 11$, L5, (G2), V1-type identification, Balaban's proofs, SU(2) | |

Shared inputs: (H0), L5, (G2), odd $L$ ($> 11$ for (H0)), and the identification of the computed $\beta_j(0)$ with Balaban's $\beta_j(0)$ (V1; in Route B also the expansion point). The uniform-in-level hypotheses ((B2) vs (U) + $(B\theta)$) are formally distinct as named hypotheses. Their content overlaps: both rest on (H2) (Section 3.7). In Route K, (H2) gives $(C1\sigma)$ (Theorem C1) and then (B2) (Theorem B2). In Route B, (H2) gives (U) with (T1), and the vertex bound behind $(B\theta)$. So a proof of (H2) serves both routes. One closed route suffices for Theorem 2w.

## 3.7 The hypothesis (B2) of Route K, and the single uniform inequality (H2)

Both routes need a bound on the coarse symbols that is uniform in the level. This section states that bound for Route K, records what is proved and what is refuted, and reduces it to the inequality (H2). The numerical values of (H2) and its certification status are in Sections 5.4–5.7.

(B2): $\sup_{j\geq 0}\sup_{K\in \mathrm{BZ}}|\sigma_j^{(2)}(K)| < \infty$. Weaker form that suffices, (B2w): $\sup_j \int_{\mathrm{BZ}}|K|^{-2}|\sigma_j^{(2)}(K)|\,d^4K < \infty$ ($\sigma^{(2)}$ enters $d_j$ only undifferentiated; `KAPPA_BOUNDED` Remark 2.3) [PROVED+VERIFIED: NOT GRADED BY NAME; IMPLIED BY `KAPPA_BOUNDED_VERIFY` §9(IV)].

$\sigma^{(2)}$ obeys the one-step recursion (`U3_AW` (1.2)): $\sigma_j^{(2)} = L^{-2}\mathcal{A}_j\sigma_{j-1}^{(2)} + Q_j[\sigma_{j-1}^{(1)}] + g_j^{(2)}$, with $\mathcal{A}_j s = L^{-2}H_j^{\dagger}(\oplus_l s(k_l))H_j$ and $H_j(K)$ the one-step gauge-fixed minimizer map. Notation: $\rho_1 := \sup_{j,K}\|H_j(K)\|/L$ (one step); $H_{(2),j}(K) := (\oplus_l H_{j-1}(k_l))H_j(K)$ (two steps), $\rho_2 := \sup_{j\geq 2}\sup_{K\ \mathrm{real}}\|H_{(2),j}(K)\|/L^2$.

### 3.7.1 Proved

- $K = 0$ bound [PROVED+VERIFIED]: $\sup_j|\sigma_j^{(2)}(0)| \leq C(|e|C_A + e^2)$. Theorem A3 + scaling give $|\partial^2\sigma_j^{(1)}(0)| \leq 2C_A$ (polarization; `KAPPA_BOUNDED_VERIFY` §7), and the slaving identity (W-2) expresses $\sigma_j^{(2)}(0)$ through these second derivatives.

- Each fixed $j$ [PROVED+VERIFIED]: $\sigma_j^{(2)}$ is analytic on a fixed strip (Prop. A2), hence bounded. The content of (B2) is uniformity in $j$.

- Prop. 4.1 [PROVED+VERIFIED, CONDITIONAL]: if (h) $h := \sup_{j,K}\|H_j(K)\| < L^2$ and $(C1\sigma)$ $\sup_j\sup_K|\partial_K\sigma_j^{(1)}(K)| < \infty$, then (B2) holds. Proof: in the form norm $\pm s \leq N(\Delta + 1)$, $N_j \leq (h^2/L^4)N_{j-1} + Z_2$, and $Z_2$ is finite by Theorem A3, $(C1\sigma)$, and fixed $f^0$ vertices.

- Why the Lemma A4 mechanism alone fails [PROVED+VERIFIED, REASON CORRECTED BY `KAPPA_BOUNDED_VERIFY` §7]: the obstruction is the missing uniform $C^1$ bound on $\sigma^{(1)}$, which

enters twice: in $Q_j$ (term $L^{-3}\partial_K \sigma^{(1)}_{j-1}$) and in the longitudinal part of $\sigma^{(2)}$ given by (W-2). Theorem A3 is $C^0$ only.

### 3.7.2 Refuted, and the failed one-step bound

- (HS) $h < L\sqrt{L-1}$ (named (H′) in `U3_AW`) [REFUTED AT $L = 3$]. Measured $\max_K \|H_j(K)\|/L = 1.652, 1.736, 1.757, 1.760$ for $j = 1, 2, 3, \infty$ (`U3_AW_VERIFY` §6); the threshold is $\sqrt{2} = 1.414$. Gauge choice cannot help: $H_j$ is already minimal-norm in its gauge orbit.

- One-step sup-norm $C^1$ recursion at $L = 3$ [NUMERICAL, REPRODUCED; `B2_TWOSTEP_VERIFY` GRADE 2]. Its factor is $\rho_1^2/L = 0.909, 1.0046, 1.0289, 1.0325$ for $j = 1, 2, 3, \infty$, so it exceeds 1 for $j \geq 2$ ($\rho_1 > \sqrt{3} = 1.7321$). At $K = 0$ the factor is $L^{-1}$; the bad region is the face $K_3 = K_4 = \pi$, $K_1 = K_2 \approx 1.86$–$1.92$. This shows that the one-step sup-norm bound fails at $L = 3$. It does not show that a $C^1$ bound fails; a weighted norm is not excluded.

### 3.7.3 (H2) and the two-step theorems [PROVED+VERIFIED, CONDITIONAL ON (H2)]

Source: `B2_TWOSTEP.md` §1–§3; `B2_TWOSTEP_VERIFY.md` grades 1, 3, 4 (CONFIRMED).

$$\text{(H2)} \quad \rho_2 = \sup_{j\geq 2} \sup_{K\in \mathrm{BZ\ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L.$$

(H2) is a plain operator-norm inequality on real $K$. It uses no strip, no $\Delta$-relative weight, and only $n = 2$.

**Theorem 3.10** (Theorem C1 of `B2_TWOSTEP`). *(H2) ⇒ (C1σ), with contraction factor $q_1 = \rho_2^2/L^2$ on the derivative.*

**Theorem 3.11** (Theorem B2 of `B2_TWOSTEP`). *(H2) ⇒ (B2), hence (B2w), in plain sup norm, with factor $q_2 = \rho_2^2/L^4$.*

Here (C1σ) is $\sup_j \sup_K |\partial_K \sigma_j^{(1)}(K)| < \infty$ (Section 3.7.1). The verified points are these.

- Two-step map. $\mathcal{A}_j\mathcal{A}_{j-1}s = L^{-4}H^\dagger_{(2)}(\oplus s)H_{(2)}$, with $dk_{l,l'}/dK = L^{-2}$ [CONFIRMED, grade 1].

- Theorem 3.10. The chain-rule gain is exactly $L^{-2}$ per two steps on real $K$. So the threshold is $\rho_2 < L$, not $\sqrt{L^2-1}$: no strip-Cauchy loss enters. The large constants ($b(L)$ of O6, $\eta_1$, $\eta_2$, $G_1$) enter only the source term, not the contraction factor [CONFIRMED, grade 3].

- Theorem 3.11. This contraction needs only $\rho_2 < L^2$, which (H2) implies [CONFIRMED, grade 3]. In one step it is Prop. 4.1's (h).

- (H2) vs (H″). (H2) is the plain operator norm of the same matrix as `U3_AW` (H″) at $n = 2$. It is weaker: threshold $L$ (3 at $L = 3$) against $\sqrt{L^2-1}$ (2.828), real $K$ only, $n = 2$ only [CONFIRMED, grade 4]. (H″) ⇒ (B2) stays [ARGUED] and is no longer needed.

- Form of $Q_j$ (G3) [CONFIRMED by read-off of `U3_AW` (1.2)/(3.3) and P10]: $\partial\sigma^{(1)}$ enters linearly. `U3_LEMMA51` (3.3) itself was not re-derived from the Moyal rule by either party.

- Not covered: $C^2$ and $C^3$ bounds on $\sigma^{(1)}$ under (H2) (`B2_TWOSTEP` §5, "look routine", not written, not graded). They are not needed for (B2).

- Submultiplicativity: $\rho_2 \leq \rho_1^2$ always. At $L = 3$ the gain of the two-step map is that this is strict by a factor 1.50 (2.064 vs 3.098) [CONFIRMED].

#### 3.7.4 One-step sufficiency [ARITHMETIC PROVED; INPUTS NUMERICAL]

$\|H_{(2),j}(K)\| \le \max_l \|H_{j-1}(k_l)\| \cdot \|H_j(K)\| \le L^2\rho_1^2$. So $\rho_1^2 < L$ implies (H2), and then the one-step $C^1$ recursion of `B2_TWOSTEP` (2.2) contracts with factor $\rho_1^2/L$. At $L = 3$ the measured $\rho_1^2$ exceeds $L$, so the one-step route fails and the two-step map is needed; at $L = 5$ and $L = 7$ the measured $\rho_1^2/L$ is below 1. The values are in Section 5.5.

#### 3.7.5 The precise remaining inequality

At a fixed odd $L \ge 3$ the single uniform hypothesis is (H2).

- Route K: (H2) $\Rightarrow$ ($C1\sigma$) (Theorem C1) $\Rightarrow$ (B2) (Theorem B2) [PROVED+VERIFIED, CONDITIONAL ON (H2)].
- Route B: (H2) + (T1) $\Rightarrow$ (U), and (H2) $\Rightarrow$ the vertex bound for $\gamma_{rec}$ in ($B\theta$) (`BETA_FP_U` Theorem 2.4, Prop. 3.2) [PROVED, UNVERIFIED].
- Status of (H2): [NUMERICAL AT $L = 3, 5, 9$; CERTIFICATION OPEN]. At $L = 9$ there is one point only. No value exists at $L > 11$ (Balaban's range), so the numbers are supportive only.
- Kept as alternatives: ($C1\sigma$) + (h) $\Rightarrow$ (B2) [PROVED+VERIFIED]; (H″) $\Rightarrow$ (B2) [ARGUED]; (B2w) directly. No per-momentum $d = 4$ constant-field code exists, so (B2) itself has not been scanned numerically (`KAPPA_BOUNDED_VERIFY` §8).

### 3.8 Conclusion of the reduction

Theorem 2w $\Leftarrow$ (Route K inputs) or (Route B inputs). Precisely:

*Route K.* (B2) [OPEN; REDUCES TO (H2) (VERIFIED REDUCTION, SECTION 3.7.3); (H2) NUMERICAL AT $L = 3, 5, 9$, CERTIFICATION OPEN] + (PI), (F), ($M_j$), V1, (G2) [FIXED-LEVEL] + (H0) [BALABAN, ODD $L > 11$] + L5 [OPEN] $\Rightarrow$ Theorem K ($R = \kappa_*$) $\Rightarrow$ (5.1) $\Rightarrow$ Lemma D″ $\Rightarrow$ Theorem 2w. Status of the implication: [PROVED+VERIFIED, CONDITIONAL].

*Route B.* (U) + ($B\theta$) with $\theta_* < 1$ + LW (5.7) + bounded $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ + identification of $\beta$ (uniform in $k$) + (G2) + (H0) [ODD $L > 11$] + L5 $\Rightarrow$ Theorem $\mathrm{B_{PW}}$ ($R = C_B$) $\Rightarrow$ (5.1) $\Rightarrow$ Lemma D″ $\Rightarrow$ Theorem 2w. Status: Theorem $\mathrm{B_{PW}}$ logic [PROVED+VERIFIED, CONDITIONAL]; inputs [PROVED, UNVERIFIED]. With `BETA_FP_U` (Section 3.6.3): (T1) + (H2) + (DF) + LW (5.7) $\Rightarrow$ Corollary 4.1 ($R = C_B$); with A6 + (H0) + L5 + (G2) $\Rightarrow$ Theorem 2w. Status: [PROVED, UNVERIFIED, CONDITIONAL] (the inputs (T1), (DF), A6 and the `BETA_FP_U` reductions are unverified).

*Single uniform hypothesis.* Both routes need (H2). A certified (H2) at one odd $L > 11$ would discharge (B2), the only uniform-in-level hypothesis of Route K (Theorems C1, B2). In Route B it would reduce (U) to (T1); (DF) and A6 would remain.

*Then Balaban's Theorem 1.* Theorem 2w gives (b) and (c′). With $\gamma$ shrunk so that (S) holds, every one of the 19 cross-scale uses in CMP 119 and CMP 122 I–II is covered (Section 3.4.3), and the single-scale uses need only $g_k \le \gamma$. So Theorem 1 holds for the bare coupling $g_0(K, g)$, for every $K$ and every $g \in ]0, g_1]$, without Balaban's coupling assumption. This needs L5 in the sharpened form of Section 3.4.5: $\beta$ must exist on every history the shooting visits. Status: [PROVED, UNVERIFIED (SWEEP), CONDITIONAL ON ROUTE K OR ROUTE B INPUTS, ON L5, AND ON BALABAN'S PUBLISHED PROOFS].

*Order of constants.* Fix $\beta_0, p_0, p_1, r, \nu, \kappa_0$ (Balaban). Fix $L$ (odd, $> 11$), then $R$ from Route K or B, then $b = c \log L$, then $\gamma \le \min(\gamma_0, \gamma_{Bal}, b/(4C_1), \gamma_*)$ with $\gamma^{-2} \ge c_{sw} \cdot 2R$. Then $g_1 = (\gamma^{-2} + 2R)^{-1/2}$. Balaban's phrase "for a sufficiently small positive $g$" fits $g \le g_1$.

*What this does not give.*

- Literal (0.31): only for windows of length $\geq 8R/b$. Literal (0.31) at every $K$ needs (L1) or (L2) of Section 3.3, which need individual increments, not window sums. Not needed for Theorem 1.
- The value $\lim \beta_j(0)$ and any rate: Section 3.9.
- Even $L$, or odd $L \leq 11$ without (H0) as an extra hypothesis (Balaban assumes $L$ odd $> 11$ throughout CMP 109); SU($N$) with $N > 2$.
- Monotone couplings: $g_m$ can exceed $g_n$ for $m < n$ by the bounded amount $u_m \geq u_n - D'$. Section 3.4.3 shows Balaban's proof tolerates this under (S); a line-by-line read was not done.
- A continuum limit (`ASSEMBLY.md` treats the assembly separately; not claimed here).

### 3.9 The convergence route: what it adds, and its status

Not needed for Theorem 1. It adds: (i) existence of $\beta_\infty = \lim \beta_j(0)$; (ii) its value $c \log L$; (iii) a rate; (iv) eventual positivity, hence literal (0.31) for $K \geq K_0$ (Lemma D′ (d)); (v) with certified early steps, literal (0.31) at every $K$ (Section 3.3 (L2)). It also implies (B2), since convergence of $\sigma_j^{(2)}$ implies boundedness. So it needs strictly more than Route K.

#### 3.9.1 Components and status

- U3 Theorem 4.1 with O5: $|\partial_K^\alpha(\Delta_M - \Delta_\infty)| \leq CM^{-2}$, $|\alpha| \leq 2$, uniform to $K = 0$, odd $L$ [PROVED+VERIFIED] (`U3_VERIFY`, `U3_REPAIRS_VERIFY`). Numerically $|\Delta_M - \Delta_\infty| =$ 1.007, 0.129, 0.0147 at $M = 3, 9, 27$.
- O6: $\|\mathfrak{B}_j(K)^{-1}\| \leq b(L)$ for all $j$, $K$, one-step constraint, trivial background [PROVED+VERIFIED]. Crude constant ($b(3) \approx 7.5 \cdot 10^{33}$); measured corner values 103.5, 176.7, 193.0, 195.3 ($j = 0, 1, 2, \infty$, $L = 3$) use a surrogate $\Lambda$ (`U3_REPAIRS_VERIFY` §5).
- Theorem 6.1: $\beta_{j+1}(0) = \Phi(\mathcal{D}_j)$ is Lipschitz in the level data [PROVED+VERIFIED, CONDITIONAL ON V1]. The norm is the Taylor-data norm (`U3_VERIFY` P8 correction).
- Lemma 5.1 (convergence of $\sigma_j^{(1)}$, $\sigma_j^{(2)}$ in $C^2$, $|\alpha| \leq 2$):
  - neutral sector, rate $L^{-4j}$ [PROVED+VERIFIED FOR THE VALUE BOUND; THE $K$-DERIVATIVE PARAGRAPH IS PROVED, UNVERIFIED];
  - (W) [PROVED+VERIFIED] (`U3_AW` Theorem W); it removes the marginal direction of the $f^1$ recursion;
  - (A) qualitative and Theorem A3 [PROVED+VERIFIED]; (A) in full (strip/$C^3$ for $m = 1$, all of $m = 2$) $\Leftarrow$ (H″) [OPEN]; the route via (HS) is [REFUTED] at $L = 3$;
  - $\sigma^{(1)}$ convergence at rate $jL^{-2j}$ [SKETCH IN `U3_AW` §4.7, UNVERIFIED]; $\sigma^{(2)}$ convergence [OPEN, NEEDS (A) IN FULL];
  - the single-block-constraint version of Lemma 5.1 is false for $m = 2$ in $d = 4$ [PROVED+VERIFIED]; Balaban's composed constraint is not affected.

- Prop. 3.6: $|\beta_{j+1}(0) - \beta_\infty| \le C(1+j)^2 L^{-2j}$ [PROVED, UNVERIFIED AS AN ASSEMBLY; CONDITIONAL ON V1, (A) IN FULL]. (The form $Cj^2L^{-2j}$ of the first version of these notes is 0 at $j = 0$ and is corrected here.) The identification of $\sigma$-Taylor data with the data of (6.1) is cited from `U3_VERIFY` §7, not re-shown. (N) (clean rate $L^{-2j}$) is not needed and is judged plausibly false.

- Value of the limit. Given convergence, Theorem K or Theorem $\mathrm{B_{PW}}$ gives $\beta_\infty = c \log L$ at once: $S_k/k \to \beta_\infty$ by Cesàro and $S_k/k = c \log L + O(1/k)$ [PROVED+VERIFIED AS AN IMPLICATION: `THEOREM2_PROOF_VERIFY` POINT 4; `BETA_FP_PW` THEOREM 3.2(III)]. Older routes to the value: `BETA_FP_H` Theorem 3.1′ with (H) restated at the constant field, which given (W) reduces to Lemma 5.1 (1.2) alone (`BETA_FP_H_VERIFY` items (1)–(4)), plus (PI), (F), (M) [CONDITIONAL; (M) ARGUED]; and `BETA_FP_PW` Theorem 3.1 (limit form) given (U), ($C\theta$), LW (5.7) [PROVED, UNVERIFIED]. The original (H) with IR weights (`BETA_FP_LEMMA43` (4.2)) was graded UNSUPPORTED and is not used.

- Rate of $\mathcal{C}(\Lambda)$: $O(\Lambda^{-2} \log \Lambda)$ given Lemma 5.1 with rates, via `U3_AW` §3.4(b) slaving [ARGUED, `BETA_FP_H_VERIFY` §5].

### 3.9.2 Early steps and $j_*$

Under Prop. 3.6 and $\beta_\infty = c \log L$, $\beta_{j+1}(0) \ge (c \log L)/2$ for $j \ge j_* := \min\{J : C(1+j)^2 L^{-2j} \le (c \log L)/2 \text{ for } j \ge J\}$. Heuristic size [ARGUED]: $C \sim b^3$ gives $j_* \approx 110$ ($L = 3$), $\approx 70$ ($L = 13$); $C \sim b^6$ (one-step inverses inside $C_{data}$) gives $\approx 218$, $\approx 138$ (`THEOREM2_PROOF_VERIFY` Point 5(e)). With the measured $\|\mathfrak{B}^{-1}\| \approx 195$: $j_* \approx 10$ ($L = 3$), $\approx 4$ ($L = 13$). Certifying $\beta_1(0), \dots, \beta_{j_*}(0) > 0$ needs interval arithmetic at odd $L \ge 13$ (blocks $\approx 3.4 \cdot 10^5$ rows per momentum by dense methods; the umklapp code reduces this to $12 \times 12$ blocks at $L^4$ momenta). It is needed only for (L2), not for Theorem 1.

# 4 The finite-torus existence theorem

This section proves Theorem B. It takes the trajectory of Section 3 as the input block (B-T2). Section 4.1 fixes what "exists" means here. Section 4.2 states the uniform bound (H-U) that the blocks must deliver, and Sections 4.3–4.4 state the theorems. Section 4.5 proves them modulo the named blocks, and Section 4.6 gives the dependency table, which is the single status table of the paper. Section 5 then collects the numerical support for the hypotheses.

Source: `ASSEMBLY.md` (2026-10-27), verified in `ASSEMBLY_VERIFY.md`. The statements carry the verifier's corrections (Section 4.8). The theorem is a reduction to named blocks. It is not unconditional. The notation is that of Sections 2.1 and 2.5.

## 4.1 What "exists" means here

Jaffe–Witten footnote 2 [15] [VERIFIED QUOTE] (`reports/clay_ym.txt` line 619; anchor: the sentence "Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$", line 582): "We specifically exclude weak-existence (compactness) as the solution to the existence part of the Millennium problem, unless one also uses other techniques to establish properties of the limit (such as the existence of a mass gap and the axioms)."

Reading used here. "Exists" means: (i) the family $\{S_{n,\varepsilon}\}$ is relatively compact (uniform bounds); (ii) at least one limit point exists along a sequence $\varepsilon_k \to 0$; (iii) every limit point has the properties (OS0)–(OS3) below, proved on the limit, not assumed. No uniqueness is claimed. Full-sequence

convergence needs E13 (comparison of two trajectories; `SOURCES.md` (E14-d), D6), which is not in this paper.

This reading is weaker than the footnote's parenthesis. The theorem below gives no gap and only the torus versions of four axioms. It is NOT an admissible Clay solution. It is the finite-torus step of Jaffe–Witten §6.5 [VERIFIED QUOTE]: "One must then verify the existence of limits of appropriate expectations of gauge-invariant observables as the lattice spacing tends to zero and as the volume tends to infinity" (only the first limit is done here).

The OS axioms are not stated in `reports/clay_ym.txt`. It cites them as OS (CMP 31 (1973), CMP 42 (1975)) [19, 20] and requires "axiomatic properties at least as strong as those cited" (lines 279–280) [VERIFIED QUOTE]. The forms used follow the template's application of OS II (`reports/o3_continuum.txt` lines 4987–5003) [VERIFIED QUOTE]: "The linear growth condition requires a bound $|S_n(F)| \le \sigma_n |F|_{s_n}$ with fixed $s$ and $\sigma_n \le A(n!)^D$ for fixed $A$, $D$", and restriction to Schwartz tests flat on the coincidence diagonals gives the test class of that reference. OS II itself was not re-read [CITED].

### 4.2 Hypothesis delivered by the blocks

**Definition 4.1** ((H-U) uniform bound)**.** There are $s \in \mathbb{N}$ and $A, D < \infty$ such that for all $K \ge K_0$, all $n \ge 1$, and all $F \in \mathcal{D}_n$:

$$|S_{n,\varepsilon}(F)| \le \sigma_n \|F\|_{sn}, \qquad \sigma_n := A(n!)^D, \quad \varepsilon = L^{-K}.$$

The derivative order $sn$ grows linearly in $n$. The OS linear growth condition admits this. (H-U) is the output of blocks (B-T2) + (B-S) + the bridge (B-N1) of Section 4.5.

### 4.3 Theorem (finite-torus existence, compactness form)

**Theorem 4.2** (finite-torus existence, compactness form [PROVED MODULO NAMED BLOCKS])**.** *Let $G = SU(2)$, $d = 4$, $L$ odd and large, $\ell \in 2\mathbb{N}$ fixed. Let $\gamma > 0$ be Balaban's small constant and $g \in ]0, \gamma]$ small. For $K \ge K_0$ let $\varepsilon = L^{-K}$ and $g_0 = g_0(\varepsilon, g)$ be a shooting solution (Lemma 3.4) whose effective couplings satisfy $g_k \in ]0, \gamma]$ for $0 \le k \le K$ and $g_K = g$. Assume (H-U). Then:*

*(a) Compactness. Every sequence $K_m \to \infty$ has a subsequence $K_{m_i}$ such that, with $\varepsilon_i = L^{-K_{m_i}}$, the limit $S_n(F) := \lim_i S_{n,\varepsilon_i}(F)$ exists for every $n \ge 1$ and every $F \in \mathcal{D}_n$.*

*(b) For every limit family $(S_n)_{n \ge 1}$ obtained in (a):*

- *(OS0) Regularity. $S_n$ is linear on $\mathcal{D}_n$, and $|S_n(F)| \le A(n!)^D \|F\|_{sn}$. In particular $S_n$ restricted to $C_c^\infty(T^{4n} \setminus \bigcup_{i<j}\{x_i = x_j\})$ is a distribution of order $\le sn$.*

- *(OS1) Invariance. $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for every $a \in T_\ell^4$ (all continuous translations), and $S_n(F \circ R^{\times n}) = S_n(F)$ for every $R$ in the hypercubic group $B_4$ (signed permutations of the axes, acting about a lattice site).*

- *(OS2) Reflection positivity. Let $\theta(x_0, \vec{x}) = (-x_0, \vec{x})$ on $T_\ell^4$ and $T_+ := \{x : x_0 \in ]0, \ell/2[\}$. For every finite family $F_0 \in \mathbb{C}$, $F_n \in \mathcal{D}_n$ with $\operatorname{supp} F_n \subset T_+^n$ $(1 \le n \le N)$:*

$$\sum_{n,m=0}^{N} S_{n+m}(\Theta F_n \otimes F_m) \ge 0, \quad (\Theta F_n)(x_1, \dots, x_n) := \overline{F_n(\theta x_n, \dots, \theta x_1)}, \quad S_0 := 1.$$

- *(OS3) Symmetry and reality. $S_n(F \circ \pi) = S_n(F)$ for every permutation $\pi$ of the $n$ arguments, and $S_n(F) \in \mathbb{R}$ for real $F$.*

*(c) Nontriviality is a separate statement, Theorem 4.3.*

*What is claimed about measures.* No probability measure is claimed. The limit is a family of Schwinger functions on $\mathcal{D}_n$. A measure on distributions would need finite moments of smeared fields, and $\mathrm{Var}\sum_p f(x_p)\varphi_p \sim \varepsilon^{-4}$ (Section 2.5). (OS0)–(OS3) on $\mathcal{D}_n$ are the inputs of OS reconstruction.

## 4.4 Theorem (nontriviality), conditional on (B-NT)

**Theorem 4.3** (nontriviality [CONDITIONAL ON (NT) OF SECTION 4.5.9])**.** *Under the hypotheses of Theorem 4.2 and (NT), every limit family satisfies: (i) $S_2 \neq 0$: there are $f_1, f_2 \in C^\infty(T^4)$ with disjoint supports and $S_2(f_1 \otimes f_2) \neq 0$. (ii) $S_2$ is not the two-point function of a scale-invariant (in particular free Maxwell) composite: $d^8 S_2(x,y)$, $|x-y| = d$, is not asymptotically constant as $d \to 0$.*

Part (ii) is stated here in the form the proof supports (two-sided envelope only); see Section 4.8, item 3. The source stated (ii) as "behaves like $c\,g(d)^4$ with $g(d)^2 \sim (\beta \log(1/d))^{-1}$", which needs $\beta_j \to \beta_\infty$ and is not proved.

## 4.5 Proof modulo blocks

Structure. (B-T2) gives the trajectory. (B-S) gives sourced bounds uniform in $\varepsilon$. (B-N1) turns them into (H-U). (B-C) gives (a). (B-REG), (B-TI), (B-RP), and (OS3) give (b). (B-NT) gives Theorem 4.3. Only (B-C), (B-REG), (B-TI), (B-RP), and (OS3) are proved in this section; they use nothing beyond (H-U) and exact lattice symmetries.

### 4.5.1 (B-T2) Theorem 2: the running coupling

Needed. The hypothesis of Balaban's Theorem 1 (CMP 122 II p. 355), $g_k \in ]0,\gamma]$ for all $k$, quoted in Section 1.2 [VERIFIED QUOTE]. It was to be discharged by Theorem 2 of CMP 109 p. 259 [8]: for small $g$ there is $g_0(\varepsilon, g)$ with all $g_k \in ]0,\gamma]$, $g_K = g$, and (0.31): $1/g^2 + \beta \log(L^k\varepsilon)^{-1} \leq 1/g_k^2 \leq 1/g^2 + \beta' \log(L^k\varepsilon)^{-1}$. Balaban: "has not been published yet" [VERIFIED QUOTE].

Status from Section 3 (and §10 of `THEOREM2_PROOF_v4.md`). The weak form, Theorem 2w, is [PROVED+VERIFIED, CONDITIONAL] by Route K (Theorem K + Lemma D″) on (H2) and the fixed-level inputs (PI), (F), ($M_j$), V1, (G2), plus L5, the $K$-independence of $\beta_j(g)$, and (H0) at odd $L > 11$. It suffices for Theorem 1 by the cross-scale sweep [PROVED, UNVERIFIED]. Literal (0.31) is not obtained (it needs eventual positivity of $\beta_j(0)$). Route B (Theorem B of `BETA_FP_PW`) is a formally distinct second route; its logic is [PROVED+VERIFIED, CONDITIONAL], its inputs (U), (DF), (T1) are [PROVED, UNVERIFIED] and rest on (H2). The reduction pieces are Lemma D′, Lemma D″, and Lemmas D, E (the one-argument forms in Lean).

Conclusion. (B-T2) is a [HYPOTHESIS] in the sense that (H2) is certified at no $L$ yet; see the dependency table (Section 4.6).

### 4.5.2 (B-S) Sourced bounds E14-a..g

Needed. A version of Balaban's Theorem 1 with the (C1) source, i.e. with the local bare coupling $w_p = 1/g_0^2 - f(x_p)$, $f$ real, $|\partial^m f| \leq r$ for $m \leq 5$ (class $F_r$), uniformly in $\varepsilon$. Output form (`SOURCES.md` (N1)): $\log Z_\varepsilon[f] - W_\varepsilon[f]$ is $C^\infty$ in source amplitudes $z$ with $|\partial_z^\alpha(\log Z_\varepsilon[f] - W_\varepsilon[f])| \leq C^{|\alpha|}\alpha!\, r^{-|\alpha|}$ for $f = \sum_i z_i f_i$ with disjoint supports. $W_\varepsilon[f]$ is a local field-independent contact functional (allowed to diverge; it contributes zero to every mixed derivative of order $\geq 2$ for disjoint supports, `SOURCES_VERIFY` item 5(a)).

Estimates needed (SOURCES.md §2.5): (E14-a) sourced small-field step; (E14-b) sourced large-field step and R operation; (E14-c) derivative bounds in $f$. Not needed: (E14-d) fixed-torus convergence (= E13); (E14-g) tensor sources. (E14-e) RP and symmetry transfer is done in this section (Sections 4.5.6, 4.5.7). (E14-f) nontriviality at a fixed layer is needed only for Theorem 4.3. The current status of each sub-block is in the dependency table (Section 4.6); (B-S) remains a [HYPOTHESIS].

### 4.5.3 (B-N1) Bridge: source-form bounds ⇒ (H-U) [OPEN]

SOURCES_VERIFY item 5(b) [VERIFIED QUOTE]: the source form "does not give $|S_n(F)| \le C^n n! \|F\|_{s,n}$ for $F$ flat on diagonals with a fixed $s$ ... Mark N1 for general $F$ as a gap." This paper does not close the gap. It gives a route [DERIVED SKETCH], which shows that the gap is removed by the OS index growth $sn$.

1. Moments from cumulants. $\langle \prod_i \phi(f_i) \rangle = \sum_{\text{partitions}} \prod_{\text{blocks}} \kappa(\cdot)$. Singleton cumulants vanish ($\varphi$ is centred). For disjoint supports the source form bounds every cumulant by $C^{|B|}|B|!\, r^{-|B|}$, so the moment by $C^n n!\, r^{-n}$.

2. Scale-$\delta$ tests. A bump of side $\delta \le 1$ and amplitude $a$ has $|\partial^5 f| \sim a\delta^{-5}$; it lies in $F_r$ if $a \le cr\delta^5$. So $n$ disjoint unit-amplitude scale-$\delta$ bumps have $|\text{moment}| \le C^n n!\, \delta^{-5n}$. This is lossy but polynomial in $\delta^{-1}$.

3. Whitney decomposition. Split $T^{4n}$ by a smooth dyadic partition in $\rho(x) := \min_{i<j} |x_i - x_j| \in [\delta, 2\delta]$, $\delta = 2^{-m} \ge \varepsilon$. On each piece cover by cubes of side $\delta/100$; cube tuples are pairwise disjoint. Expand $F\cdot$(cutoffs) in a Fourier series on each cube tuple; the coefficients decay faster than any power; each term is a product of disjointly supported scale-$\delta$ bumps. Flatness gives $|\partial^\beta F(x)| \le C_k \|F\|_{k+|\beta|} \rho(x)^k / k!$. The count is $\le C^n \delta^{-4n}$ cube tuples. Derivatives of the cutoffs cost at most $(Cn^2/\delta)^M$; with $M = O(n)$ this is $\le C^n (n!)^c \delta^{-M}$.

4. Bound per dyadic scale: $\le C^n (n!)^{D'} \delta^{k-9n-M} \|F\|_{k+M}$. Choose $k = sn - M$ with $s$ large enough ($s \ge 20$ suffices on this count) so the $\delta$-exponent is $\ge n$; the sum over dyadic $\delta \in [\varepsilon, 1]$ converges uniformly in $\varepsilon$.

5. Sub-lattice region $\rho(x) \lesssim \varepsilon$: $\le C^n \varepsilon^{-4n}$ plaquette tuples, each moment $\le 2^n$, and $|F| \le \|F\|_k \varepsilon^k$; this is $\le C^n \|F\|_{sn}$ for $k = sn \ge 4n$.

Result (if the steps hold): (H-U) with some fixed $s$ and $D$. To check: uniformity of the source bound in the position of the bumps and in $\varepsilon$; the $C^\infty$ real-source form of (E14-c) with $n$-th derivatives for all $n$; the cutoff-derivative count in step 3. The verifier lists further gaps: complex Fourier modes against real sources (a factor $2^n$), the extra derivatives of Fourier summation (absorbed in $s$), and the smooth min-distance partition. Later work (SOURCES_D4D5, graded in SOURCES_D4D5_VERIFY) confirms "($S_{\text{src}}$) ⇒ (H-U) with $s = 6$, $D > 1$" given (FX) (see Section 4.6). ASSEMBLY_N1.md closes the three gaps given (S) (confirmed in SOURCES_HHIST_N1_VERIFY.md).

### 4.5.4 (B-C) Compactness [PROVED HERE, GIVEN (H-U)]

**Proposition 4.4.** *Under (H-U), every sequence $\varepsilon_m \to 0$ has a subsequence along which $S_{n,\varepsilon}(F)$ converges for every $n$ and every $F \in \mathcal{D}_n$. The limit $S_n$ is linear and satisfies $|S_n(F)| \le \sigma_n \|F\|_{sn}$.*

*Proof.* 1. Separability. $C^{sn}(T^{4n})$ with $\|\cdot\|_{sn}$ is separable (trigonometric polynomials with rational coefficients are dense; Fejér means converge in $C^k$). A subset of a separable metric space is separable. So $\mathcal{D}_n$ has a countable subset $Q_n$ dense for $\|\cdot\|_{sn}$. 2. Diagonal extraction. $Q := \bigcup_n Q_n \times \{n\}$ is countable; enumerate it as $(F^{(1)}, n_1), (F^{(2)}, n_2), \dots$. Each sequence $S_{n_t,\varepsilon_m}(F^{(t)})$ is bounded by $\sigma_{n_t}\|F^{(t)}\|_{sn_t}$ (H-U), so it has a convergent subsequence. Extract successively and take the diagonal subsequence $\varepsilon_i$. 3. Extension. Let $F \in \mathcal{D}_n$, $\eta > 0$, and $G \in Q_n$ with $\|F - G\|_{sn} \le \eta$. By (H-U) and linearity, $|S_{n,\varepsilon_i}(F) - S_{n,\varepsilon_j}(F)| \le 2\sigma_n\eta + |S_{n,\varepsilon_i}(G) - S_{n,\varepsilon_j}(G)|$. The last term $\to 0$. So the sequence is Cauchy; call the limit $S_n(F)$. 4. Linearity passes to pointwise limits, and $|S_n(F)| = \lim |S_{n,\varepsilon_i}(F)| \le \sigma_n\|F\|_{sn}$. □

Remarks. (i) The same subsequence serves all $n$; this matters for (OS2). (ii) Nothing here needs E13. (iii) The bounded normalizations $B_K$ and $c_\varepsilon$ can be included in the extraction.

#### 4.5.5 (B-REG) Regularity (OS0) [PROVED HERE, GIVEN (H-U)]

By step 4 of the proof of Proposition 4.4 the limit satisfies $|S_n(F)| \le \sigma_n\|F\|_{sn}$, $\sigma_n = A(n!)^D$. On the compact torus the Schwartz seminorm is replaced by $\|F\|_p$ (the weights control decay at infinity on $\mathbb{R}^{dn}$; the torus has none). This is the quoted OS II form. Distribution property: for $F \in C_c^\infty$ of the off-diagonal open set, $F \in \mathcal{D}_n$ and the bound is continuity in the $C^{sn}$ topology, so $S_n$ is a distribution of order $\le sn$ there.

Caveat. OS II is formulated on $\mathbb{R}^{dn}$. On $T^{4n}$ the axioms are used in their torus form (periodic test functions). This is a change of setting, not a consequence of OS II.

#### 4.5.6 (B-TI) Translation and hypercubic invariance (OS1) [PROVED HERE, GIVEN (H-U)]

*Lattice, exact.* Put the origin at a lattice site.

- Translations $a \in \varepsilon\mathbb{Z}^4$. The map $U \mapsto U^a$, $U^a(b) := U(b-a)$, permutes the links, preserves the product Haar measure and the Wilson action, and $P_p(U^a) = P_{p-a}(U)$. So $\langle\prod_i \varphi_{p_i+a}\rangle_\varepsilon = \langle\prod_i \varphi_{p_i}\rangle_\varepsilon$ and $S_{n,\varepsilon}(F \circ \tau_a^{\times n}) = S_{n,\varepsilon}(F)$. The coupling $g_0(\varepsilon)$ is one number, so nothing in $\mu_\varepsilon$ depends on position.

- Hypercubic $R \in B_4$ (signed permutation matrices, $|B_4| = 384$). $R$ maps sites to sites, links to links (possibly reversed), plaquettes to plaquettes, and $x_{Rp} = Rx_p$. Define $(RU)(b) := U(R^{-1}b)$ with $U(-b) = U(b)^{-1}$. Haar measure is invariant under $U \mapsto U^{-1}$ and $\operatorname{Re}\operatorname{tr} W^{-1} = \operatorname{Re}\operatorname{tr} W$ for $W \in SU(2)$, so $A$ and the measure are invariant and $P_{Rp}(RU) = P_p(U)$. Hence $S_{n,\varepsilon}(F \circ R^{\times n}) = S_{n,\varepsilon}(F)$.

- These identities hold at every $\varepsilon$. $F \circ \tau_a^{\times n}$ and $F \circ R^{\times n}$ are again in $\mathcal{D}_n$. They pass to the limit along the subsequence of Proposition 4.4: $S_n(F \circ R^{\times n}) = S_n(F)$, and $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for $a \in \varepsilon_i\mathbb{Z}^4$ for all large $i$.

*Continuous translations.* Let $a \in T^4_\ell$. Pick $a_i \in \varepsilon_i\mathbb{Z}^4$ with $|a - a_i| \le \varepsilon_i$ (componentwise rounding gives $\le \varepsilon_i\sqrt{4}/2 = \varepsilon_i$). Write $F_a := F \circ \tau_a^{\times n}$. The mean value theorem along the diagonal direction gives $\|F_a - F_{a_i}\|_{sn} \le 4n|a - a_i|\,\|F\|_{sn+1}$. Then

$$|S_n(F_a) - S_n(F)| \le |S_n(F_a) - S_{n,\varepsilon_i}(F_a)| + |S_{n,\varepsilon_i}(F_a) - S_{n,\varepsilon_i}(F_{a_i})| + |S_{n,\varepsilon_i}(F_{a_i}) - S_{n,\varepsilon_i}(F)| + |S_{n,\varepsilon_i}(F) - S_n(F)|.$$

The first and last terms $\to 0$ by Proposition 4.4. The third is 0 by lattice invariance. The second is $\le \sigma_n \cdot 4n\,\varepsilon_i\|F\|_{sn+1}$ by (H-U) and linearity. Let $i \to \infty$. $F$ is smooth, so $\|F\|_{sn+1} < \infty$. Rotations outside $B_4$ are not available (Section 4.7).

### 4.5.7 (B-RP) Reflection positivity (OS2)

*Lattice theorem* [CITED]: Osterwalder–Seiler [21], Ann. Phys. 110 (1978) 440–471. The paper is not in the project folder; no verbatim quote. Content as usually stated: for the Wilson action with compact gauge group and any $\beta \geq 0$, and $\theta$ the reflection in a lattice hyperplane that bisects the links orthogonal to it, $\langle \Theta(\Phi)\Phi \rangle \geq 0$ for every function $\Phi$ of the link variables in the positive half-lattice, $(\Theta\Phi)(U) := \overline{\Phi(\theta U)}$. Jaffe–Witten [15] line 546 [VERIFIED QUOTE]: "Reflection positivity holds for the Wilson approximation [36], a major advantage". The template uses the same input (`o3_continuum.txt` line 4902) [VERIFIED QUOTE]: "At every cutoff the nearest-neighbour measure is reflection positive across microscopic site and link seams." Proof idea (standard): fix the crossing links to 1 by a gauge transformation; the action splits as $A_+ + \theta A_+ + \sum_{\text{crossing}} \operatorname{Re}\operatorname{tr}(X_p \theta(X_p)^*)$, and $\exp(\beta \operatorname{Re}\operatorname{tr}(XY^*))$ is a positive-type kernel (positive combination of characters), which gives a sum of squares. On the periodic lattice there are two fixed planes, $x_0 = \varepsilon/2$ and $x_0 = \varepsilon/2 + \ell/2$; both are link planes because $N_\varepsilon$ is even ($\ell$ even, Section 2.5). The periodic two-seam version is standard [CITED]; its text was not checked. For site-plane reflections see Menotti–Pelissetto [17] (Section 4.8, item 1).

*Passage to the limit* [PROVED HERE, GIVEN (H-U) AND THE LATTICE THEOREM]. Setup. $\theta_\varepsilon(x) := \theta(x) + \varepsilon e_0 = \theta(x - \varepsilon e_0)$, fixed planes $x_0 = \varepsilon/2,\ \varepsilon/2 + \ell/2$. The positive half $\Lambda_+$ consists of links with both ends in $\{x_0 \in \{\varepsilon, 2\varepsilon, \dots, \ell/2\}\}$. Fix $F_0 \in \mathbb{C}$ and $F_n \in \mathcal{D}_n$, $\operatorname{supp} F_n \subset T_+^n$, $1 \leq n \leq N$. Supports are compact in the open set, so there is $\delta > 0$ with $x_0 \in [\delta, \ell/2 - \delta]$ on all supports.

Step 1 (lattice form). For $\varepsilon < \delta/2$ every plaquette with centre in a support has all four links in $\Lambda_+$. Define $\Phi := F_0 + \sum_{m=1}^{N} \sum_{p_1 \dots p_m} F_m(x_{p_1}, \dots, x_{p_m}) \prod_i \varphi_{p_i}$. $\Phi$ depends only on $\Lambda_+$ links. Since $\varphi_p$ is real and $P_p(\theta_\varepsilon U) = P_{\theta_\varepsilon p}(U)$, re-indexing gives $\Theta\Phi = \bar{F}_0 + \sum_n \sum_q \overline{F_n}(\theta_\varepsilon x_{q_1}, \dots, \theta_\varepsilon x_{q_n}) \prod \varphi_{q_i}$. Hence

$$0 \leq \langle \Theta\Phi\, \Phi \rangle_\varepsilon = \sum_{n,m=0}^{N} S_{n+m,\varepsilon}(\Theta_\varepsilon F_n \otimes F_m), \quad \Theta_\varepsilon F_n := (\Theta F_n) \circ \tau_{-\varepsilon e_0}^{\times n},\ S_{0,\varepsilon} := 1.$$

(Argument order inside $\Theta F_n$ is immaterial by (OS3).) Each $\Theta_\varepsilon F_n \otimes F_m$ lies in $\mathcal{D}_{n+m}$: its two factors have supports at positive distance and each factor is flat on its own diagonals.

Step 2 (seam shift). $\|\Theta_\varepsilon F_n \otimes F_m - \Theta F_n \otimes F_m\|_{s(n+m)} \leq 2^{s(n+m)} \cdot 4n\,\varepsilon\, \|F_n\|_{s(n+m)+1} \|F_m\|_{s(n+m)}$. By (H-U) the difference of the $S_{n+m,\varepsilon}$ values is $\leq \sigma_{n+m}$ times this, which $\to 0$.

Step 3 (limit). Along the subsequence of Proposition 4.4, $S_{n+m,\varepsilon_i}(\Theta F_n \otimes F_m) \to S_{n+m}(\Theta F_n \otimes F_m)$ for each of the finitely many $(n, m)$. So $\sum_{n,m} S_{n+m}(\Theta F_n \otimes F_m) = \lim_i \sum_{n,m} S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m) \geq 0$.

**Lemma 4.5** (positivity of a finite form survives pointwise limits [PROVED HERE])**.** *Let $a^{(i)} \in \mathbb{C}^{(N+1)\times(N+1)}$ with $\sum_{n,m} \bar{c}_n c_m a^{(i)}_{nm} \geq 0$ for all $c$ and all $i$, and $a^{(i)}_{nm} \to a_{nm}$ for every $(n, m)$. Then $\sum \bar{c}_n c_m a_{nm} \geq 0$ for all $c$.*

*Proof.* For fixed $c$ the left side is a finite linear combination of the entries, so it converges to $\sum \bar{c}_n c_m a_{nm}$; $[0, \infty)$ is closed. □

Step 3 is this lemma with $a^{(i)}_{nm} := S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m)$ (replace $F_n$ by $c_n F_n$), plus Step 2.

### 4.5.8 (OS3) Symmetry and reality [PROVED HERE]

$S_{n,\varepsilon}(F \circ \pi) = S_{n,\varepsilon}(F)$: the $\varphi_p$ are commuting real random variables; re-index the sum. For real $F$, $S_{n,\varepsilon}(F) \in \mathbb{R}$. Both identities are linear and pass to the limit. $S_0 = 1$ by definition.

#### 4.5.9 (B-NT) Nontriviality: what is needed

- A two-point function alone never excludes a Gaussian: every positive-definite $S_2$ is the covariance of some Gaussian generalized field.
- The plaquette density is a composite ($\approx \frac{1}{4}\operatorname{tr} F^2 \times \varepsilon^4$). Even in the free theory its connected $n \geq 3$ functions are nonzero (Wick squares). So $S_n^c \neq 0$ for some $n \geq 3$ does not prove interaction.
- The meaningful test excludes (a) the zero theory and (b) the free Maxwell composite. The free composite has, at short distance, the scale-invariant form $c_0 d^{-8}$. Asymptotic freedom predicts $c\, g(d)^4 d^{-8}$ with a running coupling.

*Free-theory prediction along this sequence.* At tree level with the bare coupling, $\langle \varphi_p \varphi_{p'} \rangle \approx c\,\varepsilon^8 g_0^4 |x-y|^{-8}$. Since $g_0(\varepsilon)^2 \leq (1/g^2 + \beta \log \varepsilon^{-1})^{-1} \to 0$ by (0.31), the bare-coupling prediction for $S_2$ at fixed separation tends to 0. The claim is that the true $S_2$ does not.

*Statement (NT) needed* [HYPOTHESIS]; owned by (E14-f), a corollary of (E14-a) at a fixed layer. There are $c > 0$ and $C$ such that for all $K$, all $f_1, f_2 \in C^\infty(T^4)$ with disjoint supports at distance $\geq d$, $d \geq L\varepsilon$, and $k(d)$ the layer with $L^k \varepsilon \leq d < L^{k+1}\varepsilon$:

$$S_{2,\varepsilon}(f_1 \otimes f_2) = c \iint f_1(x) f_2(y)\, g^4_{k(|x-y|)} |x-y|^{-8}\, dx\, dy\, (1+R), \quad |R| \leq C(g^2_{k(d)} + (d/\ell)^2).$$

$c$ is the one-loop (tree-level in the composite) coefficient: $c \propto \sum(\text{free field-strength covariance})^2$, a sum of squares, so $c > 0$ [DERIVED, STANDARD]; its value must be computed in Balaban's normalization.

*Derivation of Theorem 4.3 from (NT) and (B-T2)* [PROVED HERE, GIVEN (NT)]. (i) Take $f_1, f_2 \geq 0$, not identically 0, with supports at distance $\geq d_0$. Corrected condition (Section 4.8, item 3): the support diameters are $\ll \ell$, so that $g^2 + (|x-y|/\ell)^2$ is small on the whole support pair and $|R| \leq \frac{1}{2}$. By (0.31) at layer $k(d)$ with $d \in [d_0, \ell]$: $1/g_k^2 \leq 1/g^2 + \beta' \log(L^k\varepsilon)^{-1} \leq 1/g^2 + \beta' \log(L/d_0)$, uniformly in $\varepsilon$. So $S_{2,\varepsilon}(f_1 \otimes f_2) \geq \frac{1}{2} c (1/g^2 + \beta' \log(L/d_0))^{-2} \iint f_1 f_2 |x-y|^{-8} > 0$ uniformly in $\varepsilon$, and the limit inherits the bound. (ii) For bumps concentrated at separation $d \to 0$ (with $d$ a fixed multiple of their width), $d^8 S_2 \leq (1 + C g^2_{k(d)})\, c\, (1/g^2 + \beta \log(1/d))^{-2} \to 0$ as $d \to 0$ by the lower bound in (0.31), while $d^8 S_2 > 0$ by (i). A scale-invariant composite would give a nonzero constant limit. So the limit is neither zero nor the free composite. Only the two-sided bound (0.31) is used. Caveat. (ii) needs (NT) uniformly down to $d \geq L\varepsilon$ with the remainder controlled by $g^2_{k(d)}$, i.e. the fixed-layer expansion at every layer. If (E14-f) is proved only at a fixed layer $j$, (i) holds, and (ii) holds only in the weaker form "$d^8 S_2(d)$ is bounded between the two (0.31) envelopes for $d \in [L^{-j}, 1]$".

### 4.6 Dependency table

The table gives the status of every block of Theorem B and of every named hypothesis of Theorem A. It follows the state recorded in `PLAN.md` (through the entry `SOURCES_MA2_VERIFY.md`, 2026-10-28) and §10 of `THEOREM2_PROOF_v4.md`. Global conditions carried by every row: (GC1) Balaban's published proofs CMP 95–122 (1984–89) are correct as published; nothing in this paper re-checks them line by line. (GC2) Balaban's inductive assumptions (CMP 109 pp. 263–264) hold with the stated uniformity. (GC3) $L$ odd, $L > 11$, $M$ large, $\gamma$ small as Balaban requires; $\ell$ even (sufficient, Section 2.5). Rows "item $n$" are the named hypotheses of Section 3; $n$ is their item in Section 6. Their file columns use the short names of `THEOREM2_PROOF_v2_VERIFY` §K (KAPPA

= KAPPA_BOUNDED, H_VERIFY = BETA_FP_H_VERIFY, PW_VERIFY = BETA_FP_PW_VERIFY, v1 verifier = THEOREM2_PROOF_VERIFY).

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-T2 | Theorem 2w (Theorem 3.1): trajectory $g_0(\varepsilon, g)$, (b), (c′), (S) | [PROVED+VERIFIED, CONDITIONAL] (Route K = Theorem K + Lemma D″). Literal (0.31) not obtained | THEOREM2_PROOF_v4.md, KAPPA_BOUNDED.md | (H2), (PI), (F), $(M_j)$, V1, (G2), L5, (H0), $K$-independence of $\beta_j(g)$ |
| B-T2.a | Lemmas D, D′, D″, E (shooting) | [PROVED+VERIFIED; LEAN] (D′, D, E, D″ one-argument form; history form open) | LEMMA_DPRIME.md, lean/ | continuity of $\beta_{k+1}$ in history |
| B-T2.b | Sweep: (c′)+(S) ⇒ Balaban's Theorem 1 inputs (Section 3.4.3; item 10) | [PROVED, UNVERIFIED]; 4 of 19 rows verified (LEMMA_DPRIME_VERIFY Steps 4–5, rows 11, 12, 16, 18) | LEMMA_DPRIME_SWEEP.md | Balaban readings (unchecked list) |
| B-T2.1 | U3 Thm 4.1 (Gaussian RG, rate $L^{-2j}$) + O5 | [PROVED+VERIFIED] | U3_GAUSSIAN_RG.md, U3_REPAIRS.md | trivial background |
| B-T2.2 | U3 Lemma 5.1; (A), (W) | neutral sector [PROVED+VERIFIED]. (W) numerically exact; (A) confirmed qualitatively and quantitatively (U3_AW_VERIFY.md); (HS) [REFUTED] at $L = 3$. Not on the path of Theorem 2w after (H2); needed for the convergence route only | U3_LEMMA51.md, U3_AW.md | (A) in full, (N) (plausibly false) |
| B-T2.3 | O6 bordered-matrix invertibility | one-step [PROVED+VERIFIED]; composed constraint reduced to one-step O6 + (A) | U3_REPAIRS.md | — |
| B-T2.4 | $c = 11N^2/(12\pi^2)$, anchor-free identity Prop. 2.1, telescoping | [PROVED+VERIFIED]. $\beta_\infty = c \log L$ itself replaced by window sums (Theorem K, Route B) | BETA_FP_LEMMA43.md, BETA_FP_LEMMA43_VERIFY.md | (F) [CITED, SOURCE CHECKED] |
| B-T2.5 | $\kappa_k$ bounded (Theorem K) | [PROVED+VERIFIED, CONDITIONAL ON (B2)]; (B2) [PROVED+VERIFIED, CONDITIONAL ON (H2)] | KAPPA_BOUNDED.md, KAPPA_BOUNDED_VERIFY.md, B2_TWOSTEP.md, B2_TWOSTEP_VERIFY.md | (H2), (PI), $(M_j)$ |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-T2.H2 | (H2) uniform two-step inequality $\rho_2 < L$ (Section 3.7.3; item H) | [NUMERICAL]: $L = 3, 5, 9$ confirmed by verifier, margin 31% ($L = 3$), 58% ($L = 5$); certification open. Point enclosure $[1.9993, 2.1273] < 3$ at $L = 3$, $j = \infty$, one point [INTERVAL ENCLOSURE, VERIFIED] (Section 5.6); full certification about $1.5 \times 10^4$ core-hours, an estimate graded UNSUPPORTED; one-step route: $\rho_1(\infty) = 1.760, \ldots, 2.0895$ at $L = 3, \ldots, 13$, $\sup \rho_1^2 = 4.37 < 13$ at $L = 13$ [NUMERICAL] (no verifier); $\rho_2 \le \rho_1(j) \max_l \rho_1(j - 1, k_l)$ [PROVED+VERIFIED; LEAN]; premise P4 [PROVED, UNVERIFIED]; $L = 13$ certificate sized at about 900–1,100 core-hours after fix G1, launch approved with conditions, not launched | `H2_LARGE_L.md`, `H2_LARGE_L_VERIFY.md`, `H2_CERTIFY.md`, `H2_CERTIFY_VERIFY.md`, `H2_CERTIFY_L13.md`, `H2_CERTIFY_L13_G1.md`, `H2_CERTIFY_P4.md`; B2_TWOSTEP §2; BETA_FP_U §2.3 | interval arithmetic, tail bound $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$) |
| B-T2.H1 | one-step route at $L = 13$: $\sup \rho_1^2 < L \Rightarrow$ (H2) | $\rho_1(\infty) = 2.0895$, $\sup \rho_1^2 = 4.37 < 13$ [NUMERICAL] (no verifier; G5 open); factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j - 1, k_l)$ [PROVED+VERIFIED; LEAN] (`rho2_le`); premise P4 [PROVED, UNVERIFIED] (identity (D) [LEAN], `P4_divergence`); fix G1 sizes the certificate at about 900–1,100 core-hours over $j = 1, 2$ and M0:169; launch approved with conditions (G1 code, a verifier pass on the driver, a self-sustaining run), not launched | `H2_CERTIFY_L13.md`, `H2_CERTIFY_L13_G1.md`, `H2_CERTIFY_P4.md`, `H2_CERTIFY_VERIFY.md` | interval arithmetic; premises P1′–P4; G1 code; G5 verifier |
| B-T2.B | Route B (Theorem $B_{PW}$; item B8) | logic [PROVED+VERIFIED, CONDITIONAL] (tag decision, `PLAN.md`; PW_VERIFY §6 items 1–4); inputs (U), (DF), (T1) [PROVED, UNVERIFIED] | `BETA_FP_PW.md`, `BETA_FP_PW_VERIFY.md`, `BETA_FP_U.md` | (H2), ($B\theta$), A6, LW (5.7), (H0) |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| item 1 | (B2) / (B2w) (Section 3.7) | open; reduced to (H2) | KAPPA §2, Rem. 2.3; KAPPA_VERIFY §7–§9 | (H2) |
| item 1 | (H2) $\Rightarrow$ ($C1\sigma$) $\Rightarrow$ (B2) (Theorems 3.10, 3.11) | proved+verified, conditional on (H2) | B2_TWOSTEP §2–§3; B2_TWOSTEP_VERIFY grades 1, 3, 4 | (H2) |
| item 1 | ($C1\sigma$) + (h) $\Rightarrow$ (B2) | implication verified; ($C1\sigma$) $\Leftarrow$ (H2); (h) numerical $L = 3, 5$ | KAPPA Prop. 4.1; KAPPA_VERIFY §7 | — |
| item 1 | (H$''$) $\Rightarrow$ (A) $\Rightarrow$ (B2) | implication argued; not needed after (H2) | U3_AW §4.7; no verifier (U3_AW_VERIFY grade 5 covers (H$'$) only) | — |
| — | (HS) (Section 3.7.2) | refuted $L = 3$ | U3_AW (H$'$); U3_AW_VERIFY §6 | — |
| item H | $\rho_1^2 < L \Rightarrow$ (H2) (one step, $L \geq 5$) | arithmetic; $\rho_1^2/L = 0.771$ at $L = 5$, numerical | H2_LARGE_L §3; H2_LARGE_L_VERIFY grade 4 (measurement) | — |
| item H | certification tail $j_0$ | derivation confirmed; $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$), corrected | H2_LARGE_L §4.1; H2_LARGE_L_VERIFY grade 6 | — |
| item 2 | (PI) | argued | BETA_FP_H §3.1; H_VERIFY §5 | — |
| item 3 | (F) | cited, source checked | BETA_FP_LEMMA43; H_VERIFY §4 | — |
| item 4 | ($M_j$): (S1)–(S3), Prop. 4.4 | open / argued | BETA_FP_M §4 (no verifier); H_VERIFY §5 refuted the old step | (S1)–(S3) |
| item 5 | V1 | open | R2; U3_VERIFY C4 (order $p^2$) | — |
| item 6 | (G2) | formal | BETA_FP_LEMMA43 | — |
| item 7 | (H0), uniform in $j$ and history | [Balaban], odd $L > 11$ | CMP 109 (1.22); BALABAN_CHECK item 10 | — |
| item 8 | L5 history form and domain (Section 3.4.5) | open | LEMMA_DPRIME Rem. 3; LEMMAS_DE_VERIFY §2.1; v2_VERIFY §F | — |
| item 9 | $\beta_j(g)$ independent of $K$ | open | LEMMAS_DE_VERIFY §2 item 2; v1 verifier Point 7 | — |
| item 10 | BALABAN_CHECK item 11 readings | unchecked | BALABAN_EXCERPTS | — |
| item 11 | odd $L$; $L > 11$ in Balaban | input | CMP 109 pp. 251–252; v1 verifier Step 6 | — |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| item 12 | SU(2) | scope | — | — |
| item 13 | Lean history form | open | — | — |
| B1, B2′ | Route B: (H2) + (T1) ⇒ (U) (Theorem 2.4 of BETA_FP_U) | proved, unverified | BETA_FP_U §2.4; no verifier | (H2), (T1) |
| B1, B2′ | Route B: (U) | hypothesis; ⇐ (T1) + (H2) (BETA_FP_U); proved, unverified | BETA_FP_PW §1.2; PW_VERIFY §3 | (T1), (H2) |
| B1 | Route B: (T1) | argued | BETA_FP_U §2.3; no verifier | — |
| B3 | Route B: $(B\theta) = \gamma_{rec} + \mathfrak{d}_F$ | $\gamma_{rec}$ bounded given Thm 2.4 (proved, unverified); (DF) argued | BETA_FP_PW §3.3; BETA_FP_U §3; PW_VERIFY §6 (logic of Theorem $B_{PW}$); none for BETA_FP_U | (DF) |
| B4 | Route B: LW (5.7) | cited | cited; PW_VERIFY §3 (regime correct) | — |
| B5 | Route B: A6 ($\beta$ identification, uniform in $k$) | argued | SOURCES_D1_VERIFY 8; PW_VERIFY §3; BETA_FP_U §4.2 | — |
| B7 | Route B: $\theta_* < 1$ | trivial condition | PW_VERIFY §3 | — |
| B-T2.6 | certified early steps $\beta_j(0) > 0$ | needed only for literal (0.31) at every $K$, not for Theorem 1; [NUMERICAL], no certificate; $\beta_1(0)$ computed to $L = 11$ | `numerics/NUMERICS.md`, `UMK_BETA1.md` | $L > 11$ is Balaban's range |
| B-S | sourced Theorem 1 (E14-a,b,c) | [OPEN] | `SOURCES.md` §2.5 | B-T2 for the sourced trajectory |
| B-S.D1 | one sourced step, Gaussian level | [PROVED+VERIFIED] (with corrections) | `SOURCES_D1.md`, `SOURCES_D1_VERIFY.md` | — |
| B-S.D2 | sourced small-field step (E14-a) | (a), (b), (d), (f) confirmed; (c) rejected; (e) reduced | `SOURCES_D2.md`, `SOURCES_D2_VERIFY.md` | analyticity in input activities, $\kappa \to \kappa/2$ |
| B-S.D2C | redo of (c): frozen source per domain, Lemma A, $h \equiv 0$ | Lemma A and recursion confirmed; (I) corrected (exact only with adapted gauge-fixing weight) | `SOURCES_D2C.md`, `SOURCES_D2C_VERIFY.md` | domain hypothesis (D); (H$\beta$) |
| B-S.H$\beta$ | two-loop bound $\sup \sum_j \lvert B'_j \rvert \le \frac{1}{2}$ | [PROVED+VERIFIED, CONDITIONAL] on (I), (H-hist) | `SOURCES_HBETA.md`, `SOURCES_HBETA_VERIFY.md` | (I), (H-hist) |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-S.hist | (H-hist) history sensitivity | [VERIFIED REDUCTION] on (W4), Facts A/B, (I)/L3 and (G-gain) in both slots (`SOURCES_SM.md`; verifier: items 1–4 confirmed, 5–6 corrected); the (G-gain) chain continues in row B-S.Ggain (G3, (Q)); Lemma HH: `SOURCES_HHIST.md`, `SOURCES_HHIST_N1_VERIFY.md` | `SOURCES_SM.md`, `SOURCES_SM_VERIFY.md` | (G-gain) |
| B-S.Ggain | (G-gain): $g_k\|B\|$ gain in the small-field step | pointwise Cauchy and uniform $C_\beta$ on $]0,\gamma]$ [PROVED+VERIFIED] (small-field, incl. vacuum subtraction); Lemma G4 corrected, fix reduced; $\chi_k$ surface term = Lemma G3, proved given (Q) [PROVED+VERIFIED, CONDITIONAL] (`SOURCES_G3_VERIFY`); (Q) reduced to CMP 99 (3.132), ($\mathrm{E}_U$), and the localization lemma LOC [VERIFIED REDUCTION] (`SOURCES_Q_VERIFY`, `SOURCES_CDC_VERIFY`, `SOURCES_EU_VERIFY`, `SOURCES_LOC_VERIFY`); large field [OPEN] | `SOURCES_GGAIN.md`, `SOURCES_GGAIN_VERIFY.md` | Bałaban's $R_2$ bound (2.16), $\gamma_0$ for general $U$, kernel-moment bounds for ($\mathrm{E}_U$); not at $g = 0$ |
| B-S.G3 | Lemma G3: $\chi_k$ surface term, uniform in $g$ | [PROVED+VERIFIED, CONDITIONAL] on (Q) (Route 2 surface estimate; the B/B′ cut closed in `SOURCES_Q`) | `SOURCES_G3.md`, `SOURCES_G3_VERIFY.md` | (Q) |
| B-S.Q | (Q): covariance lower bound $c_- > 0$ | [VERIFIED REDUCTION] to CMP 99 (3.132) (printed), ($\mathrm{E}_U$), and LOC; Woodbury identity confirmed (CDC); free-case constants [NUMERICAL] (`numerics/kappa_d.py`) | `SOURCES_Q.md`, `SOURCES_Q_VERIFY.md`, `SOURCES_CDC.md`, `SOURCES_CDC_VERIFY.md`, `numerics/KAPPA_D.md` | (3.132); two [I] bridges |
| B-S.EU | ($\mathrm{E}_U$): uniform block extension | $K_0(4) = 195.19$ ($N = \infty$) [NUMERICAL, REPRODUCED]; $K \le K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$ [DERIVED SKETCH]; ($\mathrm{E}_U$) [VERIFIED REDUCTION] to kernel-moment bounds Bałaban does not print | `SOURCES_EU.md`, `SOURCES_EU_VERIFY.md`, `numerics/k0_d.py` | $k$-uniform kernel moments; unprinted $C_1'$ |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-S.LOC | LOC: localized covariance $C^{(k)}(Z_0, 0)$ | [PROVED+VERIFIED, CONDITIONAL] (short lemma; symmetry (S) not needed) | `SOURCES_LOC.md`, `SOURCES_LOC_VERIFY.md` | Bałaban's $R_2$ bound (2.16); $\gamma_0$ for general $U$; $\alpha_0, \alpha_1$ small |
| B-S.thr | (H-thr) threshold shell sum | Lemma T and trajectory sum [PROVED+VERIFIED], conditional on (H-hist2); (H-hist2) [VERIFIED REDUCTION] to the real uniform (H-hist), no new input (`SOURCES_HHIST2`, `SOURCES_HHIST2_VERIFY`) | `SOURCES_HTHR.md`, `SOURCES_HTHR2.md`, `SOURCES_HTHR2_VERIFY.md` | (H-hist2), M3 weights, frozen integers (L3) |
| B-S.hist2 | (H-hist2): mixed second history derivatives | [VERIFIED REDUCTION] to the real uniform (H-hist), no new input (Route 2(a); decay $\lambda_C^{m/2}$; the product form $\lambda^m$ [REFUTED]) | `SOURCES_HHIST2.md`, `SOURCES_HHIST2_VERIFY.md` | real uniform (H-hist) |
| B-S.D3 | sourced large-field step + R operation (E14-b) | draft verified with corrections (`SOURCES_D3_VERIFY`, `SOURCES_D3R_VERIFY`, `SOURCES_D3W8_VERIFY`; route (a), Lemma F2′ confirmed given (W4), (Eps), F1, (D)). (FX) is a genuine new hypothesis (`SOURCES_D4D5_VERIFY.md`); its state is in row B-S.FX | `SOURCES_D3.md`, `SOURCES_FX.md`, `SOURCES_FX_VERIFY.md` | (FX), (FX2) |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-S.FX | O1(b), O2, O3, O4, R2 | O1(b) [PROVED+VERIFIED] (Lemma [P], no additive constant); O2 [PROVED+VERIFIED] (the R-denominator is the bare Wilson integral (1.100)); O3 [PROVED+VERIFIED, CONDITIONAL ON N3] (N3 = the open HBETA strip bound $\|\delta w\| \le \frac{3}{2}\tau$); R2 ($\kappa_{fin} \ge \kappa_{final}$) [PROVED+VERIFIED] as a reduction, in volume form, given $R_{K+1} \ge 1$ and $C_c$ independent of $g$; O4 (unbounded components in (1.103)) avoided by keeping Balaban's main form (1.98), whose fallback was Gevrey $2+\eta$. N3 reduced to (H-hol) + real (H-hist) + (H2-hist), condition (a) settled [VERIFIED REDUCTION] (`SOURCES_HBETA_N3_VERIFY`, `SOURCES_N3A_VERIFY`); (H-hol) small field [VERIFIED REDUCTION] (`SOURCES_HHOL_VERIFY`); large field [VERIFIED REDUCTION] to (M-A), (M-B), (M-C), (H-hist2) (`SOURCES_HHOL_LF_VERIFY`); (M-B) [PROVED+VERIFIED] with $p_0 \ge 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \le 0.2$; (M-A) reduced to (H−), (M-A′) closed, (H−) rests on CMP 102 (189) [VERIFIED REDUCTION] (`SOURCES_MA_VERIFY`, `SOURCES_MA2_VERIFY`); (M-C) [OPEN] | `SOURCES_FX_O1.md`, `SOURCES_FX_O1_VERIFY.md`, `SOURCES_FX_O23.md`, `SOURCES_FX_O23_VERIFY.md`, `SOURCES_FX_R2.md`, `SOURCES_FX_R2_VERIFY.md` | (M-C); CMP 102 (189); $R_{K+1} \ge 1$, $C_c$ independent of $g$ |
| B-S.N3 | N3: HBETA strip bound $\|\delta w\| \le \frac{3}{2}\tau$ | [VERIFIED REDUCTION] to (H-hol) + real (H-hist) + (H2-hist) | `SOURCES_HBETA_N3.md`, `SOURCES_HBETA_N3_VERIFY.md` | (H-hol), (H-hist), (H2-hist) |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-S.N3A | condition (a) of Lemma N3 | settled [PROVED+VERIFIED] (slack $> 5\times$; item 1 corrected) | `SOURCES_N3A.md`, `SOURCES_N3A_VERIFY.md` | constant order $L \to$ Bałaban's constants $\to \gamma \to g \le \gamma/2 \to r_0 \to K$ |
| B-S.hol | (H-hol), small field: $\beta_j$ holomorphic on $\lvert g - x\rvert < \rho_0 x$ | [VERIFIED REDUCTION] to Bałaban's conditions with modified constants | `SOURCES_HHOL.md`, `SOURCES_HHOL_VERIFY.md` | (C1′)–(C4′), (C4″), (2.24′) |
| B-S.holLF | (H-hol), large field (frozen thresholds at $x^*$) | [VERIFIED REDUCTION] to (M-A), (M-B), (M-C), (H-hist2) | `SOURCES_HHOL_LF.md`, `SOURCES_HHOL_LF_VERIFY.md` | (M-A)–(M-D) |
| M-A | equipartition $\langle A - A_{\min}\rangle_u \le (n_i/u)(1+\eta_x)$ | [PROVED+VERIFIED] given (H−); (M-A′) closed [PROVED+VERIFIED]; (H−) [VERIFIED REDUCTION] to CMP 102 (189), an existing trust item | `SOURCES_MA.md`, `SOURCES_MA_VERIFY.md`, `SOURCES_MA2.md`, `SOURCES_MA2_VERIFY.md` | CMP 102 (189); (C1), (C2) |
| M-B | (1.80)–(1.98) bookkeeping, changed constants | [PROVED+VERIFIED] with $p_0 \ge 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \le 0.2$ | `SOURCES_MB.md`, `SOURCES_MB_VERIFY.md` | (M-A), (M-C) |
| M-C | complex small-field step | [OPEN] | `SOURCES_HHOL.md` ledger, `SOURCES_HHOL_LF.md` | — |
| M-D | frozen map vs Bałaban's map | not a convention: needs the frozen map's derivative at $x^*$, which is (H-hist) + (H-thr) [VERIFIED REDUCTION] | `SOURCES_HHOL_LF_VERIFY.md` | (H-hist), (H-hist2) |
| B-S.D4 | real-source derivative bounds (E14-c) | D4.1, D4.3, D4.4 and Lemma C (order $1+\eta$, given (FX)) [PROVED+VERIFIED]; ($\mathrm{S_{src}}$) $\Rightarrow$ (H-U), $s = 6$, $D > 1$, confirmed given (FX) | `SOURCES_D4D5.md`, `SOURCES_D4D5_VERIFY.md` | (FX), (FX2), S1 at all scales |
| B-N1 | source form $\Rightarrow$ (H-U) on $\mathcal{D}_n$ | derived given the sourced Gevrey bound (S) (`ASSEMBLY_N1.md`; confirmed, `SOURCES_HHIST_N1_VERIFY.md`); ($\mathrm{S_{src}}$) $\Rightarrow$ (H-U), $s = 6$, $D > 1$, confirmed given (FX); residual: uniformity of the D3R constants down to separation $16\varepsilon$ | this paper, `SOURCES_D4D5.md` | (FX), uniformity in bump position and $\varepsilon$ |
| B-C | compactness (diagonal extraction) | [PROVED HERE]; verifier: correct | Proposition 4.4 | (H-U) |

| Block | Content | Current tag | Files | Hypotheses carried |
|---|---|---|---|---|
| B-REG | (OS0) | [PROVED HERE]; verifier: correct | Section 4.5.5 | (H-U); torus form of OS II |
| B-TI | (OS1) all of $T^4$ and $B_4$ | [PROVED HERE]; verifier: correct | Section 4.5.6 | (H-U); origin at a site |
| B-RP | (OS2) | lattice: [CITED]; passage: [PROVED HERE]; verifier: correct | Section 4.5.7 | $\ell$ even (sufficient); (H-U) |
| OS3 | permutation symmetry, reality | [PROVED HERE] | Section 4.5.8 | — |
| B-NT | nontriviality (Theorem 4.3) | [OPEN]: needs (NT) = (E14-f); derivation from (NT) [PROVED HERE], corrected statement | Section 4.5.9 | (NT) at every layer for (ii); $c > 0$ in Balaban's normalization |
| E13 | full-sequence convergence | not needed here; [OPEN] | `SOURCES.md` (E14-d), D6 | — |

Critical path to an unconditional (modulo GC1–GC3) Theorem 4.2: (H2) certified at the actual $L$ (with (PI), (F), $(M_j)$, V1, (G2), L5, (H0)) for B-T2; then B-S: the real uniform (H-hist) (large-field case; Bałaban-level inputs of G3), (M-C), and the residual uniformity of B-N1. The blocks proved in this section are short and are not on the critical path.

## 4.7 What the theorem does not give

Each item needs a separate argument that this paper does not give.

- Rotation invariance. Only $B_4$ and the torus translations are proved. Full $O(4)$ is not a symmetry of $T^4_\ell$. `LEMMAS.md` ledger item 5 [VERIFIED QUOTE]: on the torus the inclined comparison "gives only the tilted-axis reflection symmetry $R_{2\theta s}$, never a rotation"; the template reaches rotations only after the torus $\to$ plane limit, which it controls with its mass gap.

- Infinite volume. $\ell$ is fixed. Jaffe–Witten tie the limit $T^4 \to \mathbb{R}^4$ to a volume-uniform gap [VERIFIED QUOTE]: "New ideas are needed to prove the existence of a mass gap that is uniform in the volume of space-time. Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$." No public mechanism without a gap is known to us (`LEMMAS.md` ledger item 8).

- Mass gap. Not addressed. On a finite torus a "gap" of the transfer matrix is not the Clay mass gap.

- Uniqueness. Different subsequences may give different Schwinger functions. Uniqueness needs E13, which Balaban does not have. Also not given: uniqueness of the vacuum after reconstruction (no clustering on a torus).

- Vacuum state. Time is periodic, so OS reconstruction from torus reflection positivity gives a transfer matrix on $T^3$ and a thermal state at inverse temperature $\ell$, not a vacuum state (Section 4.8, item 4).

- Also not given: clustering (OS4); a probability measure for the density field; Wilson-loop observables (class C3); the stress tensor and the operator product expansion (class C2 and

beyond); independence of the limit from the block-averaging choices of Balaban's RG (only the lattice measure enters $S_{n,\varepsilon}$, but $g_0(\varepsilon)$ is chosen through Balaban's flow).

- Status of the whole statement. Even with every block closed, the theorem is conditional on Balaban's published 1984–89 proofs (GC1). Under Jaffe–Witten footnote 2 it is a compactness ("weak-existence") result on a torus with the torus forms of OS0–OS3; it is not a solution of the Millennium problem.

### 4.8 Corrections after verification

The verifier confirmed the theorem statement, all in-chapter proofs (B-C, B-REG, B-TI, $B_4$, the RP passage, OS3, the finite-form lemma), the quotes of Jaffe–Witten, and the open status of B-N1. Four items were corrected (`ASSEMBLY_VERIFY.md`). The statements above carry the corrected wording; the original claims are listed here.

1. *$\ell$ even is sufficient, not necessary* (verifier §5, grade 3). The parity count is right: $L$ odd gives $N_\varepsilon = \ell L^K$ even iff $\ell$ even; for odd $N_\varepsilon$ every time reflection has one site seam and one link seam. Osterwalder–Seiler (1978) [21] treat link-plane (between-site) reflections. Site-plane reflections were added by Menotti–Pelissetto, CMP 113 (1987) 369–373 [17] [VERIFIED QUOTE] (abstract, per the verifier): "We extend the proof by Osterwalder and Seiler of reflection positivity of lattice gauge theories with fermions to the case in which the reflection is performed with respect to planes containing sites." and "Our proof applies to all observables and relies on the particular structure of the Wilson action." The mixed case (one site seam, one link seam on a periodic lattice) is very likely routine, since each mechanism is local to its seam, but no citation states it. So odd $\ell$ is not excluded; the hypothesis $\ell$ even only avoids the mixed case. The source's phrase "plausible but not cited" stands. The periodic two-seam link version attributed to Osterwalder–Seiler is [CITED], not [VERIFIED QUOTE]. Two minor points from the verifier: the support distance after the seam shift is $\geq 2\delta - \varepsilon$ (the source's $2\delta - 2\varepsilon$ is conservative); the factor $2^{s(n+m)}$ in Step 2 is unneeded but harmless.

2. *Reference title.* The Jaffe–Witten reference list gives the title "Gauge theories on the lattice"; the bibliography entry follows the Annals of Physics title.

3. *Nontriviality needs the two-sided envelope only and support diameters $\ll \ell$* (verifier §6, grade 6). In Theorem 4.3(i), $|R| \leq \frac{1}{2}$ needs $g^2 + (|x-y|/\ell)^2$ small on the whole support pair, so the support diameters must be $\ll \ell$, not only the minimum distance $d_0$. In Theorem 4.3(ii) the source stated the asymptotic $d^8 S_2 \sim c\, g(d)^4$ with $g(d)^2 \sim (\beta \log(1/d))^{-1}$. The proof uses only the two-sided bound (0.31); the asymptotic needs $\beta_j \to \beta_\infty$ and is not proved. The statement of Theorem 4.3(ii) above is the supported one. The overclaim was in the statement, not in the proof.

4. *Periodic time gives a thermal state* (verifier §6, grade 8). OS reconstruction from torus reflection positivity gives a transfer matrix on $T^3$ and a thermal state at inverse temperature $\ell$, not a vacuum. This remark is added to the list of non-results (Section 4.7).

## 5 Numerical results

This section collects the numerical results that support the hypotheses of Section 3. Sections 5.1–5.3 give the one-loop increments and the per-level quantities of the Route K proof. Sections 5.4–5.6 give the values of (H2), the one-step values, and one rigorous point enclosure. Section 5.7 gives the

plan for a certified (H2), whose open parts are listed again in Section 6. SU(2), infinite volume, Balaban's normalized-trace units. None of this is certified. Only odd $L$ is inside the proofs; $L = 2, 4$ use corner blocks and are outside them.

### 5.1 First increment $\beta_1(0)$

Sources: `NUMERICS` §6–§7; `UMK_BETA1.md`.

| $L$ | $\beta_1(0)$ | how | $b = c \log L$ | $\beta_1 - b$ | status |
|---|---|---|---|---|---|
| 2 | 0.630 | $N_g = 12$, $p \to 0$ fit | 0.2575 | +0.372 | [NUMERICAL, REPRODUCED] (`KAPPA_BOUNDED_VERIFY` §4: 0.6296) |
| 3 | 0.664 | $N_g = 8$, $p \to 0$ fit (0.6625–0.6661) | 0.4082 | +0.256 | [NUMERICAL, REPRODUCED]; $N_g$ $6 \to 8$ shift $\approx 0.015$ |
| 4 | 0.757 | $N_g = 6$, single $p = 0.2$; $O((Lp)^2)$ bias $\approx 0.01$ | 0.5150 | +0.242 | [NUMERICAL] |
| 5 | 0.749 ($N_g = 12$, $p = 0.1$); fit 0.743 | umklapp code | 0.5979 | +0.145 | [NUMERICAL] |
| 7 | 0.803 | umklapp code, $N_g = 8$, $Lp = 0.5$ | 0.7229 | +0.080 | [NUMERICAL] |
| 9 | 0.833 ($Lp = 0.5$); 0.846 ($Lp = 0.9$), 0.851 ($Lp = 1.0$) | umklapp code, Windows, $N_g = 8$ | 0.8163 | $\approx +0.02$ | [NUMERICAL] |
| 11 | 0.850 ($Lp = 0.5$); 0.868 ($Lp = 1.0$) | umklapp code, Windows, $N_g = 8$; $N_g = 8$ grid adequacy at $L = 11$ unchecked | 0.8908 | $\approx -0.04$ | [NUMERICAL] |

The first-step constant $\beta_1 - c \log L$ is +0.37, +0.26, +0.24, +0.15, +0.08, +0.02, −0.04 for $L = 2, 3, 4, 5, 7, 9, 11$. It decreases with $L$ and changes sign between $L = 9$ and $L = 11$. $\beta_1(0)$ itself stays positive at every $L$ computed. The umklapp code was validated against `ym_momentum` to $10^{-12}$ per $K$ by its author; no independent verifier for the $\beta_1$ values at $L \geq 5$. The queue continues to $L = 15$ on Windows. No value at $L > 11$ exists yet.

### 5.2 Later increments ($L = 2$ only)

| quantity | value | status |
|---|---|---|
| $\beta_1 + \beta_2$ (composed, $N_g = 4/6/8$) | 1.148 / 0.967 / 0.953 | [NUMERICAL]; single $p = 0.2$, $Lp$ bias label |

| | | |
|---|---|---|
| per-level $\beta_2(0)$ | $\approx 0.32$ ($0.953 - 0.630$). Per-level recursion 0.317–0.341 ($N = 8$–12, single $p = 0.2$). Symmetric Nyquist: 0.33334 ($N = 8$, $p_1 = 0.2$), 0.31679 ($N = 12$, $p_1 = 0.2$), 0.32283 ($N = 12$, $p_1 = 0.4$) (`BETA_FP_VERIFY` §7; `BETA_FP_PW_VERIFY` §5–§6). Level-1 values carry $N$-aliasing of about $\pm 5\%$. | [NUMERICAL]; per-level recursion reproduced in `BETA_FP_VERIFY` §7 |
| $\beta_3(0)$ | 0.40 ($N = 8$), 0.286 ($N = 12$), 0.261 (verifier, symmetric Nyquist, $N = 8$) | unconverged; not usable |
| abelian sector $S_k^{ab}$ | $(1 - L^{-4k})/4$, matched to all digits | [PROVED+VERIFIED]; [LEAN] counting part (`AbelianCount.lean`) |
| (U) test | $\lvert\beta_{j+1}(p) - \beta_{j+1}(0)\rvert \le 0.2p^2$ at every computed point ($L = 2$ levels 0–1, $L = 3$ level 0, $p \le 0.8$); uniformity in $j$ not testable with two levels | [NUMERICAL, RUN BY THE VERIFIER] (`BETA_FP_PW_VERIFY` §6) |

All computed increments are positive.

### 5.3 Quantities of the Route K proof

| quantity | value | status |
|---|---|---|
| $d_1 = b - \beta_1$, $L = 3$ | $-0.256 \pm 0.01$ (fits $-0.254 \cdots - 0.258$) | [NUMERICAL, REPRODUCED] |
| $d_1$, $\kappa_1$, $L = 2$ | $-0.372$, 1.114 | [NUMERICAL, REPRODUCED] |
| $d_2$, $d_3$, $L = 2$ | $\approx -0.44$, $\approx -0.45$ ($d_3$ uses unconverged $\beta_3$) | [NUMERICAL]; bounded-looking, not a test |
| $\rho_1 = \max_K \lVert H_j(K)\rVert / L$, $L = 3$, $j = 1, 2, 3, \infty$ | 1.652, 1.736, 1.757, 1.760 | [NUMERICAL, RUN BY THE VERIFIER, `U3_AW_VERIFY` §6]; refutes (HS); $h^2/L^4 \approx 0.34 < 1$; one-step $C^1$ factor $\rho_1^2/L > 1$ for $j \ge 2$ |
| $\rho_1$, $L = 5$, $j = 1, 2, \infty$ | 1.860, 1.9565, 1.963; $\rho_1^2/L = 0.771$ | [NUMERICAL]; $j = 1$ and the bound measured by the verifier (`H2_LARGE_L_VERIFY` §6) |
| $\rho_2$ ((H2)), $L = 3, 5, 7, 9$ | see Section 5.4: 2.0634, 2.108, $\ge$ 2.1131, $\ge$ 2.1142 at $j = \infty$ against thresholds 3, 5, 7, 9 | [NUMERICAL, REPRODUCED] at $L = 3, 5$; [NUMERICAL, RUN BY THE VERIFIER] at $L = 7, 9$ |
| $a_*$, $L = 5$ | 0.0107 (binding mode near $K = 0$, gauge/ghost sector) | [NUMERICAL, RUN BY THE VERIFIER] (`H2_LARGE_L_VERIFY` §6) |
| (H″), $n = 2$, $L = 3$ | $2.056 < 2.828$ | [NUMERICAL, RUN BY THE VERIFIER], coarse grid, one level pair |
| $\sigma_j^{(1)}(0)$ $J$-part, $j = 0, 1, 2$ ($d = 2$ proxy) | 0.750000, 0.75015, 0.75051 vs 0.75 | [NUMERICAL, RUN BY THE VERIFIER] supports Theorem W |

| | | |
|---|---|---|
| $\sigma_j^{(2)}(K)$ scan | not run: no $d = 4$ constant-field per-momentum code | — |

### 5.4 Values of (H2) [NUMERICAL; NONE CERTIFIED]

Code: `numerics/h2_scan.py` (umklapp Schur pieces, one colour, Gram form; `H2_LARGE_L` §1). Code structure CONFIRMED (`H2_LARGE_L_VERIFY` grade 1): the one-colour reduction is exact at $U = 1$, the $H$ and $\Delta_{out}$ blocks and the Gram identity are correct, the separable Prop. A form is exact. Sign slip in `H2_LARGE_L` §4 ($H = +[\mathfrak{B}^{-1}]_{f,c}$); harmless.

| $L$ | $j$ | $\rho_2$ (scan max) | threshold $L$ | status |
|---|---|---|---|---|
| 3 | 2 | 2.00857 | 3 | [NUMERICAL, REPRODUCED] (`B2_TWOSTEP_VERIFY` §4; `H2_LARGE_L_VERIFY` §1) |
| 3 | 3 | 2.05601 | 3 | [NUMERICAL, REPRODUCED] (spot value 2.05599 at the argmax) |
| 3 | $\infty$ | 2.0634 (2.06332–2.06377 over $l_{max}$ 2–5) | 3 | [NUMERICAL, REPRODUCED]; margin 31%; $q_1 \leq 0.473$, $q_2 \leq 0.053$ |
| 5 | 2, 3, 4, $\infty$ | 2.098, 2.108, 2.108, 2.108 | 5 | [NUMERICAL, REPRODUCED]: independent dense code gives 2.098283 at $j = 2$; the verifier's own search gives 2.107921 at $j = \infty$, same argmax; margin 58% |
| 7 | $\infty$ | $\geq$ 2.1131 (face diagonal only) | 7 | [NUMERICAL, RUN BY THE VERIFIER] (spot 2.11309) |
| 9 | $\infty$ | $\geq$ 2.1142 (one point, $(2.6, 2.6, \pi, \pi)$) | 9 | [NUMERICAL, RUN BY THE VERIFIER]; the $j = 2$ point of `H2_LARGE_L` gives $\geq 2.1104$ |

- Shape. The argmax lies on the face $K_3 = K_4 = \pi$, on the diagonal $K_1 = K_2$: $\approx 2.22 \to 2.40$ ($L = 3$, $j = 2 \to \infty$), $\approx 2.54$ ($L = 5$). Off-face points are lower: best off-face 1.995 ($L = 3$, $j = 2$), 2.004 ($L = 3$, $j = \infty$), 2.088 ($L = 5$, $j = \infty$). The corner $(\pi, \pi, \pi, \pi)$ stays below the face peak, but the gap shrinks: 0.0070, 0.0028, $\geq 0.0025$, $\geq 0.0023$ ($L = 3, 5, 7, 9$). The corner is the obvious place for a competing maximum at large $L$.

- Level dependence. At $L = 5$, $\rho_2(\infty) - \rho_2(j) = 9.64 \cdot 10^{-3}$, $4.0 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ ($j = 2, 3, 4$); ratios 24.1, 25.0, against $L^2 = 25$. At $L = 3$ the ratios are 7.5, 8.7. Levels $j = 4, \ldots, \infty - 1$ are not sampled at $L = 3$.

- $L$-trend. $j = \infty$: 2.0633, 2.1079, $\geq 2.1131$, $\geq 2.1142$ ($L = 3, 5, 7, 9$). This is consistent with a bounded $\rho_2$ near 2.11–2.14. The rate claim "at least as fast as $L^{-2}$" is CORRECTED (`H2_LARGE_L_VERIFY` grade 5): it used a lower bound at $L = 7$ as if it were the sup. "(H2) gets easier with $L$" is a measured trend for $L \leq 9$, not a result for $L > 11$.

- All values are lower estimates of a sup (grid plus local search).

### 5.5 One-step values

The arithmetic of Section 3.7.4 shows that $\rho_1^2 < L$ implies (H2). The measured one-step values are as follows.

- $L = 3$: $\rho_1^2 = 3.10 > 3$. The one-step route fails (Section 3.7.2); the two-step map is needed.
- $L = 5$: $\rho_1 = 1.860$ ($j = 1$, Wilson fine kernel, dense, measured by the verifier), 1.9565 ($j = 2$), 1.963 ($j = \infty$). So $\rho_1^2 \le 3.853 < 5$ and $\rho_1^2/L = 0.771$ [NUMERICAL, CONFIRMED AS A MEASUREMENT, UNCERTIFIED; `H2_LARGE_L_VERIFY` GRADE 4].
- $L = 7$: $\rho_1 \ge 2.032$ (diagonal only), $\rho_1^2/L \approx 0.59$ [NUMERICAL].

So at $L \ge 5$ the one-step map suffices for Theorem C1, as far as measured. For certification this replaces $L^8$ blocks per $K$ (two-step map) by $L^4$ blocks per $K$ (625× fewer at $L = 5$).

### 5.6 A rigorous point enclosure at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED]

Source: `H2_CERTIFY.md` with the prototype `numerics/h2cert_proto.py`; verifier `H2_CERTIFY_VERIFY.md`. At $L = 3$, $j = \infty$, and the face point $K^* = (2.4035, 2.4035, \pi, \pi)$ near the argmax of Section 5.4, ball arithmetic (python-flint) gives

$$\rho_2(\infty, K^*) \in [1.9993,\ 2.1273] < 3.$$

The width comes from the umklapp tail and scales as $N^{-4}$ in the truncation $N$. The verifier reran the code at $N = 12$ and obtained [1.5475, 2.5792], which contains the floating-point value 2.063361 and confirms the $N^{-4}$ scaling; at $N = 6$ the code stops with an assertion instead of returning a wrong interval. The verdict is CONFIRMED given the premises P1–P3 of `H2_CERTIFY_VERIFY`; P1 (the loop-family coefficients of the (0.4) row are exact rationals $k/648$, $k$ even) is proved there, and the fidelity of the loop family to Bałaban's paper stays inside P3. Further results of the same two files:

- One-step factorization: $\rho_2(j, K) \le \rho_1(j, K) \cdot \max_l \rho_1(j-1, k_l)$, exact [PROVED+VERIFIED]. So two-step work is needed only where the right side is $\ge L$, near the face diagonal.
- Oishi–Rump bound $\|\mathfrak{B}^{-1}\| \in [42.7,\ 43.6]$ at $K^*$ (bounds CONFIRMED). The values 177 and 195 of Section 3.9 use the full $\Delta$ instead of $\frac{1}{4}\Delta$; this factor reconciles them. No $\|\mathfrak{B}^{-1}\|$ enters $\rho_2$.
- The umklapp tail enclosure is CONFIRMED for $M = \infty$. For finite $M$ it is CORRECTED: its lower bound must be 0, with class-wise factors $(\pi/2)^{12}$ for two or more large components.
- A uniform Lipschitz grid is refuted as a route: $\Lambda_K \approx 2.8 \cdot 10^4$ (CONFIRMED) would need about $10^{13}$ cells. The replacement is a per-cell invertibility test with a mean-value enclosure.
- Cost: `H2_CERTIFY` estimates about $1.5 \cdot 10^4$ core-hours for a full certification at $L = 3$. The verifier grades this estimate UNSUPPORTED: at the tested sizes the arb overhead dominates, and the box-sum speedup is overstated.

This is one point at one level. It does not certify the supremum over the Brillouin zone or over $j$.

### 5.7 Certification plan and $j_0$ [DESIGNED, NOT DONE]

`B2_TWOSTEP_VERIFY` grade 6 (CORRECTED in size): certifying (H2) is a computer-assisted proof project, not a check. `H2_LARGE_L` §4 gives the design; `H2_LARGE_L_VERIFY` grades 6–8 check it. Three parts.

(a) Tail $j \geq j_0$. From the resolvent identity, $\rho_2(j) \leq R_\infty + (\rho_1^2/a_*)(\delta_{j-2} + \delta_{j-1})$, where $R_\infty = \rho_2$ at $j = \infty$, $a_* = \inf_K a_j(K)$ with $\|[\mathfrak{B}^{-1}]_{f,f}\| = 1/a_j(K)$ exactly, and $\delta_m = \sup_K \|\Delta_{L^m} - \Delta_\infty\|$ (O5 decay) [derivation CONFIRMED]. Inputs CORRECTED by the verifier: the 2-norm sup of $\delta_m$ is at the corner, $\delta = 179$, 92.0, 18.7, 2.39, 0.317 ($M = 1, 3, 9, 27, 81$), 45–90× the assumed value at small $m$; $a_*(L = 5) = 0.0107$, not 0.05. With these inputs, $j_0 \approx 8$ ($L = 3$) and $\approx 6$ ($L = 5$). With the O6 coercivity constant, $j_0 = 25$, which cannot be used in practice (`B2_TWOSTEP_VERIFY` §5 estimated $j \gtrsim 45$ with $\|\mathfrak{B}^{-1}\| \leq b(L) \sim 10^{19}$).

(b) Levels $2 \leq j < j_0$ and $j = \infty$. A certified sup over the BZ of $\|H_{(2),j}(K)\|$ (at $L \geq 5$: of $\|H_j(K)\|$, Section 3.7.4). Plan: interval arithmetic with local mean-value enclosures on boxes in the sorted chamber (1/384 of the BZ), $4 \times 4$ and $5 \times 5$ verified inverses, one $4 \times 4$ $\lambda_{max}$ per $K$; about $2.5 \cdot 10^3$ boxes at half-width 0.05 at $L = 3$; the analytic form of $\Delta$ near $K = 0$, where $\rho_2(0) = 1$. At $j = \infty$ also a certified tail bound for the Prop. A sum ($l_{max}$ truncation; observed effect $4 \cdot 10^{-4}$). There is no explicit global Lipschitz constant today: the A0/O5 strip width is existential, and the constant that can be made explicit is $\sim 10^{40}$. Monotonicity in $j$ is not proved and is not used.

(c) Inputs to certify first: $\|\mathfrak{B}_j(K)^{-1}\| \leq B_{num}$ for all $j$, $K$ (the O6 constant $b(L)$ is unusable), the O5 constant $C_0$ explicitly (or certified $\delta_m$), certified $a_*$, $\rho_1$, $R_\infty$.

The run plan (`H2_LARGE_L_VERIFY` grade 8) uses atomic JSON writes and pre-copied profile caches; the profile build time at $L \geq 11$ is not budgeted. No certification run has been launched.

## 6 What remains

This section lists every open item of the paper in one place. The items are grouped by the kind of work they need: certification, fixed-level and structural inputs of Theorem 2w, items of the sources block, and verification. The item labels H, 1–13, and B1–B8 are those of the dependency table (Section 4.6). For Theorem 2w ⇒ Theorem 1 the list is complete in the sense of §10 of `THEOREM2_PROOF_v4.md`.

### 6.1 Certification

Single uniform hypothesis (both routes):

H. (H2) $\rho_2 = \sup_{j \geq 2} \sup_{K \text{ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L$ [NUMERICAL AT $L = 3, 5, 9$ (TWO-STEP) AND $L = 13$ (ONE-STEP, $\sup \rho_1^2 = 4.37 < 13$, UNVERIFIED); CERTIFICATION OPEN]. It replaces ($C1\sigma$) + (h) and (H″) of the earlier reductions as the one uniform-in-level inequality. Route K: (H2) ⇒ ($C1\sigma$) ⇒ (B2) [PROVED+VERIFIED, `B2_TWOSTEP_VERIFY` GRADE 3]. Route B: (H2) + (T1) ⇒ (U) [PROVED, UNVERIFIED, `BETA_FP_U` THEOREM 2.4]. Values ($j = \infty$, uncertified lower estimates of the sup): 2.0634 ($L = 3$, threshold 3, margin 31%), 2.108 ($L = 5$, threshold 5, margin 58%), $\geq 2.1131$ ($L = 7$), $\geq 2.1142$ ($L = 9$, one point). At $L \geq 5$ the one-step condition $\rho_1^2 < L$ suffices ($\rho_1^2/L = 0.771$ at $L = 5$). Certification (Section 5.7): a computer-assisted proof project with tail level $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$), certified $\|\mathfrak{B}^{-1}\|$, and interval arithmetic over the BZ; At $L = 13$ the one-step certificate is sized at about 900–1,100 core-hours after fix G1; launch approved with conditions, not launched (`H2_CERTIFY_L13_G1.md`).

One point is certified so far: at $L = 3$, $j = \infty$, and $K^* = (2.4035, 2.4035, \pi, \pi)$, $\rho_2 \in [1.9993, 2.1273]$ [INTERVAL ENCLOSURE, VERIFIED] (Section 5.6). Certified early steps $\beta_1(0), \dots, \beta_{j_*}(0) > 0$ at odd $L > 11$ are needed only for literal (0.31) at every $K$ (Section 3.9.2).

### 6.2 Fixed-level and structural inputs of Theorem 2w

Route K hypothesis set (all must hold):

1. (B2) $\sup_j \sup_K |\sigma_j^{(2)}| < \infty$ (or (B2w)). Reduced to H [VERIFIED REDUCTION]. Older reductions kept: $(C1\sigma)$ + (h) [PROVED+VERIFIED]; (H″) [ARGUED]. (HS) is refuted.
2. (PI) probe independence, at each fixed level [ARGUED].
3. (F) lattice = continuum log coefficient at constant field [CITED SOURCE CHECKED; LATTICE TRANSFER STANDARD, NOT PROVED HERE].
4. $(M_j)$ formal = true $f^2$ coefficient at each fixed level: (S1) $\mathcal{K}_t(f) \geq -C_1|f|$, (S2) density of states $\leq C_2|f|$ near 0, (S3) relative bound of the level difference, all on the lattice [OPEN]; Prop. 4.4 [ARGUED].
5. V1 order-$g^0$ completeness of CMP 109 (1.3)–(1.5): $\beta_{j+1}(0)$ is the Maxwell coefficient of the per-step Gaussian determinant [OPEN; SUPPORTED AT ORDER $p^2$].
6. (G2) per-cell infinite-volume densities [FORMAL].

Route B hypothesis set (alternative to 1–5; per `BETA_FP_U` §4; all [PROVED, UNVERIFIED] as implications, no verifier file for `BETA_FP_U`):

B1. (T1) $\sup_j \sup_{q \text{ real}} |c_{T,j}^{(1)}(q)| < \infty$, the plane-wave form of Theorem A3 [ARGUED; THE LANDAU-TO-SYMMETRIC GAUGE CONVERSION AND STEPS 2–5 OF A3 FOR (2.1) ARE NOT WRITTEN].

B2′. (H2) = item H.

B3. (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$, the gauge-fixing difference at the plane wave [HYPOTHESIS, ARGUED]. With the exact split $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ (`BETA_FP_U` Prop. 3.1) there is no $Y$-defect; all non-stationarity sits in $\mathfrak{d}_F$. This replaces the earlier $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ items for this route (Section 3.6.2).

B4. Lüscher–Weisz (5.7) [CITED].

B5. A6: the plane-wave increment equals Balaban's $\beta$ up to the expansion point, uniformly in $k$ [ARGUED]. A uniform-in-$k$ input, of the same kind as (DF).

B6. (H0) = item 7 (Lemma D″ (i)).

B7. $\theta_* < 1$, so that $c_2 \ln(1/\theta_*^2) \geq 0$ in $C_B$ [TRIVIAL CONDITION ON THE CHOICE OF $\theta_*$].

Route B also needs 6 ((G2)) and the items 8–13. It does not need (PI), $(M_j)$, V1; it needs (DF) and A6 instead.

Common to both routes:

7. (H0) Balaban's smoothness in $g$ with uniform bounds, $|\beta_j(g) - \beta_j(0)| \leq C_1 g$ uniform in $j$ and history [BALABAN; CMP 109 INDUCTIVE ASSUMPTION AT ODD $L > 11$; NO EXPLICIT PROOF OF THE $g$-DERIVATIVE BOUND LOCATED IN CMP 116].

8. L5 history dependence [OPEN]. (a) The clamp proof of Lemmas D′/D″ is valid only if $\beta_{k+1}(g_0, \dots, g_k)$ is defined and continuous on the whole cube $]0, \gamma]^{k+1}$; Balaban does not supply this. (b) Balaban's $\beta_{k+1}$ exists only on histories where his induction has run; that domain must carry (c′) + (S), not only $g_i \in ]0, \gamma]$. Clamped histories can violate the growth side of (c′). (c) True trajectories with all $u_i \geq \gamma^{-2}$ inherit (c′) from (a″). A nested-interval variant of Lemma D″ on these trajectories would avoid the gap [NOT WRITTEN]; the variant is written only for Lemma D (`LEMMAS_DE_VERIFY` §1.5). Lean covers one-argument $\beta$ only (Lemmas D′, D″, D, E).

9. $K$-independence of $\beta_j(g)$ [OPEN]. The shooting needs $\beta_j(g)$, $\gamma$, and the constants independent of $K = -\log_L \varepsilon$. After rescaling, each step is the same unit-lattice block step, so $\beta_j$ is $K$-independent only in the infinite-volume limit; a finite volume brings $K$-dependence back. This infinite-volume identification is not proved (`LEMMAS_DE_VERIFY` §2 item 2; `THEOREM2_PROOF_VERIFY` Point 7). It is used implicitly through (G2) and (H0).

11. Odd $L$. All one-loop proofs need odd $L \geq 3$ (centered blocks: U3 Lemma 1.2, O5/Prop. A, evenness, Lemma 4.1 of LEMMA43). Balaban assumes $L$ odd $> 11$ throughout CMP 109, so (H0) and $\beta_j(g)$ at $g > 0$ exist only there. Theorem 2w is stated for odd $L > 11$, or with (H0) at this $L$ as a hypothesis. Numerics: $\beta_1(0)$ up to $L = 11$ (Section 5.1); (H2) at $L = 3, 5$ (scans), 7 (face diagonal), 9 (one point); $\rho_1$ at $L = 3, \dots, 13$. Only $L = 13$ is inside Bałaban's range, and it is floating point.

12. SU(2) only. The charged/neutral split of Lemma 5.1 and (W) are SU(2) statements; $c = 11N^2/(12\pi^2)$ is general but the chain is not checked for $N > 2$.

### 6.3 Items of the sources block

These items stand between the trajectory of Section 3 and the bound (H-U) of Section 4.2. Tags and files are those of the dependency table.

- (B-S) The sourced version of Bałaban's Theorem 1, (E14-a, b, c), is [OPEN] as a whole.

- (H-hist) via (G-gain). The small-field step is done: pointwise Cauchy and uniform $C_\beta$ on $]0, \gamma]$ [PROVED+VERIFIED]. Lemma G3 (the $\chi_k$ surface term) holds given (Q); (Q) is a verified reduction to CMP 99 (3.132), ($E_U$), and LOC. Second derivatives: (H-hist2). Open: the large-field case and the Bałaban-level inputs of (Q).

- (H-hist2), on which the threshold-shell bound (H-thr) depends: [VERIFIED REDUCTION] to the real uniform (H-hist), no new input; with it the M3 history weights and the frozen-integer convention (L3).

- (FX). O1(b), O2, and R2 are [PROVED+VERIFIED]; O3 is [PROVED+VERIFIED, CONDITIONAL ON N3]; O4 is avoided by Bałaban's main form (1.98). N3, the HBETA strip bound $|\delta w| \leq \frac{3}{2}\tau$, is reduced to (H-hol) + real (H-hist) + (H2-hist) (condition (a) settled); (H-hol) is a verified reduction to (M-A) (via (H−) and CMP 102 (189)), (M-B) (verified), and (M-C) [OPEN]. Also the two minor inputs of R2 ($R_{K+1} \geq 1$, $C_c$ independent of $g$). (FX2) is conditional on O3, so also on N3.

- (B-S.D2C, B-S.H$\beta$) The domain hypothesis (D), and the conditions (I) and (H-hist) of the two-loop bound.

- (B-N1) The bridge from the source form to (H-U) derived given (S) (`ASSEMBLY_N1.md`, confirmed); residual: uniformity of the D3R constants down to separation $16\varepsilon$. Given (FX), the step $(\mathrm{S}_{\mathrm{src}}) \Rightarrow$ (H-U) with $s = 6$, $D > 1$ is confirmed.

- (NT) = (E14-f), for nontriviality (Theorem 4.3); at every layer for part (ii). The constant $c > 0$ must be computed in Bałaban's normalization.

- E13, full-sequence convergence: not needed for Theorem B; [OPEN].

### 6.4 Verification debt

10. Balaban readings: the sweep (Section 3.4.3) is grep plus page images, not line-by-line, with 4 of 19 rows checked by a second reader; "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) not located; $\beta_0$ must be fixed before $\gamma$, while Balaban says $\beta_0$ "can be chosen arbitrarily small, if $g$ is sufficiently small" (not checked line by line); `BALABAN_EXCERPTS_CHECK` item 11 lists unchecked readings (II (4.37)/(5.1), CMP 119 (2.42), (2.12)/(2.23), the "$L$ odd" page, Sect. 3 estimates, 1989 bound lists); Theorem 1 itself rests on Balaban's published proofs.

B8. Verification: `BETA_FP_U` has no verifier file. `BETA_FP_PW_VERIFY` §6 items 1–4 grade the logic of Theorem B [PROVED+VERIFIED, CONDITIONAL] (tag decision, `PLAN.md`); the inputs (U), (DF), (T1) stay [PROVED, UNVERIFIED].

13. Lean formalization of the history form of Lemmas D′/D″ [OPEN]. Lemma D″ in the one-argument form is done (Section 3.4.2).

- Results tagged [PROVED, UNVERIFIED] need a verifier pass; Appendix B lists them. They include the sweep, the step Theorem 2w ⇒ Theorem 1, the `BETA_FP_U` reductions, `BETA_FP_M` Prop. 4.3, the Prop. 3.6 assembly, and the unverified files H2_CERTIFY_L13 and H2_CERTIFY_P4.

- The paper is not verified as one document. Its statements carry the tags of the project files.

### 6.5 Open for stronger statements

Not needed for Theorem 1, open for the stronger statements: (A) in full and Lemma 5.1 for $m = 2$ (convergence, rate); $C^2$, $C^3$ of $\sigma^{(1)}$ under (H2) (not written); certified $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at odd $L > 11$ (literal (0.31) at every $K$); explicit $j_*$; (N).

## A Lean formalization

The shooting lemmas of Section 3.4, the arithmetic of Theorem K and Theorem C1, and the bridge lemmas of the certifier are formalized in Lean 4 with Mathlib (17 files in `lean/YmLean/`). The inventory is as follows.

[LEAN] (no sorry; axioms propext, Classical.choice, Quot.sound)

- Lemma D′ (a)–(d) (`YmLean.lemma_DPrime`), Step 8 (`lemma_DPrime_step8`), Lemma D, Lemma E ($Cg^2$ form), Lemma D+E (`lean/YmLean/LemmaDPrime.lean`, `LemmaDE.lean`). One-argument $\beta$ only.

- Lemma D″ (`YmLean.lemma_DDoublePrime`, `lean/YmLean/LemmaDDoublePrime.lean`, 146 lines): (a″) bracket $(3b/4)(K-k) - 2R \le T(k,K) \le (5b/4)(K-k) + 2R$; (b) shooting for $g \in ]0, g_1]$ with the clamp inactive; (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, for every trajectory with couplings in $]0, \gamma]$; (d) for $n - m \ge 8R/b$, lower constant $b/2$. Hypotheses weaker than the note (continuity, any admissible $\gamma$, $R \ge 0$ derived). One-argument $\beta$ only; L5 not formalized.

- Full inventory (`lean/README.md`; file, main theorems, source note):

  1. `LemmaDPrime`: `lemma_DPrime`; `LEMMA_DPRIME.md` §1–§2.
  2. `LemmaDE`: `lemma_D`, `lemma_E`, `lemma_DE`, `lemma_DPrime_step8`; `LEMMAS.md`.
  3. `LemmaH`: `gronwall_telescope`, `lemma_H`; `SOURCES_HBETA.md` Lemma H.
  4. `DetIdentities`: `det_bordered`, `det_restrict_add_gauge`; `numerics/NUMERICS.md` §1.
  5. `Telescoping`: `telescoping_det`, `telescoping_log`; `R2_TELESCOPING.md` Prop. 3.1.
  6. `AbelianCount`: `su2_count`, `abelian_share`; `U1_INFINITE_LATTICE.md` Prop. 9.1 (used here only as a cross-check, Section 5.2).
  7. `LemmaDDoublePrime`: `lemma_DDoublePrime`; `KAPPA_BOUNDED.md` §5.3.
  8. `TheoremK`: `telescoping_bound`, `window_of_partial` (the window bound (5.1) from partial sums); `KAPPA_BOUNDED.md` §5.
  9. `Gronwall`: `discrete_gronwall`, `contraction_bound` ($N_{j+1} \le qN_j + Z$); `KAPPA_BOUNDED.md` Prop. 4.1.
  10. `CubeCount`: `cube_count_sharp`, `cube_count_lower` (Lemma O1-R′ arithmetic); `SOURCES_FX_O1.md`.
  11. `BlockNorm`: `rho2_le`, the exact factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j-1, k_l)$; `H2_CERTIFY.md`.
  12. `OishiRump`: `oishi_rump`, `oishi_rump_eta`, `frobenius_power_bound`; `H2_CERTIFY.md` §5.
  13. `Envelope`: `envelope_nontrivial`, `envelope_not_scale_invariant` ((B-NT) in log form); `ASSEMBLY.md` §2.9.
  14. `Divergence`: `profile_divergence`, `P4_divergence` (identity (D) of premise P4); `H2_CERTIFY_P4.md`.
  15. `TheoremC1`: `c1_two_step`, `c1_kappa_bound`, `c1_window`; `B2_TWOSTEP.md` §2.
  16. `StripBound`: `strip_bound_linear`, `strip_bound_quadratic` (the $\frac{3}{2}\tau$ bootstrap); `SOURCES_HBETA_N3.md` §4.
  17. `CertBridge`: `cell_invertible`, `cell_sup_bound`, `vertex_bound` (interval-to-norm bridge of the cell certifier); `numerics/h1cert_cell.py`, `h1cert_cell2.py`.

  Not formalized: interval arithmetic itself, the enclosure premises P1′–P4, the identity $\rho_1^2 L^2 = \lambda_{\max}(G)$, and the history form of the shooting lemmas.

# B Index of results by status

This appendix indexes the results of the paper by status tag; the tags match the verifier files. Here v1–v4 denote the successive versions `THEOREM2_PROOF.md`, `THEOREM2_PROOF_v2.md`, `THEOREM2_PROOF_v3.md`, and `THEOREM2_PROOF_v4.md` of the Theorem 2 notes.

[PROVED+VERIFIED]

- Lemma D′ (a)–(d), $K_0$, Remark 4 (`LEMMA_DPRIME_VERIFY` grades 1, 2, 5).
- Lemma D (corrected interval), Lemma E ($Cg^2$ form), nested-interval variant of Lemma D (`LEMMAS_DE_VERIFY`). Lemma E $C^1$ form: trivial variant, same proof.
- Lemma D″ (tail-sum form, no $J$) (`KAPPA_BOUNDED_VERIFY` §6).
- Telescoping $S_k = \sum \beta_j(0)$ for the $p \to 0$ coefficient (`BETA_FP_VERIFY` item 1, corrected scope).
- Prop. 2.1 anchor-free identity; exact at the constant field (`BETA_FP_LEMMA43_VERIFY` item 1; `BETA_FP_H_VERIFY` item 1).
- $c = 11N^2/(12\pi^2)$, normalization chain; (F) source (Savvidy eq. (1.4)) (`BETA_FP_LEMMA43_VERIFY` item 2; `BETA_FP_H_VERIFY` §4).
- KAPPA Prop. 1.1 as an identity given its fixed-level inputs; Lemma 2.1 (integration by parts); Prop. 2.2; level-independence of the degree $-4$ part; Theorem K and Theorem 2w as conditional reductions (`KAPPA_BOUNDED_VERIFY` (i)–(v)).
- $\sup_j |\sigma_j^{(2)}(0)| < \infty$; Prop. 4.1 algebra (`KAPPA_BOUNDED_VERIFY` (vi), reason for A4 failure corrected: obstruction = $\partial\sigma^{(1)}$ in $Q_j$ and in (W-2), §7 there).
- `B2_TWOSTEP` two-step recursion; Theorem C1 ((H2) ⇒ ($C1\sigma$), factor $\rho_2^2/L^2$) and Theorem B2 ((H2) ⇒ (B2), factor $\rho_2^2/L^4$), conditional on (H2); threshold $\rho_2 < L$ (no strip loss); (H2) weaker than (H″) at $n = 2$ (`B2_TWOSTEP_VERIFY` grades 1, 3, 4). Form of $Q_j$ by read-off (grade 8; `U3_LEMMA51` (3.3) not re-derived).
- `h2_scan` code structure, one-colour reduction, $H$ and $\Delta_{out}$ blocks, Gram identity, separable Prop. A form (`H2_LARGE_L_VERIFY` grade 1); tail-bound derivation (4.1)–(4.3) (grade 6, inputs corrected).
- Logic of Theorem B and Lemma 1.3 of `BETA_FP_PW`, as conditional implications (`BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED: doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B; Lemma 4.1, conditional). Its inputs (U), (DF), (T1) are listed under [PROVED, UNVERIFIED] and [ARGUED].
- Prop. E evenness, Theorem W0/W, Theorem A3, Lemma A4, A0, Prop. A2 (`U3_AW_VERIFY` grades 1–4).
- U3 Theorem 4.1 + O5, Prop. A, Prop. B, O6, Theorem 6.1 (given V1; Taylor-data norm), Lemma 5.1 neutral value bound, single-block $m = 2$ failure in $d = 4$, one-step recursion, R2 midpoint identity (`U3_VERIFY`, `U3_REPAIRS_VERIFY`, `U3_LEMMA51_VERIFY`, `U3_LEMMA51_R23_VERIFY`).
- Exact one-step factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j-1, k_l)$ (`H2_CERTIFY_VERIFY`; Lean `rho2_le`).
- Sources block: Lemma G3 given (Q) (`SOURCES_G3_VERIFY`); LOC (`SOURCES_LOC_VERIFY`); (M-A) given (H−) and (M-A′) (`SOURCES_MA_VERIFY`, `SOURCES_MA2_VERIFY`); (M-B) with $p_0 \ge 17/(\gamma_0 A_1^2)$ (`SOURCES_MB_VERIFY`); condition (a) of Lemma N3 (`SOURCES_N3A_VERIFY`); the sourced large-field draft B-S.D3, with corrections (`SOURCES_D3_VERIFY`, `SOURCES_D3R_VERIFY`, `SOURCES_D3W8_VERIFY`); (B-N1) given (S) (`SOURCES_HHIST_N1_VERIFY`).

- Cesàro: given convergence, $\kappa_k = o(k) \Leftrightarrow \beta_\infty = c \log L$ (equivalence, not a weaker form) (`THEOREM2_PROOF_VERIFY` Point 4).

[PROVED, UNVERIFIED]

- `BETA_FP_U`, no verifier file: Theorem 2.4 ((U) $\Leftarrow$ (T1) + (H2)), Prop. 3.1 ($\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$), Prop. 3.2 ($\gamma_{rec}$ bounded), Corollary 4.1, Theorem 2w by the plane-wave route given (T1), (H2), (DF), LW (5.7), A6, (H0); the argued step "(H2) closes ($C1\sigma$)" of §3.5 there is superseded by `B2_TWOSTEP` Theorem C1 [PROVED+VERIFIED].

- `LEMMA_DPRIME_SWEEP`: 19 cross-scale uses covered by (b) + (c′) + (S) (4 of 19 table rows, 11, 12, 16, 18, verified on page images by `LEMMA_DPRIME_VERIFY` comparisons A–E).

- Step "Theorem 2w ⇒ Balaban's Theorem 1" (depends on the sweep and on L5 in the sharpened form of Section 3.4.5).

- Prop. 3.6 assembly with $C(1+j)^2 L^{-2j}$; neutral-sector $K$-derivative paragraph; $\sigma^{(1)}$ convergence sketch $jL^{-2j}$.

- Necessity of individual positivity for literal (0.31) at every $K$ (elementary).

- `BETA_FP_M` Prop. 4.3 ($\log|\det|$ bound in the N–O window, given (S1)–(S3)); no verifier file.

- Premise P4 of the one-step certificate (`H2_CERTIFY_P4`; identity (D) in Lean `P4_divergence`); the $L = 13$ certificate sizing after fix G1 (`H2_CERTIFY_L13_G1`).

[NUMERICAL]

- $\rho_1(\infty) = 1.760, \ldots, 2.0895$ at $L = 3, \ldots, 13$; at $L = 13$, $\sup \rho_1^2 = 4.37 < 13$, so (H2) holds in floating point by `rho2_le`; no verifier (`H2_CERTIFY_L13`).

[VERIFIED REDUCTION]

- (Q) to CMP 99 (3.132), ($E_U$), and LOC (`SOURCES_Q_VERIFY`, `SOURCES_CDC_VERIFY`); ($E_U$) to kernel-moment bounds Bałaban does not print (`SOURCES_EU_VERIFY`; the bound $K \le K_0(1 + \ldots)$ is a [DERIVED SKETCH]).

- N3 to (H-hol) + real (H-hist) + (H2-hist) (`SOURCES_HBETA_N3_VERIFY`).

- (H-hol), small field, to Bałaban's conditions with modified constants (`SOURCES_HHOL_VERIFY`); large field to (M-A), (M-B), (M-C), (H-hist2) (`SOURCES_HHOL_LF_VERIFY`); (M-D) to (H-hist) + (H-thr).

- (H−) to CMP 102 (189), an existing trust item (`SOURCES_MA2_VERIFY`).

- (H-hist2) to the real uniform (H-hist), no new input (`SOURCES_HHIST2_VERIFY`).

[INTERVAL ENCLOSURE, VERIFIED]

- Point enclosure $\rho_2(\infty, K^*) \in [1.9993, 2.1273] < 3$ at $L = 3$; P1 (loop-family coefficients exact rationals); Oishi–Rump bound on $\|\mathfrak{B}^{-1}\|$ (`H2_CERTIFY_VERIFY`).

[ARGUED]

- (PI); $(M_j)$ (Prop. 4.4 of `BETA_FP_M`); (U) as a bare hypothesis (now reduced, see above); (T1) (plane-wave form of A3, `BETA_FP_U` §2.3); (DF) (`BETA_FP_U` §3.4); A6 = plane-wave vs constrained-minimizer expansion point (uniform in $k$); $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ bounds; (H″) ⇒ (A) in full ⇒ (B2) (`U3_AW` §4.7 sketch); $C^2$, $C^3$ of $\sigma^{(1)}$ under (H2) (not written); $j_*$ size; the rate $O(\Lambda^{-2}\log\Lambda)$.

[OPEN]

- (H2) (certification; numerical at $L = 3, 5, 9$; one-step at $L = 13$). (M-C), the complex small-field step. (B2) is reduced to it [VERIFIED REDUCTION]. (T1), (DF), A6 for Route B. V1. (S1)–(S3). L5 (domain of $\beta$; nested-interval D″ [NOT WRITTEN]). $K$-independence of $\beta_j(g)$. (A) in full. See Section 6.

[CORRECTED BY VERIFIERS SINCE V3]

- `B2_TWOSTEP` G1: the certification of (H2) is a computer-assisted proof project, not a check (`B2_TWOSTEP_VERIFY` grade 6).
- `H2_LARGE_L`: $L$-trend rate claim (grade 5); $j_0$ inputs, $\delta_m$ 45–90× larger, $a_*(L = 5) = 0.0107$ (grade 6), giving $j_0 \approx 8$ ($L = 3$), $\approx 6$ ($L = 5$).
- `BETA_FP_PW`: comparison and numerics (`BETA_FP_PW_VERIFY` grades 5, 6).

[REFUTED]

- (HS) $\sup \|H_j\| < L\sqrt{L-1}$ at $L = 3$.
- Not a refuted statement, but a failed method: the one-step sup-norm $C^1$ recursion at $L = 3$ (factor 1.005–1.033 for $j \geq 2$; Section 3.7.2).
- `U3_LEMMA51` Theorem 4.1 for $m = 2$ with a single block-$M$ constraint in $d = 4$.
- v1 claims withdrawn: "every early step must be certified for Theorem 1" (false: Lemma D′/D″); "(H′) is weaker than $\beta_\infty = c\log L$" (it is equivalent); "the whole $\log\Lambda$ sits in the fine piece" as a fact (it is equivalent to the open statement); "one-loop parts hold for SU($N$)" (Lemma 5.1's charged/neutral split is the SU(2) Cartan split); (3.2) with $Cj^2L^{-2j}$ (0 at $j = 0$); the (M) resolvent bound $\|\mathcal{K}_t^{-1}\| \leq C/|f|$ (Nielsen–Olesen zero); (H) with IR weights as stated (UNSUPPORTED).

[BALABAN]

- (H0) smoothness and uniform bounds of $\beta_{j+1}$ in $g$ (CMP 109 (1.22), pp. 263–264), at odd $L > 11$ only; Theorem 1 and the inductive bounds of CMP 109, 116, 119, 122 and the propagator papers CMP 95, 96, 98, 99, 102. Uniform level bounds partly cited from CMP 95/96/99 (O6 replaced the main citation).

[VERIFIED QUOTE]

- Theorem 2 and Theorem 1 wording, "separate paper", p. 298 history sentence, (0.1) with $1/g_k^2$ and $\chi_k$, (1.3)–(1.5) (print misprint $E^{(j+k)}$ read as $E^{(j+1)}$), (2.6)–(2.9) of CMP 119, items A–E of CMP 122 (`BALABAN_EXCERPTS_CHECK`; `LEMMA_DPRIME_VERIFY`; `LEMMA_DPRIME_SWEEP` page images). `BALABAN_EXCERPTS_CHECK` items 10 (CMP 116 has no $C^m$-in-$g$ statement) and 11 (unchecked readings) are unchanged since v3.

[NUMERICAL] see Section 5.

# C Label map

The numbered environments of this paper carry the names used in the project files. The first table maps them. The second table maps the section numbers of `THEOREM2_PROOF_v4.md` ("v4") and `ASSEMBLY.md` to the sections of this paper; section references inside the project files use the old numbers.

| this paper | name in the project files | source |
|---|---|---|
| Theorem A | Route K summary: Theorem 2w ⇒ Theorem 1 | `THEOREM2_PROOF_v4` §6, §10 |
| Theorem B | Theorem 1.3 (summary) | `ASSEMBLY` §1.3 |
| Theorem 3.1 | Theorem 2w | `THEOREM2_PROOF_v4` §1.3 |
| Lemma 3.2 | Lemma D′ | `LEMMA_DPRIME`; v4 §2.1 |
| Remark 3.3 | Remark 4 | `LEMMA_DPRIME`; v4 §2.1 |
| Lemma 3.4 | Lemma D″ | `KAPPA_BOUNDED` §5.3; v4 §2.2 |
| Lemma 3.5 | Lemma D | `LEMMAS`; v4 §2.4 |
| Lemma 3.6 | Lemma E | `LEMMAS`; v4 §2.4 |
| Proposition 3.7 | Prop. 2.1 (anchor-free identity) | `BETA_FP_LEMMA43`; v4 §3.2 |
| Theorem 3.8 | Theorem K | `KAPPA_BOUNDED`; v4 §4.1 |
| Theorem 3.9 (Theorem $B_{PW}$) | Theorem B = Theorem 3.2 | `BETA_FP_PW`; v4 §4.2 |
| Theorem 3.10 | Theorem C1 | `B2_TWOSTEP`; v4 §5.3 |
| Theorem 3.11 | Theorem B2 | `B2_TWOSTEP`; v4 §5.3 |
| Definition 4.1 | (H-U) | `ASSEMBLY` §1.2 |
| Theorem 4.2 | Theorem 1.3 | `ASSEMBLY` §1.3 |
| Theorem 4.3 | Theorem 1.4 | `ASSEMBLY` §1.4 |
| Proposition 4.4 | Proposition of (B-C) | `ASSEMBLY` §2.4 |
| Lemma 4.5 | finite-form lemma of (B-RP) | `ASSEMBLY` §2.7 |
| (GC1)–(GC3) | (G1)–(G3) of the dependency table | `ASSEMBLY` §3 |
| ($S_{src}$) | (S), final sourced bound | `SOURCES_D4D5` |
| items H, 1–13, B1–B8 | items of the open list | `THEOREM2_PROOF_v4` §10 |

| old section | this paper |
|---|---|
| v4 §1.1 | Section 1.2 |
| v4 §1.2 | Section 3.1 |
| v4 §1.3 | Section 3.2 |
| v4 §1.4 | Section 3.3 |
| v4 §2 (notation) | Section 2.3 |
| v4 §2.1 | Section 3.4.1 |
| v4 §2.2 | Section 3.4.2 |

| old section | this paper |
|---|---|
| v4 §2.3 | Section 3.4.3 |
| v4 §2.4 | Section 3.4.4 |
| v4 §2.4 (L5) | Section 3.4.5 |
| v4 §3.1 | Section 3.5.1 |
| v4 §3.2 | Section 3.5.2 |
| v4 §3.3 | Section 3.5.3 |
| v4 §3.4 | Section 3.5.4 |
| v4 §3.5 | Section 3.5.5 |
| v4 §4 | Section 3.6 |
| v4 §4.1 | Section 3.6.1 |
| v4 §4.2 | Section 3.6.2 |
| v4 §4.2a | Section 3.6.3 |
| v4 §4.3 | Section 3.6.4 |
| v4 §5 | Section 3.7 |
| v4 §5.1 | Section 3.7.1 |
| v4 §5.2 | Section 3.7.2 |
| v4 §5.3 | Section 3.7.3 |
| v4 §5.4 | Section 5.4 |
| v4 §5.5 (arithmetic) | Section 3.7.4 |
| v4 §5.5 (values) | Section 5.5 |
| v4 §5.6 | Section 5.7 |
| v4 §5.7 | Section 3.7.5 |
| v4 §6 | Section 3.8 |
| v4 §7 | Section 3.9 |
| v4 §8.1 | Section 5.1 |
| v4 §8.2 | Section 5.2 |
| v4 §8.3 | Section 5.3 |
| v4 §9 (Lean) | Appendix A |
| v4 §9 (other tags) | Appendix B |
| v4 §10 (open list) | Section 6 |
| v4 §10 (hypothesis table) | Section 4.6 |
| v4 §11 | Section 1.5 |
| v4 preamble (tag legend) | Section 1.4 |
| v4 preamble (range of $L$) | Section 2.2 |
| v4 change logs | Appendix D |
| `ASSEMBLY` §0 (notation) | Section 2.1 |
| `ASSEMBLY` §0 (observables) | Section 2.5 |
| `ASSEMBLY` §1.1 | Section 4.1 |
| `ASSEMBLY` §1.2 | Section 4.2 |
| `ASSEMBLY` §1.3 | Section 4.3 |
| `ASSEMBLY` §1.4 | Section 4.4 |
| `ASSEMBLY` §2 | Section 4.5 |
| `ASSEMBLY` §2.1 | Section 4.5.1 |
| `ASSEMBLY` §2.2 | Section 4.5.2 |
| `ASSEMBLY` §2.3 | Section 4.5.3 |
| `ASSEMBLY` §2.4 | Section 4.5.4 |
| `ASSEMBLY` §2.5 | Section 4.5.5 |
| `ASSEMBLY` §2.6 | Section 4.5.6 |
| `ASSEMBLY` §2.7 | Section 4.5.7 |
| `ASSEMBLY` §2.8 | Section 4.5.8 |
| `ASSEMBLY` §2.9 | Section 4.5.9 |

| old section | this paper |
|---|---|
| ASSEMBLY §3 | Section 4.6 |
| ASSEMBLY §4 | Section 4.7 |
| ASSEMBLY corrections | Section 4.8 |

# D Revision history

- *Theorem 2 notes, v3.* Built on THEOREM2_PROOF_v2.md with the corrections of THEOREM2_PROOF_v2_VERIFY (cited there as v2_VERIFY): the range of $L$ (odd, $L > 11$, or (H0) at this $L$ as a hypothesis); L5 sharpened (the clamped shooting leaves Bałaban's domain; the nested-interval variant of Lemma D″ is [NOT WRITTEN]); (H″) ⇒ (B2) tagged [ARGUED]; "independent" routes replaced by "formally distinct", with the points $\theta_* < 1$, $k$-uniform $\beta$ identification, and the first-order vertex smoothness of (U); the independence record completed (ONLINE_SWEEP §2 as the first channel; the O(3) paper cited); open list extended (item 9, B5, item 10, hypothesis table); sweep count "4 of 19 table rows"; $\gamma_*$ depends on $R$ through $B'$. At that time BETA_FP_PW_VERIFY.md had no grade, and Route B stayed [PROVED, UNVERIFIED]. The grades of LEMMA_DPRIME_VERIFY and KAPPA_BOUNDED_VERIFY §9(iv) were cited explicitly.

- *Theorem 2 notes, v4.* Added the material verified after v3: B2_TWOSTEP.md with B2_TWOSTEP_VERIFY.md, H2_LARGE_L.md with H2_LARGE_L_VERIFY.md, lean/README.md (Lemma D″ in Lean), BETA_FP_U.md (Route B inputs), and numerics/NUMERICS.md §7. The section on (B2) was rewritten around (H2) (Theorems C1, B2; threshold $\rho_2 < L$; one-step sufficiency at $L \geq 5$; certification plan; corrected $L$-trend and $j_0$ inputs). The ledger and open list were updated, with item H as the single uniform hypothesis of both routes. KAPPA_BOUNDED_VERIFY states the (B2) obstruction in its §7 and §9(vi). BALABAN_EXCERPTS_CHECK had no change since v3.

- *Tag change in the LaTeX version of v4.* The logic of Theorem $\mathrm{B_{PW}}$ is tagged [PROVED+VERIFIED, CONDITIONAL], citing BETA_FP_PW_VERIFY §6 items 1–4, per the tag decision in PLAN.md (2026-10-28). The Markdown v4 kept [PROVED, UNVERIFIED]. The inputs (U), (DF), (T1) from BETA_FP_U stay [PROVED, UNVERIFIED].

- *Assembly.* ASSEMBLY.md with the corrections of ASSEMBLY_VERIFY.md (Section 4.8). The verifier also found the dependency table stale in five rows (grade 7; no overclaim): B-T2.2, B-T2.3, B-T2.4, B-S.D2C, B-S.D3 lagged PLAN.md. The table was rebuilt with a tag for every block and the sub-blocks added later (Route K, Route B, (H2), (H-thr), (G-gain), (FX)). Minor items: SOURCES.md §1.3 gives $c_j = 1 + O(g_j^4)$, while this paper uses $O(g^3)$ per SOURCES_VERIFY item 8 (the corrected value); the variance estimate drops a harmless $g_0^4$ (logarithmic) factor.

- *This version.* The Theorem 2 chapter and the assembly chapter are merged into one article: one notation section, one status legend, numbered environments (Appendix C), one numerical section, one list of open items, and the hypothesis table of v4 §10 merged into the dependency table. Three names were changed to avoid collisions (Section 2.6). The dependency table and the open list were brought to the current state of PLAN.md: (FX) is reduced to N3 (O2 and R2 [PROVED+VERIFIED], O3 [PROVED+VERIFIED, CONDITIONAL ON N3], O4 avoided by the main form; previously O2, O3 [PROVED, UNVERIFIED] and R2, O4 [OPEN]), and the point enclosure of (H2) at $L = 3$ is verified (Section 5.6). No theorem, lemma, or proof changed.

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